



**Business Information
Systems Faculty of
Commerce and Business
Administration Helwan
University**



Final Year (Graduation) Project

Second Era

Project No. (102)

Team Members

1.	Ali Yasser Ali Ali	512292440
2.	Ahmed Mohamed Nasser Eldin Mohamed	512292207
3.	Ziad Ramadan Abdelmaqsoud Ali	512292201
4.	Abanoub Hanna Mikhail	512292150
5.	George Wasim Malak Aziz	512292255
6.	Mohamed Abdelkader Mohamed Elsayed	512292439

Supervisors

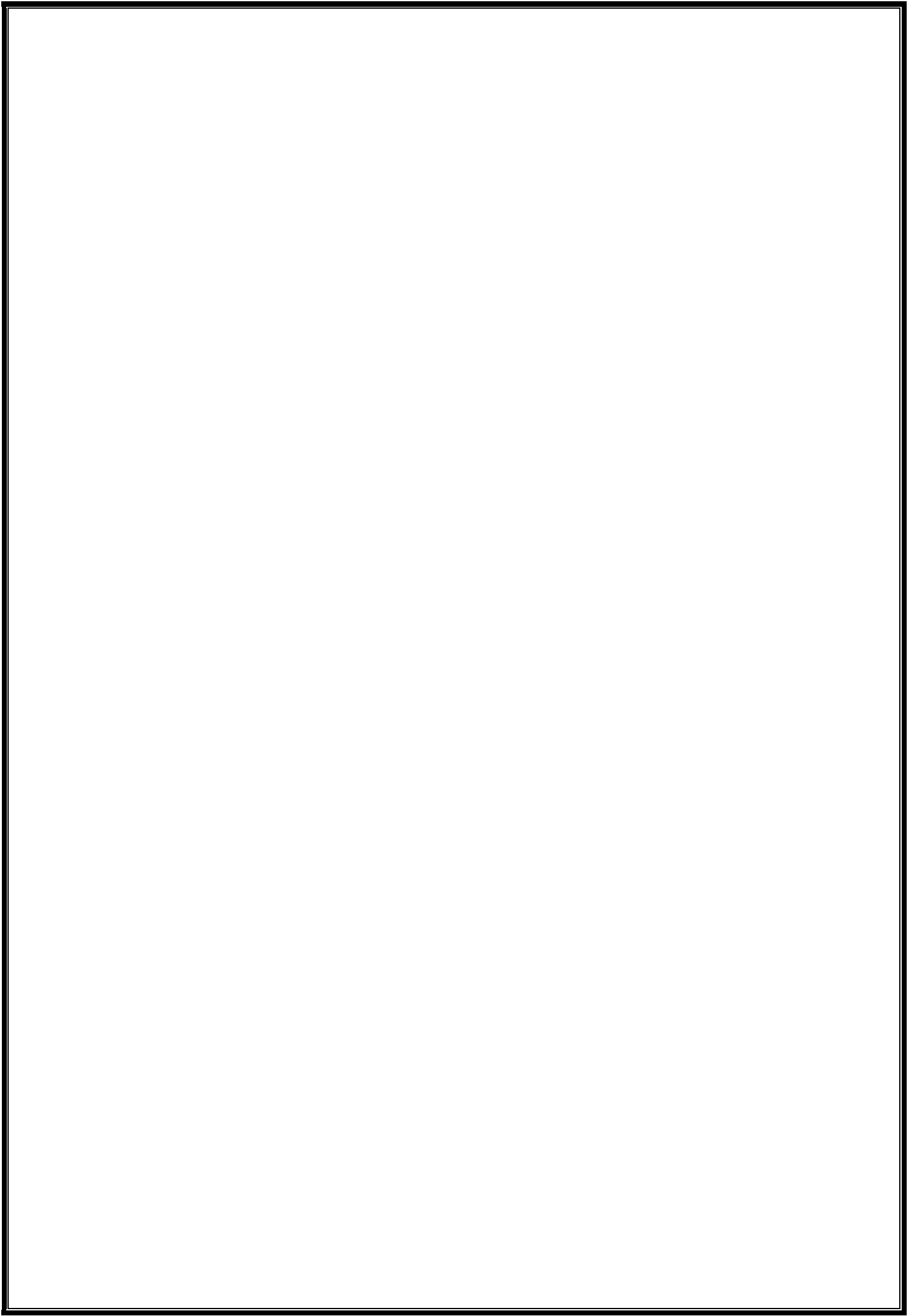
(IT)

Prof. Dr. Ahmed El Seddawy

(Business)

Prof. Dr. Hanan Obayah

2025



DECLARATION

I / we hereby certify that this material is entirely my own work, that I have exercised reasonable care to ensure that the work is original and does not to the best of my knowledge breach any law of copyright and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of my work.

Signed: Team Members

Registration No.:

Date: 15/06/2025

ABSTRACT

Our business idea is to collaborate with companies in the market to provide new graduates and final-year university students with the practical experience and knowledge that typical college courses often lack. We offer courses and hands-on assessments designed to enhance their skills, ultimately reducing both the duration and cost of internships at the companies they aspire to join.

Second Era is a leading provider of innovative educational and training solutions that aim to achieve excellence in skill development and empower young people to achieve their full potential. Founded with a clear vision to provide a distinguished and stimulating educational experience that helps young people grow personally and professionally,

Second Era is an innovative educational and training initiative that aims to empower young people to acquire the skills and knowledge necessary to enter the labor market and succeed in it. This platform is considered a comprehensive educational environment that combines specialized educational content with practical training, which enables young people to develop their personal and professional skills and achieve excellence in various fields.

Second Era also provides professional training and workshops, which are essential elements in the journey of skill development and performance improvement in multiple fields. These two processes are related to providing individuals with the knowledge and skills necessary to succeed in the competitive labor market.

Table of Contents

Chapter One: Introduction	12
1.1 Introduction (Aim of the Chapter).....	12
1.2 Project Idea and Scope	12
1.3 System Objectives	13
1.4 System Features	13
1.5 Related Works (Analysis of Previous Works + Determination of the Existing Problem)	14
1.6 System Requirements	15
1.7 System Users	16
1.8 System Methodology	16
1.9 SDG Alignment and Contributions	17
1.9.1 SDG 4: Quality Education.....	17
1.9.2 SDG 8: Decent Work and Economic Growth	18
1.9.3 SDG 9: Industry, Innovation, and Infrastructure.....	18
1.9.4 SDG 10: Reduced Inequalities	18
1.9.5 SDG 17: Partnerships for the Goals	19
1.9.6 Our Vision for Sustainability	19
1.10 Project Time Plan	19
Chapter Two: Bussnies Plan	22
2.1 Executive Summary.....	22
2.2 Vision & Mission.....	23
2.2.1 Our Vision	23
2.2.2 Our Mission	23
2.2.3 Our Values.....	23
2.3 Business Idea & Market	24
2.4 Marketing Plan.....	26
2.4.1 Competitor Analysis	26
2.4.2 Description of the market	31
2.5 Legal Form.....	51
2.6 Start-up Capital & Sources of Start-up Capital.	52

2.6.1 Start-Up Capital.....	52
2.6.2 Sources of Start-up Capital.....	52
2.7 Financial Plan	53
2.7.1 Organization and Staff.....	53
2.7.2 Business Operation and Costs	54
2.7.3 Expected Revenues.....	55
2.7.4 Expected Expenses	57
2.7.4 Income Statement	58
2.7.5 Cashflow	58
2.7.6 Balance Sheet	59
2.7.7 Owner Equity.....	60
Chapter Three: System Analysis	62
3.1 Introduction.....	62
3.2 User Requirements	62
3.2.1 Functional Requirements.....	62
3.2.2 Non-functional Requirements	65
3.3 System Requirements	66
3.3.1 Software Requirements	66
3.3.2 Hardware Requirements	66
3.4 System Diagrams	67
3.4.1 UML Use Case Diagram	67
3.4.2 UML Sequence Diagram.....	68
3.4.3 UML Activity Diagram	71
Chapter Four: System Design	81
4.1 Introduction:	81
4.2 Class Diagram:.....	81
4.3 Database Structure (ERD):	83
4.4 Database Mapping:	85
Chapter Five: System Implementation	89
5.1 Introduction.....	89
5.2 System screenshots	89

5.2.1 Home page.....	89
5.2.2 About Page	91
5.2.3 Plans Page.....	93
5.2.4 Contact & Support Page	94
5.2.5 Help Page.....	95
5.2.6 Register Page	97
5.2.7 login Page:	101
5.2.8 Dashboard Page:	103
5.2.9 Home Page (After Login):.....	106
5.2.10 Student Only Pages:	109
5.2.11 Instructor Only Pages:	116
5.2.12 Company Representer Only Page:	120
5.2.13 Admin Pages:.....	122
5.2.14 System code & database.....	128
Chapter Six: Testing Methodology.....	133
6.1 Introduction.....	133
6.2 Testing Methodology.....	133
6.3 Login Validation	134
6.3.1 Validations related to the login process include:	134
6.4 Sign Up Validation	135
6.4.1 Validations covering the user registration process include:....	135
6.5 Main Function Validation.....	137
6.5.1 Validations encompassing core application features beyond login and sign up:	137
6.5.2 Jumping Attack Validation.....	137
Chapter Seven Conclusion and Future Work	140
7.1 Introduction (Aim of the Chapter).....	140
7.2 Conclusion	140
7.3 Future Work.....	141
References.....	142

Table Of Figures

FIGURE 1. PROJECT MILESTONE	20
FIGURE 2. PROJECT TIMELINE	20
FIGURE 3. SWOT	37
FIGURE 4. OUR OFFICE.....	45
FIGURE 5. COMPANY HIERARCHY	53
FIGURE 6. USE CASE DIAGRAM.....	67
FIGURE 7. SEQUENCE DIAGRAM (STUDENT)	68
FIGURE 8. SEQUENCE DIAGRAM (INSTRUCTOR)	69
FIGURE 9 SEQUENCE DIAGRAM (ADMIN).....	69
FIGURE 10. SEQUENCE DIAGRAM (COMPANY).....	70
FIGURE 11. ACTIVITY DIAGRAM (REGISTRATION)	71
FIGURE 12. ACTIVITY DIAGRAM (COURSE REGISTRATION)	72
FIGURE 13. ACTIVITY DIAGRAM (SUBMIT ASSIGNMENT)	73
FIGURE 14. ACTIVITY DIAGRAM (APPLY TO JOB).....	74
FIGURE 15. ACTIVITY DIAGRAM (CREATE COURSE).....	75
FIGURE 16. ACTIVITY DIAGRAM (ADD ASSESSMENT).....	76
FIGURE 17. ACTIVITY DIAGRAM (POST JOB).....	77
FIGURE 18. ACTIVITY DIAGRAM (REVIEW JOB)	78
FIGURE 19. CLASS DIAGRAM	82
FIGURE 20. ERD	84
FIGURE 21. MAPPING	86
FIGURE 22. HOME PAGE (HERO SECTION).....	89
FIGURE 23. HOME PAGE (INFO SECTION)	90
FIGURE 24. HOME PAGE (CTA SECTION, FOOTER)	90
FIGURE 25. ABOUT PAGE (1).....	91
FIGURE 26. ABOUT PAGE (2).....	91
FIGURE 27. ABOUT PAGE (3).....	92
FIGURE 28. ABOUT PAGE (4).....	92
FIGURE 29. ABOUT PAGE (5).....	93
FIGURE 30. PLANS PAGE (1).....	93
FIGURE 31. PLANS PAGE (2).....	94
FIGURE 32. CONTACT & SUPPORT PAGE (1).....	94
FIGURE 33 CONTACT & SUPPORT PAGE (2).....	95
FIGURE 34. HELP PAGE (FAQ).....	95
FIGURE 35. HELP PAGE (FAQ).....	96
FIGURE 36. HELP PAGE (FAQ).....	96
FIGURE 37. REGISTER PAGE	97
FIGURE 38. REGISTER PAGE (STUDENT).....	98

FIGURE 39. REGISTER PAGE (INSTRUCTOR)	99
FIGURE 40 . REGISTER PAGE (COMPANY REPRESENTER)	100
FIGURE 41. LOGIN PAGE (PUBLIC & STUDENT (AFTER REGISTRATION)) ..	101
FIGURE 42. LOGIN PAGE (INSTRUCTOR & COMPANY REPRESENTER (WAIT UNTIL ADMIN APPROVE) (AFTER REGISTRATION))	102
FIGURE 43. DASHBOARD PAGE (STUDENT)	103
FIGURE 44. DASHBOARD PAGE (INSTRUCTOR).....	104
FIGURE 45. DASHBOARD PAGE (COMPANY REPRESENTER)	105
FIGURE 46. HOME PAGE (AFTER LOGIN) (STUDENT)	106
FIGURE 47. HOME PAGE (AFTER LOGIN) (INSTRUCTOR).....	107
FIGURE 48. HOME PAGE (AFTER LOGIN) (COMPANY REPRESENTER)	108
FIGURE 49. ENROLL PAGE.....	109
FIGURE 50. MY COURSES PAGE	110
FIGURE 51. COURSE DETAILS PAGE	111
FIGURE 52. ASSESSMENT DETAILS PAGE	112
FIGURE 53. FIND JOB PAGE	113
FIGURE 54. PAYMENT PAGE	113
FIGURE 55. JOB DETAILS PAGE	114
FIGURE 56. APPLY FOR JOB SECTION	115
FIGURE 57. CREATE COURSE PAGE	116
FIGURE 58. MY COURSES PAGE (INSTRUCTOR).....	117
FIGURE 59. NEW ASSIGNMENT PAGE	118
FIGURE 60. NEW ASSIGNMENT NOTIFICATION	118
FIGURE 61. COURSES DETAILS PAGE (INSTRUCTOR).....	119
FIGURE 62. POST NEW JOB PAGE	120
FIGURE 63. MY LISTING JOB PAGE	121
FIGURE 64. VIEW APPLICATION JOBS.....	121
FIGURE 65. ADMIN (DASHBOARD)	122
FIGURE 66. ADMIN (DASHBOARD)	122
FIGURE 67. ADMIN (REPORTS PAGE)	123
FIGURE 68. USER MANAGEMENT PAGE.....	124
FIGURE 69. USER MANAGEMENT PAGE.....	124
FIGURE 70. ADMIN (SUBSCRIPTION PAGE)	125
FIGURE 71. PENDING PAGES (COMPANY REPRESENTER).....	126
FIGURE 72. PENDING PAGES (INSTRUCTOR)	126
FIGURE 73. PENDING PAGES (REJECT SECTION).....	127
FIGURE 74. CODE BACKEND	128
FIGURE 75 CODE BACKEND	128
FIGURE 76 CODE BACKEND	129
FIGURE 77 CODE FRONTEND	129

FIGURE 78 CODE FRONTEND	130
FIGURE 79 CODE FRONTEND	130
FIGURE 80. DATABASE.....	131
FIGURE 81. LOGIN VALIDATION.....	134
FIGURE 82. LOGIN VALIDATION.....	134
FIGURE 83. SIGN UP VALIDATION.....	136
FIGURE 84. SIGN UP VALIDATION.....	136
FIGURE 85. SIGN UP VALIDATION.....	136

List Of Tables

TABLE 1. LIST OF ACRONYMS/ABBREVIATIONS	10
TABLE 2. PROJECT TIME PLAN	19
TABLE 3. START-UP CAPITAL.....	52
TABLE 4. SOURCES OF START-UP CAPITAL	52
TABLE 5. ORGANIZATION AND STAFF	53
TABLE 6. BUSINESS OPERATION AND COSTS.....	54
TABLE 7. EXPECTED REVENUES.....	55
TABLE 8. EXPECTED REVENUES (SUMMARY).....	56
TABLE 9. ASSUMPTIONS.....	56
TABLE 10. EXPECTED EXPENSES.....	57
TABLE 11. INCOME STATEMENT.....	58
TABLE 12. CASHFLOW	58
TABLE 13. BALANCE SHEET.....	59
TABLE 14. OWNER EQUITY	60

List Of Acronyms/Abbreviations

Acronym	Definition Of Acronym
SWOT	Strengths, Weaknesses, Opportunities, and Threats
STP	Segmentation, Targeting and Positioning
PESTEL	P - Political, E - Economic, S - Social, T - Technological, E - Environmental, L - Legal

Table 1. List Of Acronyms/Abbreviations

Chapter One

1. INTRODUCTION

- 1.1 Introduction (Aim of The Chapter)**
- 1.2 Project Idea and scope**
- 1.3 System Objectives**
- 1.4 System Features**
- 1.5 Related Works (analysis of previous works + determination of the existing problem)**
- 1.6 System Requirements**
- 1.7 System Users**
- 1.8 System methodology**
- 1.9 Project Time Plan**

Chapter One: Introduction

1.1 Introduction (Aim of the Chapter)

This chapter introduces the *"Second Era"* educational platform, a pioneering EdTech initiative designed to bridge the critical gap between academic education and practical industry requirements. The aim is to develop a comprehensive online system that facilitates skill development, industry collaboration, and personalized learning experiences for students, graduates, and companies.

By providing tailored training programs, assessments, and interactive content, the project aspires to enhance employability, reduce internship costs, and foster continuous professional growth. This chapter lays out the foundational concepts, scope, objectives, and the strategic approach for developing this innovative platform.

1.2 Project Idea and Scope

Project Idea

"Second Era" is envisioned as an innovative educational ecosystem that collaborates with industry partners to offer practical, industry-relevant courses and assessments to university students and recent graduates. Unlike traditional e-learning platforms, this system emphasizes real-world skills, hands-on experience, and direct engagement with companies to prepare learners for the workforce.

Scope

- **Target Audience:** University students in their final years, recent graduates, and industry professionals seeking upskilling.
- **Core Offerings:** Customized courses, live workshops, practical assessments, and mentorship opportunities.
- **Geographical Reach:** Initially focusing on Egypt, with plans for international expansion into regional markets.
- **Functional Modules:** User registration, course management, trainer and company portals, assessment tools, communication channels, and feedback systems.

- **Technological Scope:** Development of a web and mobile-compatible platform, integrating multimedia content, real-time communication, and secure data management.
- **Partnership Scope:** Collaboration with universities, industry players, government agencies, and training providers to ensure relevance and sustainability.

1.3 System Objectives

- **Enhance Skill Development:** Provide practical, industry-relevant training that complements academic curricula.
- **Facilitate Industry Collaboration:** Build strong partnerships with companies to tailor training programs that meet labor market needs.
- **Reduce Internship Duration and Cost:** Offer efficient, targeted assessments and training to shorten internship periods and lower associated costs.
- **Improve Employability:** Equip learners with the skills, certifications, and confidence needed to secure employment.
- **Enable Continuous Learning:** Support lifelong learning through accessible resources, ongoing mentorship, and community engagement.
- **Create a Scalable Platform:** Ensure the system can grow with increasing users, expanding course offerings, and integrating new technologies.
- **Data-Driven Decision Making:** Collect and analyze user data to refine programs, enhance user experience, and demonstrate impact.

1.4 System Features

- **User Management:** Registration, profile customization, progress tracking, and certification issuance.
- **Course Catalog:** Diverse offerings including technical skills, soft skills, industry-specific training, and workshops.
- **Interactive Content:** Live webinars, recorded lectures, quizzes, assignments, and simulations.
- **Assessment Tools:** Practical evaluations, project submissions, peer reviews, and real-time feedback.
- **Trainer and Industry Partner Portals:** Dedicated interfaces for trainers to upload content and schedule sessions, and for companies to propose projects and provide feedback.

- **Communication & Collaboration:** Chat rooms, forums, direct messaging, and virtual meetups for active engagement.
- **Analytics & Reporting:** Dashboards to monitor learner progress, course completion rates, and assessment outcomes.
- **Admin Panel:** Manage users, content, partnerships, and platform settings efficiently.
- **Mobile Compatibility:** Fully responsive design for access via smartphones and tablets to maximize reach.
- **Security & Privacy:** Data encryption, secure login, role-based access control, and compliance with data protection laws.

1.5 Related Works (Analysis of Previous Works + Determination of the Existing Problem)

Many existing online education platforms (e.g., Coursera, Udemy, LinkedIn Learning) provide valuable content but often lack direct industry engagement, personalized assessments, and tailored pathways aligned with local labor market demands. Several issues persist:

- **Limited Industry Collaboration:** Most platforms operate independently of industry needs, resulting in skills mismatches.
- **Lack of Practical Assessments:** Many offerings focus on theoretical knowledge without hands-on evaluation or real-world projects.
- **Insufficient Customization:** Content delivery often fails to adapt to individual learner needs or regional market requirements.
- **Weak Engagement & Interaction:** Minimal real-time interaction diminishes learner motivation and practical understanding.
- **Inadequate Feedback & Certification:** Limited mechanisms for feedback and recognition that truly validate skills for employers.

Existing Problems Addressed by "Second Era":

- Creating an industry-integrated, interactive platform that emphasizes practical assessments.
- Offering customized programs aligned with employer needs.
- Facilitating ongoing communication between learners, trainers, and companies.
- Providing real-time analytics for continuous improvement.

This system aims to fill these gaps by delivering industry-specific, customizable, interactive, and assessment-driven training solutions.

1.6 System Requirements

Hardware

- Cloud hosting servers with scalable infrastructure.
- User devices such as desktops, laptops, tablets, and smartphones.

Software

- Web application developed using modern frameworks (e.g., React, Angular, or Vue.js for frontend; Node.js or Django for backend).
- Database systems (MySQL, PostgreSQL, or MongoDB).
- Content Management System (CMS) for easy content updates.
- APIs for third-party integrations (e.g., payment gateways, LMS tools).

Network & Security

- Reliable internet connection with high bandwidth.
- Secure SSL certificates, user authentication protocols, and role-based access control.
- Data encryption and compliance with GDPR or local data protection laws.

Other Requirements

- Content creation tools for trainers.
- Training and support for platform administrators and trainers.
- Continuous system testing and updates for security and functionality.

1.7 System Users

- **Students & Final-Year Students:** Primary beneficiaries engaging with courses, assessments, and certifications.
- **Recent Graduates:** Users seeking upskilling opportunities to improve employability.
- **Industry Partners & Companies:** Collaborate for curriculum input, project proposals, and recruitment.
- **Trainers & Educators:** Content developers and facilitators delivering courses and assessments.
- **Platform Administrators:** Manage system operations, user management, content approval, and partnership relations.
- **Technical Support Staff:** Responsible for maintaining system health, troubleshooting, and updates.

1.8 System Methodology

The project will adopt an **Agile Development Methodology**, emphasizing flexibility, iterative progress, and stakeholder engagement.

Phases Include:

- **Requirement Gathering:** Conduct workshops and interviews with stakeholders to define detailed needs.
- **Design & Prototyping:** Develop wireframes and prototypes to visualize user interfaces and workflows.
- **Development & Integration:** Build core modules, integrate multimedia content, and establish APIs.
- **Testing & Validation:** Conduct usability, security, and performance testing, involving real users for feedback.
- **Deployment:** Launch the platform in phases—initial pilot followed by full deployment.
- **Training & Support:** Provide user training sessions and ongoing technical support.
- **Feedback & Iteration:** Collect user feedback post-launch to refine features and add functionalities.

1.9 SDG Alignment and Contributions

Second Era is committed to integrating sustainable development principles into its mission to empower youth and foster innovation in education. By aligning our initiatives with the United Nations Sustainable Development Goals (SDGs), we aim to create lasting social, economic, and environmental impacts that contribute to a more equitable and sustainable future.



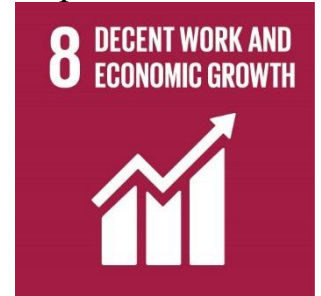
1.9.1 SDG 4: Quality Education

- **Our Commitment:**
 - Deliver high-quality, practical training programs that complement academic curricula, ensuring students acquire relevant skills for today's job market.
 - Develop accessible online educational platforms that enable learners from diverse backgrounds to access training, regardless of geographic or socio-economic barriers.
- **Impact:**
 - Enhances employability and entrepreneurial skills among youth, reducing skills mismatches and unemployment rates.
 - Promotes lifelong learning and continuous professional development.



1.9.2 SDG 8: Decent Work and Economic Growth

- **Our Commitment:**
 - Facilitate smoother transitions from education to employment by providing industry-relevant training and assessments.
 - Partner with companies to offer internships and practical experiences that create pathways to decent work.
- **Impact:**
 - Boosts youth employment opportunities and reduces unemployment.
 - Supports local economic growth by cultivating a skilled, ready-to-work workforce.



1.9.3 SDG 9: Industry, Innovation, and Infrastructure

- **Our Commitment:**
 - Leverage innovative digital platforms to deliver scalable, high-impact training solutions.
 - Incorporate emerging technologies like AI, data analytics, and e-learning tools to modernize vocational education.
- **Impact:**
 - Promotes digital infrastructure development and technological innovation in education.
 - Fosters a culture of continuous innovation aligned with industry standards.



1.9.4 SDG 10: Reduced Inequalities

- **Our Commitment:**
 - Design inclusive training programs accessible to marginalized and underserved communities, including rural areas and socio-economically disadvantaged groups.
 - Implement scholarship schemes or subsidized courses to ensure equitable access to quality education.
- **Impact:**
 - Reduces educational and employment inequalities, promoting social mobility.
 - Empowers marginalized youth to participate actively in the economy.



1.9.5 SDG 17: Partnerships for the Goals

- **Our Commitment:**

- Build strategic partnerships with universities, private sector companies, government agencies, and NGOs to expand the reach and impact of our programs.
- Foster collaborative projects, joint research, and knowledge-sharing initiatives to enhance educational quality and innovation.

- **Impact:**

- Strengthens multi-stakeholder cooperation, facilitating resource sharing and best practices.
- Drives sustainable development through collective efforts and shared expertise.



1.9.6 Our Vision for Sustainability

At **Second Era**, sustainability is at the core of our strategy. We believe that investing in youth education and skill development is fundamental to building resilient economies and inclusive societies. By actively contributing to these SDGs, we strive to inspire a new generation of professionals who are not only skilled but also committed to sustainable growth and social responsibility.

1.10 Project Time Plan

Phase	Duration	Activities
Requirement Analysis	0.5 month	Meetings, detailed specifications, scope confirmation.
System Design	0.5 months	UI/UX design, architecture planning, database schema creation.
Prototype Development	1 week	Initial prototypes for key modules, Supervisor review.
Core Development	1 months	Coding, content creation, API integrations, preliminary testing.
Testing & Quality Assurance	0.5 week	Usability testing, security audits, fixing bugs.
Pilot Deployment	Ongoing	Limited rollout, user onboarding, collecting feedback.
Full Deployment	Ongoing	System launch, marketing, and user training.
Post-Launch Support	Ongoing	Continuous monitoring, updates, feature enhancements based on feedback.

Table 2. Project Time Plan

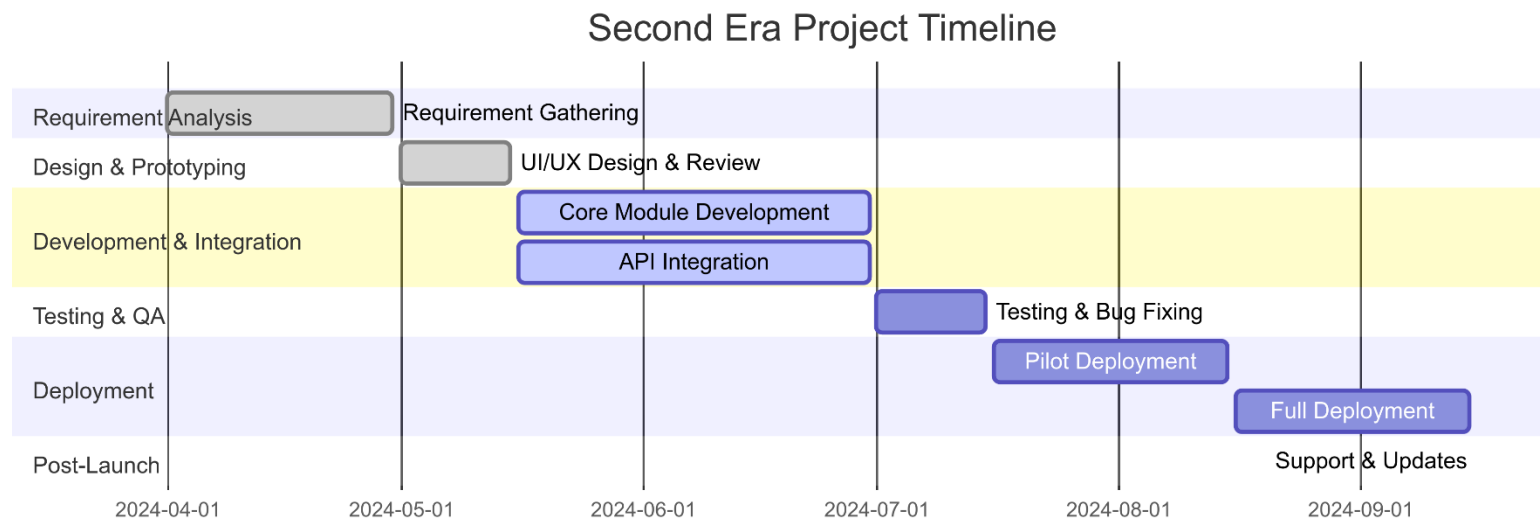


Figure 2. Project Timeline

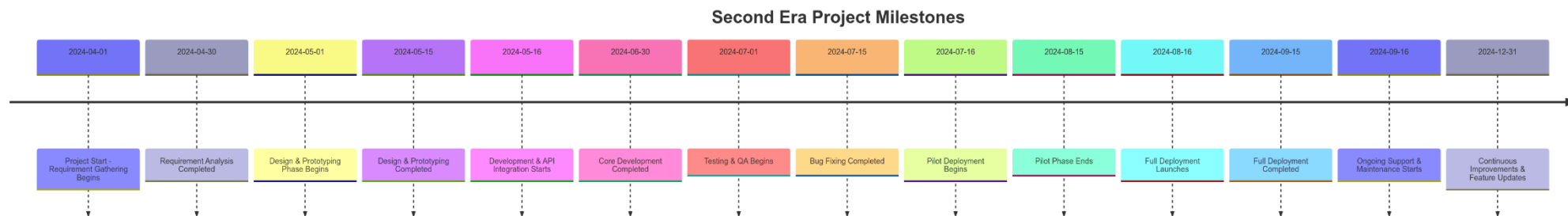


Figure 1. Project Milestone

Chapter Two

2. Bussnies Plan
 - 2.1. Executive Summary.
 - 2.2. Vision & Mission.
 - 2.3. Business Idea & Market
 - 2.4. Marketing Plan.
 - 2.5. Legal Form.
 - 2.6. Start-up Capital & Sources of Start-up Capital.
 - 2.7. Financial Plan.

Chapter Two: Bussnies Plan

2.1 Executive Summary

Name Of Business: “Second Era”

Legal Form: General partnership

Contact Address:

Tel: 01152293752

E-mail: sec.era@info.com

Website: Secound-Era.com

Type of Business: EdTech, service

Brief Description of the business idea:

Our business idea is to collaborate with companies in the market to provide new graduates and final-year university students with the practical experience and knowledge that typical college courses often lack. We offer courses and hands-on assessments designed to enhance their skills, ultimately reducing both the duration and cost of internships at the companies they aspire to join.

Customers: Students, Fresh Graduate, Companies

Owner(s):

- Ali Yasser (CEO)
- Ziad Ramadan (Operations Manager)
- Abanoub Hanna (Marketing Specialist)
- Ahmed Nasser (Customer Support Representative)
- George Wasim (Content Developer)
- Mohamed Abd El-Kader (IT & Platform Administrator)

Number of jobs to be created: 8

Start-Up Capital:

- **Investment: 200,000 EGP**
- **Working Capital: 700,000 EGP**
- **Total: 900,000 EGP**

Source of Capital:

- **Own Savings: 100,000 EGP**
- **Partner: 150,000 EGP**
- **Family: 50,000 EGP**
- **Loan: 100,000 EGP**
- **Bank Loan: 300,000 EGP (5% interest Rate)**

2.2 Vision & Mission

2.2.1 Our Vision

“To envision a world where young people can reach their full potential through education and training.”

2.2.2 Our Mission

“Second Era empowers young people to achieve their full potential by providing them with access to high-quality educational and training programs that are tailored to their needs. We strive to provide an educational and training environment that encourages continuous learning and enables young people to build a promising and bright professional future.”

2.2.3 Our Values

Continuous Learning: We believe in the importance of continuous learning as the basis for personal and professional development.

Excellence: We always strive to provide a distinguished educational and training experience that meets the aspirations and needs of young people.

Cooperation: We work as an integrated team and encourage cooperation and participation between young people, trainers, experts, and companies in the labor market.

Personal Development: We strive to promote the personal development of young people and empower them to achieve their personal and professional goals.

Positive Impact: We aim to create a positive impact on the lives of young people by empowering and inspiring them to achieve their full potential and obtain the best job opportunities that suit their abilities.

Innovation: We encourage creative thinking and the development of new solutions that contribute to improving the performance of companies and individuals.

Contribution To Success: We strive to be an essential part of the success of the companies that we cooperate with by empowering and developing young people and recent graduates and preparing them for work.

2.3 Business Idea & Market

Description of the business idea:

Our business idea is to collaborate with companies in the market to provide new graduates and final-year university students with the practical experience and knowledge that typical college courses often lack. We offer courses and hands-on assessments designed to enhance their skills, ultimately reducing both the duration and cost of internships at the companies they aspire to join.

Second Era is a leading provider of innovative educational and training solutions that aim to achieve excellence in skill development and empower young people to achieve their full potential. Founded with a clear vision to provide a distinguished and stimulating educational experience that helps young people grow personally and professionally,

Second Era is an innovative educational and training initiative that aims to empower young people to acquire the skills and knowledge necessary to enter the labor market and succeed in it. This platform is considered a comprehensive educational environment that combines specialized educational content with practical training, which enables young people to develop their personal and professional skills and achieve excellence in various fields.

Second Era also provides professional training and workshops, which are essential elements in the journey of skill development and performance improvement in multiple fields. These two processes are related to providing individuals with the knowledge and skills necessary to succeed in the competitive labor market.

What We Offer:

❖ Variety Of Educational Programs:

We offer a variety of courses, workshops, and training in different fields, starting from academic specializations to modern practical and professional skills that the labor market needs inside and outside Egypt.

❖ Interactive Educational Experience:

We believe that the best learning comes from interactive educational experiences, so we offer rich and exciting content that includes live and indirect lectures and interactive materials.

❖ **Expert And Professional Trainers:**

We collaborate with a team of expert and professional trainers who combine academic knowledge with practical and professional experience to ensure the provision of high-quality content and increase job opportunities.

❖ **A Space for Communication and Sharing:**

We provide an interactive environment that enables subscribers to communicate and interact with each other and with trainers, which contributes to the exchange of knowledge and experience.

Why We Are Unique:

❖ **Customization Of Programs**

We offer customized programs according to the individual's needs, whether it is to improve current skills or acquire new skills.

❖ **The Balance Between Theoretical and Practical**

We always strive to achieve a perfect balance between theoretical content and practical application to give subscribers a deep understanding and immediate practical skills.

❖ **Sustainable Learning**

We are interested in promoting sustainable learning as we help individuals continue developing their skills after the end of the programs through resources, guidance, and training.

2.4 Marketing Plan

2.4.1 Competitor Analysis

1. Edraak Platform

Summary:

Edraak is an Arab-based online platform that offers free courses in collaboration with renowned universities and academic institutions. Its primary focus is on delivering academic educational content and professional training across a wide range of disciplines. It aims to facilitate access to quality education in the Arab world.

Strengths:

- Majority of courses are offered free of charge, making education accessible to a broad audience.
- Strong partnerships with leading international universities and institutions, ensuring credible and high-quality content.
- Dedicated support for academic content in Arabic, promoting inclusivity and regional accessibility.

Weaknesses:

- Heavy emphasis on theoretical and academic knowledge, resulting in limited practical, hands-on training opportunities.
- Less diversity in fields that require applied skills, such as modern technology sectors (e.g., AI, cybersecurity), which demand experiential learning and real-world applications.
- Limited focus on soft skills and professional development that are crucial in the current labor market.

2. MaharaTech (Skills Platform)

Summary:

Managed under the supervision of the Egyptian Ministry of Communications, MaharaTech offers training courses designed to equip learners with digital and technical skills needed in the evolving job market. It provides both free and paid courses focusing mainly on technical competencies.

Strengths:

- Specialized in high-demand technological skills, including programming, artificial intelligence, and data analysis.
- Government backing enhances credibility and trust among users and stakeholders.
- Focus on skills directly aligned with labor market requirements.

Weaknesses:

- Predominantly technical focus, with minimal emphasis on soft skills such as leadership, communication, or interpersonal skills.
- Lack of integration of broader professional development modules or personalized pathways tailored to individual career goals.
- Limited content diversity outside core technical fields.

3. Al-Tahrir Academy

Summary:

Al-Tahrir Academy provides free online educational content, emphasizing life skills, personal development, and some academic subjects. Its goal is to foster holistic growth through engaging and accessible content.

Strengths:

- Offers comprehensive educational resources that include self-improvement and essential life skills.
- Features an interactive and engaging teaching style, making learning accessible and appealing.

- User-friendly platform tailored to a broad demographic, including youth and early professionals.

Weaknesses:

- Content scope is somewhat limited in specialized, practical, and professional fields relevant to current market demands.
- Insufficient focus on applied skills and industry-specific competencies required for immediate employability.
- Needs expansion in technical, vocational, and career-oriented training modules.

4. Udemy

Summary:

A global e-learning platform providing a vast array of paid courses across numerous disciplines. Its flexibility, affordability, and extensive course catalog make it a popular choice in Egypt and worldwide.

Strengths:

- Extensive variety of courses covering almost every field, from arts to technology.
- Affordable pricing models and frequent discounts make courses accessible.
- Self-paced learning with on-demand access enhances flexibility.

Weaknesses:

- Limited Arabic-language content, which may hinder accessibility for non-English speakers in the region.
- Course quality varies, and content is not always tailored to local market needs or industry standards.
- Insufficient focus on practical, hands-on training, especially in emerging fields that require experiential learning.

5. Rwaq Platform

Summary:

Rwaq is an Arabic platform offering free courses across various fields, with a particular emphasis on open education. It aims to make quality education accessible to Arab learners through diverse disciplines.

Strengths:

- Provides free courses in multiple fields, promoting educational inclusivity.
- Content available in Arabic facilitates user engagement and regional relevance.
- Wide range of disciplines, from academic to professional fields.

Weaknesses:

- Limited focus on practical applications and skills training necessary for immediate employment.
- Courses tend to be more theoretical, with less emphasis on modern, industry-relevant competencies.
- Needs to incorporate more experiential learning modules and industry partnerships.

2.4.1.2 Strategic Recommendations Based on Competitor Insights

- **Prioritize Practical, Applied Training:**
Develop tailored workshops and hands-on programs in high-demand sectors like fintech, software development, cybersecurity, and essential soft skills such as leadership, negotiation, and problem-solving. These should be designed to address the gap between theoretical knowledge and real-world application.
- **Forge Strategic Partnerships with Companies:**
Establish collaborations with local and international firms to facilitate internships, practical training, and employment pathways for learners. This will enhance the platform's attractiveness by directly linking education to employment opportunities.
- **Personalized Learning Pathways:**
Leverage AI and data analytics to offer customized curricula based on continuous assessments. This approach ensures learners receive

targeted guidance aligned with their career aspirations and skill gaps, increasing engagement and success rates.

- **Enhance Interactive and Immersive Content:**

Utilize innovative technologies such as virtual reality, gamification, and interactive simulations to foster active participation. This addresses the current gap in experiential learning and creates a more engaging user experience.

- **Support Continuous Education:**

Provide ongoing support, updates, and free content post-training to promote lifelong learning and professional growth, encouraging learners to stay updated and competitive even after program completion.

2.4.2.2 Potential Challenges and Mitigation Strategies

- **Pricing and Accessibility:**

Given that competitors like Edraak and Rwaq offer free content, it is essential to balance free and paid offerings strategically. Consider free foundational modules complemented by premium, specialized workshops to generate revenue while maintaining accessibility.

- **Building Trust and Brand Recognition:**

Effective marketing campaigns and establishing credibility through partnerships, certifications, and success stories are vital to gain trust among youth and employers.

- **Aligning Content with Market Needs:**

Continuous market research and industry engagement will ensure our offerings remain relevant and responsive to evolving labor market demands.

2.4.2.3 Conclusion

To successfully compete in the regional online education landscape, our platform should focus on delivering practical, interactive, and personalized learning experiences that bridge the gap between education and employment. Building strategic alliances with industry players and leveraging innovative pedagogical techniques will position us as a leader in equipping youth with the skills they need to thrive in the modern workforce.

2.4.2 Description of the market

2.4.2.1 PESTLE Analysis

Political Factors

- **Government Education Initiatives:** Egypt's government has been pushing for reforms in **higher education and vocational training**. Collaborating with **Egyptian ministries, universities, and tech hubs** could provide funding opportunities and legitimacy for our platform.
- **Regulatory and Licensing:** Online education and training services may require **approval from Egypt's Ministry of Higher Education** or partnerships with accredited institutions. Ensuring compliance early will help avoid regulatory issues.
- **Political Stability & Economic Policies:** Egypt has been attracting **foreign direct investment (FDI)** in education and tech. Our startup could **benefit from grants, incubators, and accelerators** that support educational technology (EdTech).
- **Digital Education Policies:** The growth in e-learning during the COVID-19 pandemic highlights government support for digital education, reflecting a commitment to building infrastructure that facilitates technology use in education. [\[1\]](#)
- **Empowering Target Groups:** Supportive policies aimed at increasing educational participation among various demographics—especially women in Upper Egypt—demonstrate e-learning's potential as a tool for political empowerment. [\[2\]](#)
- **Impact of Political Stability:** A stable political environment strengthens the government's ability to implement effective long-term educational policies, fostering confidence in the sustainability of digital education efforts.
- **Regulations and Legislation Clarity:** The establishment of clear laws regarding e-learning, including quality standards and data protection, enhances user trust and promotes institutional adoption.
- **International Relations Influence:** Trade policies and international relationships can significantly impact your ability to collaborate with global e-learning platforms and access external educational content, broadening the scope of resources available to learners.

Understanding these factors will be essential for navigating the EdTech landscape in Egypt and leveraging current government initiatives for growth and impact.

Economic Factors

- **Youth Unemployment & Demand for Skills:** Egypt has a **high youth unemployment rate (~24%)**, making **job-ready training programs highly attractive** to graduates. Companies also seek **cost-effective hiring solutions**, which aligns with our startup's mission.
- **Subscription Pricing & Affordability:** The average income level in Egypt means pricing should be affordable. Offering **tiered pricing, installment plans, or corporate-sponsored training** could attract more students.
- **Startup Funding & Investors:** The **Egyptian EdTech and startup ecosystem** is growing, with investors interested in **education and workforce development**. Consider **joining incubators** like Flat6Labs or Falak Startups for funding and mentorship.
- **Economic Crises and Inflation:** Currency devaluation and inflation have increased costs for training materials and certifications, affecting affordability for consumers and businesses.
- **Informal Economy:** A significant portion of the workforce is employed in the informal sector, where traditional training and certification hold less perceived value, complicating the marketability of training programs.
- **Global Economic Trends:** The importance of sectors like tourism and manufacturing necessitates a focus on specific skill sets, while global economic downturns can influence demand for training services in Egypt.
- **Market Size & Growth:**
 - EdTech market value: \$3.5B (2023)
 - Projected CAGR: 16.5% (2024-2028)
 - Private education spending: 4.1% of GDP
- **Foreign Investment Impact:**
 - FDI in education sector: \$250M (2023)
 - International partnerships increasing by 25% annually.
 - Currency fluctuation affecting pricing strategies.

Social Factors

- **Shift Toward Practical Education:** Growing demand for hands-on, career-focused learning (e.g., internships, certifications) as students prioritize employability over theoretical knowledge. Second Era aligns with this trend by offering real-world training and assessments.
- **Language Preferences:** While English dominates professional education, offering Arabic-language courses and subtitles can broaden accessibility, especially for students from non-elite universities or rural areas.
- **Population Growth & Youth Bulge:** Egypt's high youth population (60% under 30) and rising university enrollment create a competitive job market. Second Era helps graduates stand out with job-ready skills.
- **Age Distribution:** The target audience (ages 20–28) is career-driven and digitally native, preferring flexible, tech-enabled learning formats.
- **Degree vs. Skills Debate:** While some employers still prioritize degrees, students increasingly value skills. Positioning Second Era as an “Internship Accelerator” (complementing, not replacing degrees) helps balance traditional and modern expectations.
- **Proactive Career Mindset:** Students actively seek direct pathways to employment. Second Era's industry partnerships and structured training cater to this demand.
- **Cost Sensitivity:** Many students face financial barriers to unpaid internships. Second Era reduces reliance on costly, prolonged corporate training, offering a budget-friendly alternative.
- **Work-Life Balance:** Students prioritize flexible learning models (online/hybrid courses) to maintain mental health and productivity. Second Era's adaptable formats support this preference.
- **Digital Connectivity:** Tech-savvy students expect seamless digital experiences. Second Era's platform aligns with their preference for self-paced, interactive learning.
- **Demand for Job-Ready Talent:** Companies increasingly value practical skills over academic credentials. Second Era bridges this gap by delivering graduates pre-trained in tools like CRM, SEO, and sales pipelines.

- **Educational Demographics:**
 - University enrollment: 3.1M students
 - Female participation: 51.8%
 - Online learning adoption: 65% growth post-COVID
- **Cultural Considerations:**
 - Family influence on educational decisions
 - Social media impact on learning preferences
 - Professional networking culture

Technological Factors

- **Digital Learning Adoption:** Egypt's **internet penetration rate (71%)** and growing mobile usage make an **online platform highly viable**. However, ensuring **mobile-friendly learning experiences** is crucial.
- **AI & Gamification in Learning:** Many EdTech platforms use **AI-powered personalized learning** and **gamified assessments** to improve engagement. Implementing **automated skill tracking and certificates** could differentiate our service.
- **Integration with Hiring Platforms:** Collaborating with **LinkedIn, Wuzzuf, or local job portals** could make Our platform a direct pipeline for employment, boosting **our value proposition to both students and employers**.
- **Alignment with ICT 2030 Strategy:** The initiatives outlined in Egypt's Vision 2030, particularly in developing ICT infrastructure, fostering digital inclusion, and transitioning to a knowledge-based economy, offer strong support for Our business.
- **Digital Egypt's Pillars:** Focusing on digital transformation, improving digital skills, and fostering innovation aligns well with our business objectives. Support from the government in building digital infrastructure and a conducive legislative framework enhances our potential for growth.
- **Digital Infrastructure:**
 - 5G rollout status and coverage
 - Cloud service availability
 - Cybersecurity landscape
- **Emerging Technologies:**
 - VR/AR adoption in education
 - Blockchain for credentials
 - IoT integration possibilities

- **Key Advantages for Our Business:**

- Strong government support for tech development aligns with Our goals.
- Contribution to national objectives enhances Our value proposition.
- Encouragement of new learning methodologies contributes to skill development across various sectors.

By capitalizing on these factors, our business can effectively contribute to Egypt's digital transformation while navigating the evolving educational landscape. Emphasizing partnerships, leveraging technology, and aligning with national strategies will be crucial for success. [[1](#), [2](#), [3](#)]

Legal Factors

- **Educational Regulations:**

- Higher Education Law No. 72/2019
- E-learning regulatory framework
- Quality assurance requirements

- **Business Operations:**

- Company registration requirements
- Tax implications for EdTech.
- Employment law compliance

- **Intellectual Property:**

- Copyright protection for educational content
- Trademark registration process.
- Patent considerations for technology.

- **Data Protection:**

- Personal Data Protection Law requirements (similar to GDPR)
- Cross-border data transfer regulations
- Student privacy requirements

- **Accreditation & Certification Requirements:** Egypt has specific regulations around **educational training providers**. Ensure **our certifications are recognized** or partnered with reputable companies for credibility.

- **Employment & Internship Regulations:** Companies may have legal obligations regarding **internships and trainee programs**.

Partnering with firms that see **our platform as a pre-internship preparation tool** can help navigate these challenges.

Environmental Factors

- **Sustainability Initiatives:**
 - Green technology adoption in education
 - Environmental certification requirements
 - Sustainable business practices
- **Infrastructure Challenges:**
 - Power grid stability
 - Internet connectivity issues
 - Digital divide between urban/rural areas
- **Climate Considerations:**
 - Impact on physical infrastructure
 - Energy consumption management
 - Environmental policy compliance
- **Green Technology:**
 - Energy-efficient data centers
 - Sustainable device policies
 - Carbon footprint reduction strategies
- **Sustainability in Education:** Online learning reduces **paper waste and commuting emissions**, which could be a **CSR selling point for companies** looking to improve their environmental impact.
- **Energy & Connectivity Challenges:** Some areas in Egypt experience **power outages or slow internet speeds**. Offering **offline content downloads or lightweight mobile apps** can improve accessibility.

2.4.2.2 SWOT Analysis

(Bridging the Skills Gap for Egyptian Graduates)



Figure 3. SWOT

Strengths (Internal)

- **Unique Value Proposition:**
 - Combines practical training with academic theory, reducing internship costs for companies and accelerating employability for students.
 - Customizable programs and a balance of theoretical/practical learning.
- **Strategic Partnerships:**
 - Collaboration with industry experts, companies, and universities ensures a job-relevant curriculum and credibility.
- **Government Alignment:**
 - Aligns with Egypt's Vision 2030 and ICT strategies, opening doors to grants and institutional partnerships.
- **Technological Edge:**
 - Plans to integrate AI, gamification, and partnerships with job portals (e.g., Wuzzuf) for personalized learning and direct hiring pipelines.
- **Social Impact Focus:**
 - Targets youth unemployment (24%) and empowers women in underserved regions (e.g., Upper Egypt).

Weaknesses (Internal)

- **Regulatory Hurdles:**
 - Requires compliance with Egypt's strict educational licensing (Ministry of Higher Education) and data privacy laws.
- **Dependence on Partnerships:**
 - Relies heavily on corporate partnerships for internships and university collaborations for accreditation.
- **Resource Constraints:**
 - As a startup, limited funding and brand recognition compared to established EdTech competitors.

- **Informal Economy Challenge:**
 - Certifications may hold less value in Egypt's informal sector (40% of workforce).

Opportunities (External)

- **Market Growth:**
 - Egypt's EdTech market is growing at 16.5% CAGR (\$3.5B in 2023), driven by 3.1M university students.
- **Government Support:**
 - Access to grants (e.g., Digital Egypt Innovators Initiative) and incubators (Flat6Labs, Falak Startups).
- **Arabic-Language Demand:**
 - Localized Arabic content can tap into non-elite universities and rural areas.
- **Tech Adoption:**
 - Rising internet penetration (71%) and 5G rollout enable scalable online/hybrid learning.
- **Corporate CSR Trends:**
 - Companies seek ESG-compliant partners; Second Era's online model reduces carbon footprint (paperless learning).

Threats (External)

- **Economic Volatility:**
 - Inflation (36% in 2023) and currency devaluation may reduce student affordability.
- **Competition:**
 - Rivalry from global EdTech platforms (Coursera, Udemy) and traditional universities expanding online programs.
- **Regulatory Risks:**
 - Changing e-learning laws (e.g., Higher Education Law No. 72/2019) could increase compliance costs.
- **Infrastructure Gaps:**
 - Power outages and low internet speeds in rural areas may limit accessibility.

- **Cultural Bias:**
 - Some employers still prioritize degrees over skills, undermining Second Era’s “Internship Accelerator” positioning.

2.4.2.3 STP

Segmentation

Student Segmentation

Demographic:

- **Age:** 20–28 years (final-year students and recent graduates).
- **Education:** Pursuing or holding degrees in business, IT, marketing, or related fields.
- **Income:** Students from middle-income households (monthly household income: 5,000–25,000 EGP) or self-funded through part-time jobs.
- **Career Stage:** Actively seeking internships/jobs in customer service, digital marketing, sales, or tech.

Psychographic:

- **Attitudes:**
 - Frustrated by the gap between academic theory and workplace skills.
 - Motivated to upskill for competitive job markets.
- **Lifestyles:**
 - Juggling part-time jobs (e.g., customer service roles) with studies.
 - Active on professional platforms (LinkedIn, Wuzzuf).
- **Personality Traits:**
 - Ambitious, pragmatic, and anxious about job readiness.
 - Prefer hands-on learning over theoretical coursework.

Geographic:

- **Primary:** Greater Cairo (Cairo, Giza, 6th of October City).
- **Secondary:** Alexandria and Mansoura (cities with high university density and corporate presence).

Behavioral:

- **Learning Habits:** Seek short, practical courses (4–8 weeks) with certifications.
- **Job-Seeking Behavior:** Actively apply for internships but lack technical skills (e.g., CRM tools, SEO, sales pipelines).
- **Price Sensitivity:** Willing to pay for programs with guaranteed internship opportunities.

Company Segmentation

Demographic:

- **Industry:**
 - High-demand sectors: Customer service (e.g., telecom, e-commerce), digital marketing agencies, tech startups.
- **Company Size:**
 - Mid-sized firms (50–500 employees) with structured internship programs.
 - Startups (3+ years old) scaling rapidly and needing cost-effective talent.
- **Revenue:** Companies with annual budgets for training/L&D (1M+ EGP revenue).

Psychographic:

- **Attitudes:**
 - Value ROI on employee development.
 - Frustrated by the time/cost of training interns.
- **Innovation Orientation:** Early adopters of new talent-sourcing models.

Geographic:

- **Primary:** Greater Cairo (corporate hubs like Nasr City, New Cairo).
- **Secondary:** Alexandria (growing tech/outsourcing ecosystem).

Behavioral:

- **Hiring Patterns:** High internship turnover (e.g., 10+ interns/year).
- **Pain Points:**
 - Spend 3–6 months training interns on basic skills.
 - Struggle to retain interns due to skill gaps.

Targeting

Student Targeting

- **Primary Audience:**
 - Final-year students and recent graduates in Cairo/Alexandria pursuing careers in customer service, digital marketing, or sales.
 - Part-time workers (e.g., Teleperformance employees) seeking to upskill for higher-paying roles.
- **Secondary Audience:**
 - University career centers looking to improve graduate employability.

Company Targeting

- **Primary Audience:**
 - Mid-sized e-commerce platforms, digital marketing agencies, and tech startups in Cairo struggling with internship costs.
 - Companies with recurring internship programs (e.g., telecom giants like Vodafone/Etisalat).
- **Secondary Audience:**
 - SMEs in Alexandria/Mansoura seeking ready-to-work talent.

Positioning

Student Positioning

Tagline:

“From Classroom to Career: Master Real-World Skills in 8 Weeks.”

Key Message:

“We equip you with the tools companies actually need—CRM software, sales pipelines, SEO—so you can skip the ‘training phase’ and start adding value on Day 1 of your internship.”

Company Positioning

Tagline:

“Cut Internship Costs by 50% with Job-Ready Talent.”

Key Message:

“Our graduates arrive pre-trained in your industry’s tools and workflows, reducing onboarding time and ensuring faster ROI on your internship programs.”

2.4.2.4 Marketing Mix (7Ps)

1. Product

❖ Core Offerings:

The main product is skill-enhancement training programs tailored for final-year university students and recent graduates, helping them bridge the gap between academic knowledge and professional skills. These programs focus on reducing the time and cost of internships for both students and companies.

❖ Key Features:

○ Customized Educational Programs:

- Practical courses and hands-on assessments in fields like Business, IT, Marketing, and modern professional skills (e.g., CRM, SEO, sales pipelines).
- Programs balance theoretical knowledge with real-world application, tailored to individual skill gaps.

○ Interactive Learning Formats:

- Live lectures, gamified assessments, AI-powered skill tracking, and downloadable resources for offline access.
- Workshops led by industry experts with academic and professional experience.

○ Job-Readiness Tools:

- Certifications recognized by partner companies.
- Direct integration with job platforms (e.g., LinkedIn, Wuzzuf) for seamless internship/job applications.

○ Sustainable Learning Ecosystem:

- Post-program resources (e.g., mentorship, webinars) to ensure continuous skill development.

❖ Benefit to Customers:

- For students: Faster employability, practical experience, and easier access to job-ready skills.
- For companies: pre-trained potential employees, reducing time and cost associated with developing new hires through internships or corporate training programs.

2. Place

❖ Operating Model:

Second Era operates as an entirely online platform accessible to students across Egypt via desktop and mobile devices, ensuring flexibility and nationwide reach, especially for those in rural and underserved areas.

❖ Accessibility:

- Providing a *mobile-friendly platform* that works effectively even in regions with slower internet speeds.
- Option to download offline course materials for areas with connectivity issues.
- Subtitled and localized Arabic content to cater to students in non-elite universities and remote regions.

❖ Collaborations:

Working with Egyptian universities, corporate companies, and government bodies to ensure greater market penetration and program credibility.

- Examples: Partnering with hubs like Falak Startups or Flat6Labs and local job platforms like Wuzzuf for an employment pipeline.

❖ Our Office:

- Cairo Festival City, North Investors, New Cairo City, Cairo.

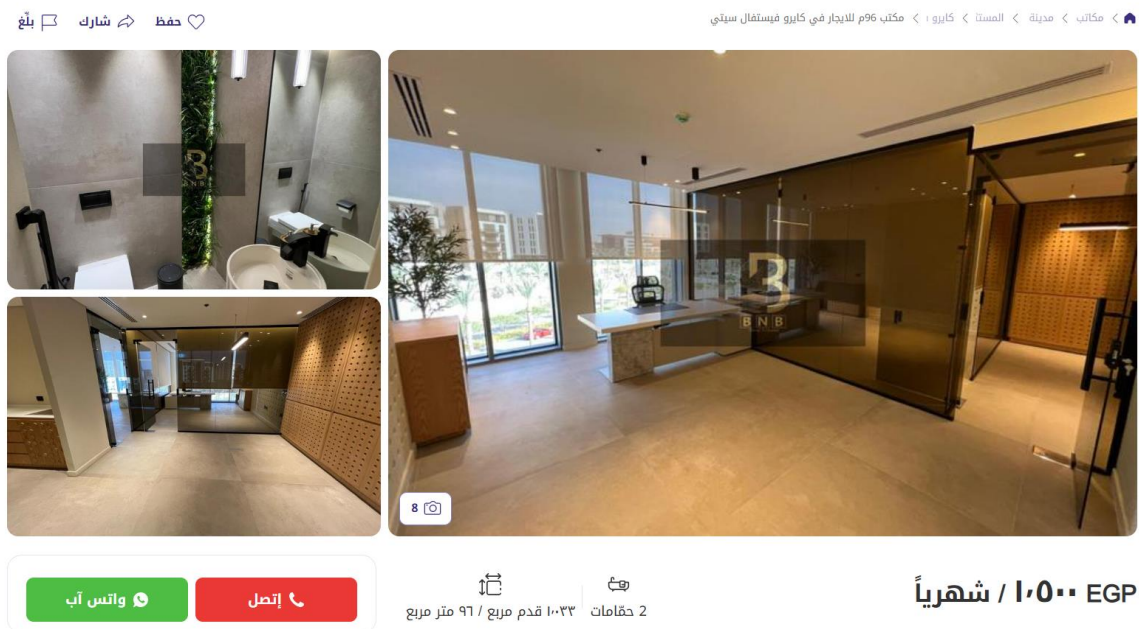


Figure 4. Our Office

3. Price

❖ **Competitive Pricing:**

Pricing structures are affordable for students, considering economic volatility and the Egyptian market's sensitivity to cost.

❖ **Tiered Pricing Model:**

- *Freemium Model:* Free introductory courses with paid certifications.
- *Basic Plan:* Includes recordings, materials, and self-paced learning (low-cost option).
- *Standard Plan:* Includes live sessions, interactive assessments, and certification (mid-tier pricing).
- *Premium Plan:* All-inclusive package with live instructor sessions, internships, and customized coaching (higher-tier pricing).

❖ **Discounts & Sponsorships:**

- Offering corporate-sponsored training programs or discounts through partnerships with large companies looking to onboard pre-trained employees.
- Discounts for early-bird subscriptions, group enrollments, or long-term packages.
- Seasonal discounts during enrollment periods (e.g., pre-academic year).

❖ **Installment Options:**

Introducing flexible payment options or installment plans to accommodate students from middle-income households.

- Installment plans (e.g., 3–6 months) via partnerships with digital wallets (e.g., Fawry, Vodafone Cash).
- Income-share agreements (ISAs) for students: Pay a percentage of salary post-placement.

4. Promotion

Targeted Campaigns:

❖ Student-Focused:

- **Social media:** TikTok/Instagram campaigns showcasing success stories, interactive quizzes, and influencer partnerships (e.g., Egyptian career coaches).
- **University Collaborations:** Workshops and webinars promoted via student unions and email newsletters.

❖ Corporate-Focused:

- **LinkedIn Marketing:** Case studies highlighting cost savings for companies (e.g., “Reduce internship costs by 40%”).
- **CSR Positioning:** Highlight environmental benefits (e.g., paperless training) to align with corporate ESG goals.

❖ Government & Community Engagement:

- Participate in national initiatives (e.g., Digital Egypt Youth Forum) for brand visibility.
- Arabic-language radio ads and community workshops in rural areas.

❖ Branding Elements:

- **Tagline:** “*Bridge the Gap: From Classroom to Career.*”
- **Testimonials:** Video interviews with placed graduates and partner companies.
- **Gamified Referrals:** Students earn discounts for inviting peers; companies get free trials for referrals.

❖ Partnership-Driven Promotion:

- Co-branded campaigns with job portals (e.g., Wuzzuf) and tech firms (e.g., Microsoft Egypt).
- Partnering with Egyptian universities to promote Second Era as a recommended training partner for final-year students.
- Collaborating with career fairs, job portals (e.g., Wuzzuf, LinkedIn), and government-backed EdTech initiatives for broader visibility.

❖ Referral Programs:

- Implementing a referral system offering discounts or incentives to existing users who recommend Second Era to their peers.

❖ Seasonal Offers:

- Providing discounts during key enrollment seasons (e.g., university graduation or summer breaks) to attract more students.

5. Physical Evidence

Even though Second Era is a digital service, physical evidence reassures customers about quality, trust, and professionalism:

Tangible Elements to Reinforce Credibility:

❖ **Online Platform Interface:**

- Professional, user-friendly design with interactive dashboards, progress trackers, and certification badges.
- Mobile-responsive layout optimized for low-bandwidth areas.

❖ **Certificates & Branded Materials:**

- Issuing digital and physical certificates co-branded with partner companies/universities.
- Branded merchandise (e.g., notebooks, USB drives) for workshop participants.

❖ **Partnership Logos & Testimonials:**

- Display logos of collaborating universities (e.g., Helwan University) and companies on the website/marketing materials.
- Video testimonials from graduates employed at partner firms.

❖ **Physical Training Centers (Future Expansion):**

- Pilot centers in Cairo and Alexandria with modern tech setups (e.g., VR labs) to showcase hands-on training quality.

6. People

Key Human Resources & Roles:

❖ **CEO (Chief Executive Officer)**

- **Tasks & Responsibilities:** Leads strategic planning, investor relations, and major partnerships. Oversees all departments and ensures long-term growth.
- **Qualifications:** 8+ years in business leadership, preferably in education or tech sectors.

❖ **Operations Manager**

- **Tasks & Responsibilities:** Manages daily workflows, coordinates across departments, and ensures training delivery meets quality standards.
- **Qualifications:** 5+ years in operations or program management; strong organizational and leadership skills.

❖ **Marketing Specialist**

- **Tasks & Responsibilities:** Plans and executes digital campaigns, manages social media, and supports student recruitment.
- **Qualifications:** 3–5 years in digital marketing, preferably in education or SaaS.

❖ **Customer Support Representative**

- **Tasks & Responsibilities:** Provides multilingual (Arabic/English) support via chat, email, and phone. Assists with technical and enrollment issues.
- **Qualifications:** Strong communication skills, customer service experience, and knowledge of EdTech platforms.

❖ **Content Developer**

- **Tasks & Responsibilities:** Designs course materials, collaborates with instructors, and ensures pedagogical quality.
- **Qualifications:** Background in instructional design or e-learning; familiarity with authoring tools and LMS.

❖ **IT & Platform Administrator**

- **Tasks & Responsibilities:** Maintains the learning platform, ensures cybersecurity, troubleshoots technical issues, and supports updates.
- **Qualifications:** 3–5 years in IT support or systems administration; experience with LMS and cloud services preferred.

❖ **Expert Trainers: (Outsource)**

- **Tasks & Responsibilities:** Recruit industry professionals with 5+ years of experience in fields like IT, marketing, and business.
- **Qualifications:** Continuous trainer development programs to align with market trends.

7. Process

The processes in delivering the service ensure smooth operation and customer satisfaction:

Efficient Service Delivery Systems:

❖ Enrollment & Learning Journey:

- Simplified online registration with AI-driven skill assessments to recommend personalized programs.
- Automated reminders for course deadlines, live sessions, and assessments.
- Seamless access to live and recorded sessions, interactive assignments, and practical projects with continuous feedback.
- Automated issuance of certificates after successful completion, with clear criteria and validity.

❖ Corporate Training Pipeline:

- Streamlined B2B onboarding: Needs analysis → customized curriculum → progress reports for companies.
- Post-training evaluations to measure skill improvement and internship readiness.

❖ Job Placement Process:

- Integration with job portals (e.g., Wuzzuf) for one-click applications.
- Monthly virtual career fairs connecting graduates with partner companies.

❖ Feedback & Improvement Loop:

- Post-course surveys for students and companies to refine content and delivery.
- Quarterly curriculum updates based on industry trends and employer feedback.

❖ CSR & Community Engagement:

- Transparent process for companies to sponsor scholarships, tracked via dashboards.

2.5 Legal Form.

General partnership: is a business arrangement by which two or more individuals agree to share responsibilities, assets, profits, and financial and legal liabilities of a jointly owned business.

a general partnership, partners agree to be personally responsible for potentially unlimited liability.

Liabilities are not capped as they would be in, say, a partnership formed as a limited liability partnership or a limited liability company.

Partners are responsible for the debts, and the seizure of an owner's assets is a possibility. Furthermore, any partner may be sued for the business's debts.

Since a general partnership is a pass-through entity where income flows straight to the owners, each partner reports their share of partnership profits or losses on their personal tax returns. The partnership itself is not taxed.

Advantages of partnership:

- ✓ A general partnership is less expensive and easier to set up than a corporation or limited liability partnership.
- ✓ Less paperwork is involved. In the United States, filing partnership paperwork with a state is generally not required, though certain registration forms, permits, and licenses may be necessary at the local level.
- ✓ As a pass-through entity, the partnership pays no taxes.
- ✓ No external financial reporting/annual report is required.
- ✓ A general partnership is simple to dissolve.

2.6 Start-up Capital & Sources of Start-up Capital.

2.6.1 Start-Up Capital

Investment	200,000 EGP
Working Capital	700,000 EGP
Total	900,000 EGP

Table 3. Start-up Capital

2.6.2 Sources of Start-up Capital

Source	Description	Amount (EGP)
1-Equity capital:		
Own savings	Initial founder contributions to launch and stabilize.	EGP 100,000
Partner	5 partners	EGP 150,000
2- Loan 1:		
Family	Informal, low-pressure financial backing from relatives.	EGP 50,000
Angel Investors	Early-stage investors interested in EdTech and employability solutions.	EGP 200,000
3-Loan 2:		
Credit cooperative	In-kind or co-funding support from educational institutions and companies.	EGP 100,000
Government scheme	Support from EdTech or youth empowerment initiatives (tax-reduce for one year).	0.5%
Bank Loan (Optional)	Backup financing for expansion or infrastructure.	EGP 300,000
Total Funding		EGP 900,000

Table 4. Sources of Start-up Capital

2.7 Financial Plan

2.7.1 Organization and Staff

Position	Tasks & Responsibilities	Qualifications
CEO	Strategy, stakeholder relations	10+ yrs leadership
Operations Manager	Daily operations, HR	5+ yrs in operations
Marketing Specialist	Campaigns, sales funnel	3+ yrs digital marketing
Customer Support Rep	Client assistance	Experience in customer service
Content Developer	Course content, assessments	EdTech background
IT & Platform Admin	Platform maintenance	IT degree, 3+ yrs
Trainer/Instructor	Deliver courses	Subject matter expert

Table 5. Organization and Staff

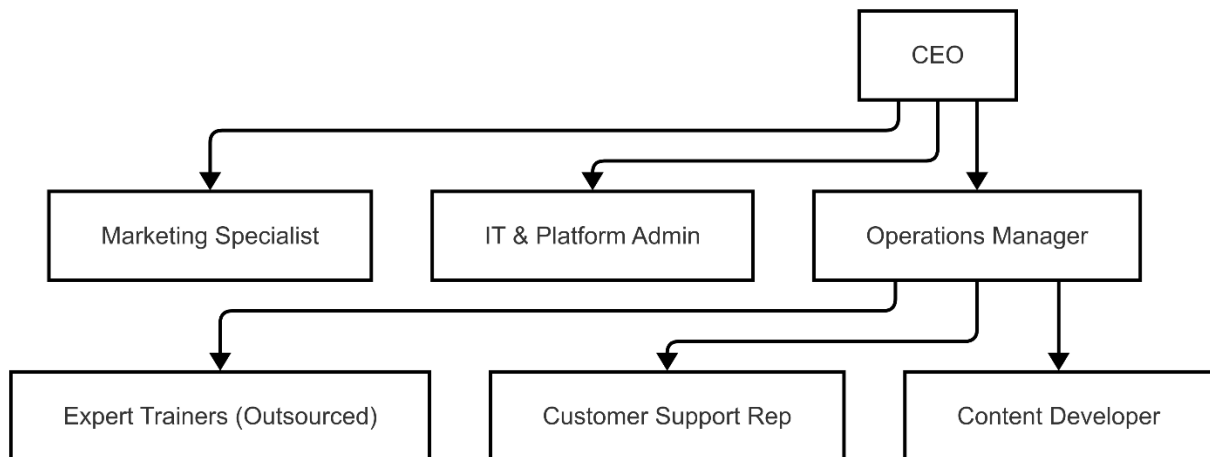


Figure 5. Company Hierarchy

2.7.2 Business Operation and Costs

Staff Positions				
Position	Num	Social security	M	Y
CEO	1	29.75%	9,083	108,990
Operations Manager	1	29.75%	7,785	93,420
Marketing Specialist	1	29.75%	6,488	77,850
Customer Support Rep	1	29.75%	7,785	93,420
Content Developer	1	29.75%	5,190	62,280
IT & Platform Admin	1	29.75%	3,893	46,710
Trainer/Instructor	2	29.75%	41,520	498,240
Total	8		EGP 81,743	EGP 980,910

Table 6. Business Operation and Costs

2.7.3 Expected Revenues

item/month	jan	feb	march	abril	may	june	july	aug	sep	oct	nov	dec	Total
Freemium Plan	0	0	0	0	0	0	0	0	0	0	0	0	EGP -
Basic Plan	0	0	0	30000	35000	92000	109000	91000	96000	46000	53000	48000	EGP 600,000
Standard Plan	0	0	0	9000	15000	22500	399000	30000	4500	9000	28500	22500	EGP 540,000
Premium Plan	0	0	0	9000	6000	60000	69000	78000	180000	174000	45000	99000	EGP 720,000
Annual Plan – Premium	0	0	0	25000	35000	0	50000	15000	75000	25000	50000	25000	EGP 300,000
Certification Fee	0	0	0	5000	8000	12000	7800	19000	21200	10600	17600	18800	EGP 120,000
Offline Access Kit	0	0	0	3750	6250	7500	20000	2500	2250	4750	3000	10000	EGP 60,000
Corporate Training (B2B)	0	0	0	0	40000	52000	68000	80000	68000	60000	36000	76000	EGP 480,000
Custom Workshop	0	0	0	0	0	0	20000	40000	20000	36000	32000	32000	EGP 180,000
Total	EGP -	EGP 6,000	EGP 10,000	EGP 75,750	EGP 145,250	EGP 246,000	EGP 742,800	EGP 350,500	EGP 466,950	EGP 365,350	EGP 265,100	EGP 326,300	EGP 3,000,000

Table 7. Expected Revenues

item/month	Total	
Freemium Plan	EGP	-
Basic Plan	EGP	600,000
Standard Plan	EGP	540,000
Premium Plan	EGP	720,000
Annual Plan – Premium	EGP	300,000
Certification Fee	EGP	120,000
Corporate Training (B2B)	EGP	480,000
Custom Workshop	EGP	180,000
Offline Access Kit	EGP	60,000
total	EGP	3,000,000

Table 8. Expected Revenues (Summary)

Assumptions:

Parameter	Value / Details
Customer growth starting	Month 4
Monthly customer growth rate	10% increase in new customers each month after month 4
Plan distribution (monthly new customers)	50% Basic, 30% Standard, 20% Premium
Average plan prices	Basic: 1,000 EGP, Standard: 1,500 EGP, Premium: 3,000 EGP
Installment options	Only on annual plans, 3-month or 6-month installment plans
Installment uptake	100% choose 12-month Plan
Revenue recognition	Spread evenly over the installment period (monthly)
Initial months (1–3)	Free trial, no revenue
B2B corporate training	10 clients/month starting month 7, each 4,000 EGP, free for 6 months, then paid afterward
Other revenues	Certification, offline kits, etc. (assumed steady, optional for simplicity)

Table 9. Assumptions

2.7.4 Expected Expenses

	Cost	Num	M	Y
F.C				
Salries				
CEO	7,000	1	9,083	108,990
Operations Manager	6,000	1	7,785	93,420
Marketing Specialist	5,000	1	6,488	77,850
Customer Support Rep	6,000	1	7,785	93,420
Content Developer	4,000	1	5,190	62,280
IT & Platform Admin	3,000	1	3,893	46,710
Trainer/Instructor	16,000	2	41,520	498,240
Total Salries			EGP 81,743	EGP 980,910
Rent Office	10,500	1	10,500	126,000
Platform Hosting & Maintenance	3,050	1	4,000	48,000
Utilities & Internet	1,500	1	2,000	24,000
Others			4,500	54,000
Total Expances			21,000	252,000
Total F.C			EGP 62,520	EGP 1,232,910
V.C				
Marketing & Advertising	20,000	1	20,000	240,000
Certification & Certification Fees	1,666	1	1,666	19,992
Office supplies & Misc.	4,050	One time	4,050	4,050
Software licenses & tools	3,050	One time	4,050	4,050
Miscellaneous & Contingency	4,000	1	4,000	4,000
Total V.C			EGP 33,766	EGP 272,092
Total Cost			EGP 96,286	EGP 1,505,002

Table 10. Expected Expenses

2.7.4 Income Statement

Item	Amount (EGP)
Revenue	3,000,000
Cost of Goods Sold (COGS)	564,000
Gross Profit	2,436,000
Operating Expenses	1,485,010
EBITDA	950,990
Taxes (approx. 0.5%)	47,550
Net Income	903,441

Table 11. Income Statement

2.7.5 Cashflow

Source	Amount (EGP)
Inflow	
Beginning Cash Balance	0
Funding from Sources	900,000
Monthly Revenue (from sales)	3,000,000
Total Cash Inflows	3,900,000
Outflow	
Operating Expenses	1,485,010
Cost of Goods Sold (COGS)	564,000
Total Outflows	2,049,010
Net Cash Flow (Inflows - Outflows)	1,850,990
Ending Cash Balance	1,850,990

Table 12. Cashflow

2.7.6 Balance Sheet

Asset Category	Amount (EGP)	Liability Category	Amount (EGP)
Current Assets		Current Liabilities	
- Cash & Cash Equivalents	1,850,990	- Accounts Payable	100,000
- Accounts Receivable	250,000	Total Current Liabilities	100,000
- Prepaid Expenses	50,000	Long-term Liabilities	
Total Current Assets	2,150,990	- Bank Loans / External Debt	300,000
Non-current Assets		Total Long-term Liabilities	300,000
- Equipment & Infrastructure	100,000	Total Liabilities	400,000
- Intangible Assets	50,000	Equity Category	Amount (EGP)
Total Non-current Assets	150,000	Owner's Capital	900,000
		Retained Earnings	1,000,990
		Total Owner's Equity	1,900,990
Total Assets	2,300,990	Total Liabilities + Equity	2,300,990

Table 13. Balance Sheet

2.7.7 Owner Equity

<i>Owner's Equity Statement</i>	
Item	Amount (EGP)
Beginning Capital	900,000
Add: Net Profit for Year	903,441
Ending Capital / Equity	1,803,441

Table 14. Owner Equity

Chapter Three

- 3. System Analysis
 - 3.1. Introduction (Aim of the chapter)
 - 3.2. User Requirements
 - i. Functional requirements
 - ii. Non-functional requirements
 - 3.3. System Requirement
 - 3.4. System Diagrams
 - i. Use case diagram.
 - ii. Sequence diagram
 - iii. Activity diagram

Chapter Three: System Analysis

3.1 Introduction

In the analysis phase, the analyst determines the functional requirements of the new system. This chapter begins with a description of the analysis phase and its main output, the system proposal.

The analysis phase derives its name from the term "*analysis*", which involves breaking down a whole into parts with the aim of understanding the nature, function, and relationships of those parts. During this phase, collaboration between the system analyst and business users is promoted to better understand user needs and expectations.

3.2 User Requirements

The following are the user requirements:

3.2.1 Functional Requirements

Functional requirements are product features or functions that must be implemented by developers to enable users to accomplish their tasks. It is crucial to ensure clarity regarding these requirements for both the development team and stakeholders.

Typically, system behavior under specific conditions is described by functional requirements.

Example: An approval request is sent by the system after personal information is entered by the user. A search feature allows courses to be browsed across various categories, whether in line with the user's interests.

3.2.1.1 Student

Students interact with the system through a variety of features aimed at enhancing their user experience:

- Registration requires essential personal details.
- Login is done via email and password.
- A course-sharing feature allows students to upload completed courses for exchange.
- Students can review and rate courses they've taken.
- Courses can be purchased in digital or physical formats.
- Tickets for offline seminars can be purchased.
- Users may add favorite quotes from courses or instructors.
- A robust search feature supports category-based course exploration.
- Feedback submissions help improve the system and services.

3.2.1.2 Job

The Job entity represents employment opportunities provided by companies.

Purpose:

- Stores job listings.
- Manages application and selection processes for students.
- Allows companies to post jobs.
- Offers students a platform to apply for roles aligned with their qualifications and interests.

3.2.1.3 Course

The Course entity covers training and academic offerings.

Purpose:

- Stores and manages educational content.
- Tracks schedules, types, and course relationships.
- Integrates with registrations and assessments.
- Organizes curriculum, supports planning, and enables student progress tracking.

3.2.1.4 Instructor

The instructor entity represents educators responsible for delivering courses and evaluating performance.

Purpose:

- Manages teaching staff information.
- Ensures proper assignment of instructors based on specialization.
- Supports course planning and delivery through resource tracking.

3.2.1.5 Assessment

The Assessment entity manages student evaluations throughout their learning journey.

Purpose:

- Organizes course and job-related assessments.
- Supports structured evaluation design.
- Helps track performance and readiness for real-world roles.
- Enhances feedback and academic quality.

3.2.1.6 Company Representative

This entity includes authorized individuals representing external companies.

Purpose:

- Manages job listings.
- Reviews student applications.
- Facilitates secure and structured interactions with the platform.
- Ensures efficient recruitment and institutional communication.

3.2.1.7 Admin

The admin entity handles system-wide operations.

Purpose:

- Full access and control over platform functions.
- Manages users, content, job approvals, and configurations.
- Ensures data accuracy, rule enforcement, and overall system stability.

3.2.2 Non-functional Requirements

- **Performance:** Describes system responsiveness to user interactions.
- **Security:** Protects against unauthorized access to data.
- **Availability:** Measures system uptime and access reliability.
- **Data Integrity:** Ensures accuracy and consistency of data across its lifecycle.
- **Reliability:** Measures system function stability under defined conditions.

3.3 System Requirements

System requirements specify configurations necessary for optimal hardware and software performance. Failure to meet them may lead to installation or performance issues.

3.3.1 Software Requirements

- **Server OS:** Linux (Ubuntu 20.04+)
- **Web Server:** Apache 2.4+
- **Backend:** Python 3.12.5
 - With frameworks Flask
- **Database:** MySQL 8.0+
- **Frontend:** HTML5, CSS3, JavaScript
 - With Bootstrap for enhanced UX.
- **Version Control:** Git
- **API Testing:** Postman
- **Security:** SSL certificates, firewall, CAPTCHA, and optional 2FA

3.3.2 Hardware Requirements

- **Server:**
 - Quad-core processor (Intel Xeon or AMD Ryzen)
 - Minimum 32 GB RAM (64 GB recommended)
 - 2 Tera SSD
 - High-speed internet (16 Gbps)
- **Client Machines (Users):**
 - Intel i3 processor (or equivalent)
 - 4 GB RAM
 - 1366×768 screen resolution
 - Internet: Minimum 2 Mbps bandwidth
 - Browser: Chrome, Firefox, Edge, Safari
- **Additional Needs:**
 - Cloud storage (e.g., AWS S3 or Google Cloud Storage)
 - Backup hardware for large file handling

3.4 System Diagrams

The following are the system diagrams:

3.4.1 UML Use Case Diagram

- Describes main system functions (functional requirements)
- Identifies system users.
- Provides a high-level overview of system functionality and user interactions.

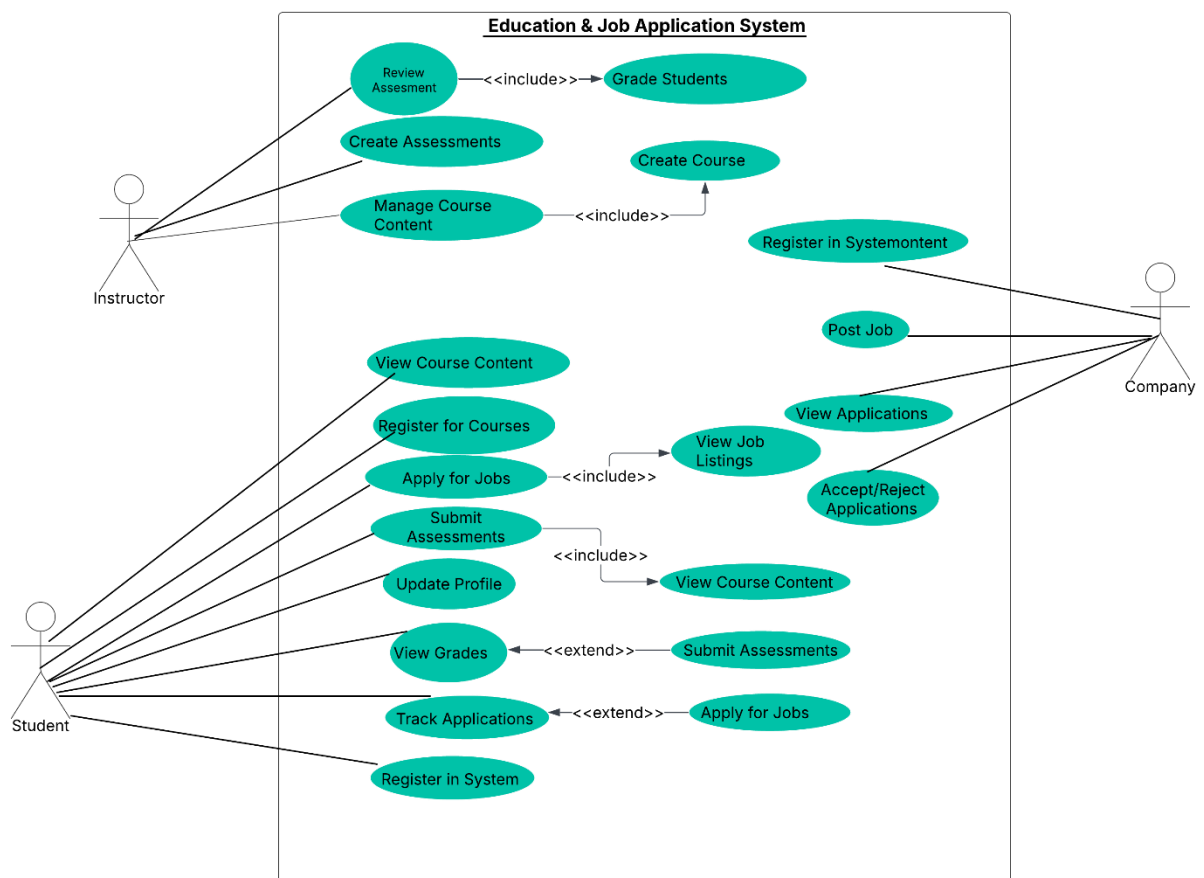


Figure 6. Use Case Diagram

3.4.2 UML Sequence Diagram

- Details how operations are executed through object interaction.
- Represents interaction order using a time axis.
- Entities: Student, Instructor, Assessment, Course, Job

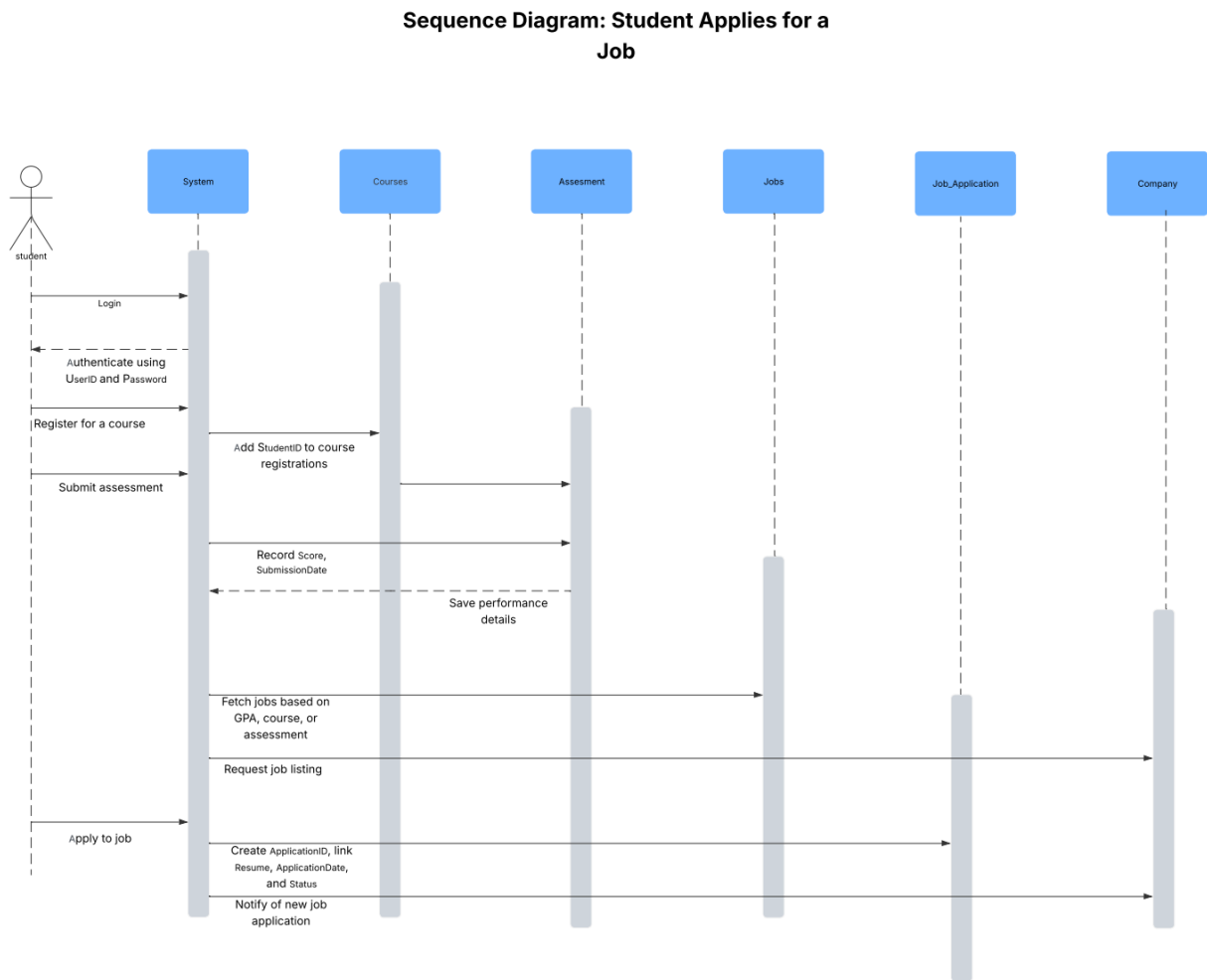


Figure 7. Sequence Diagram (Student)

**Admin Sequence
Diagram**

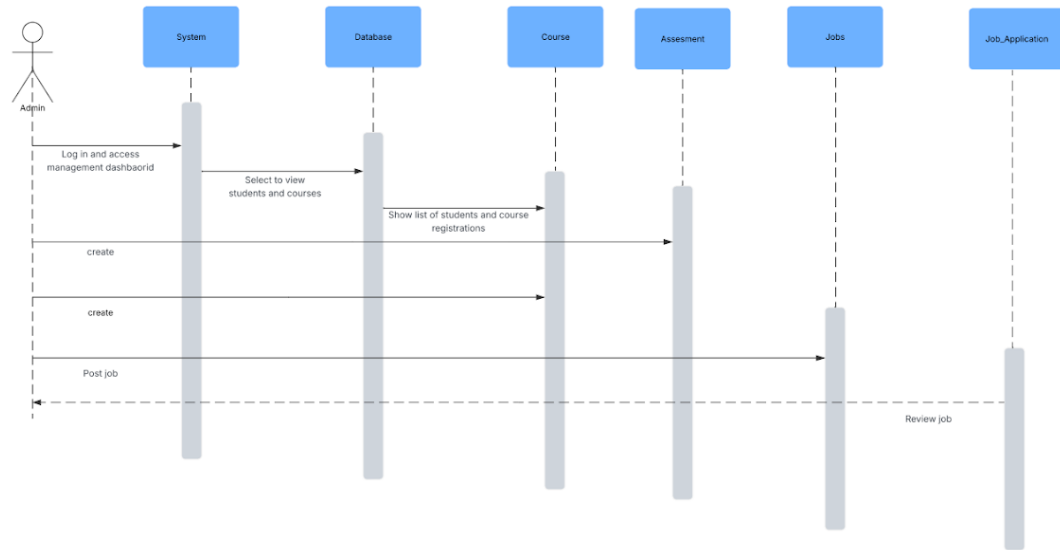


Figure 9 Sequence Diagram (Admin)

Sequence Diagram

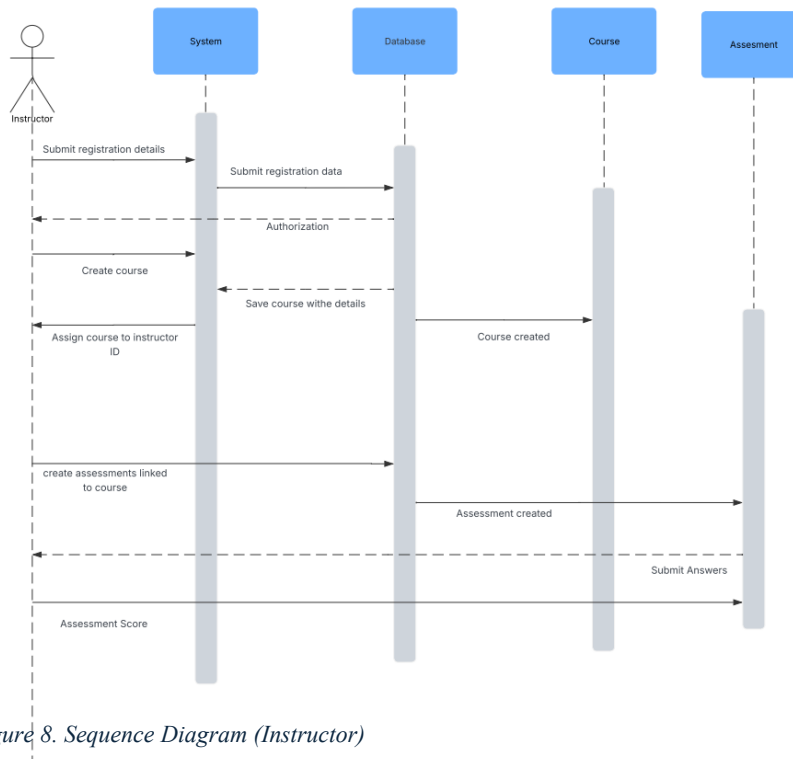


Figure 8. Sequence Diagram (Instructor)

Company Sequence Diagram

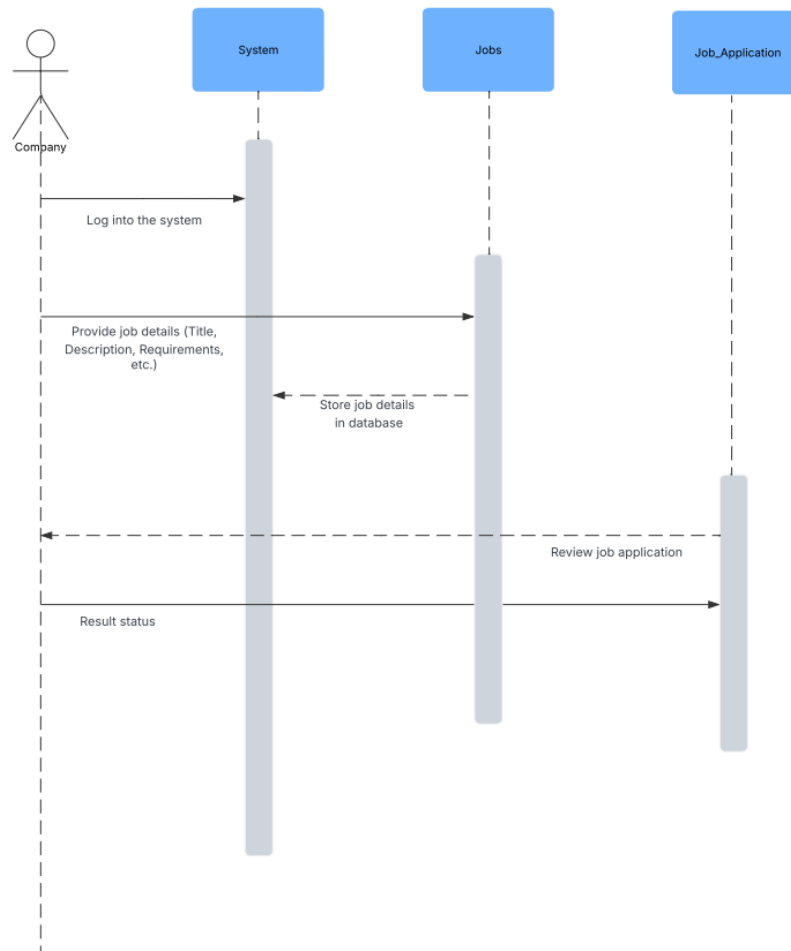


Figure 10. Sequence Diagram (Company)

3.4.3 UML Activity Diagram

- Advanced flowchart modeling transitions between activities
- Describes coordination of tasks to deliver services.
- Highlights event dependencies, overlaps, and business workflows across use cases.

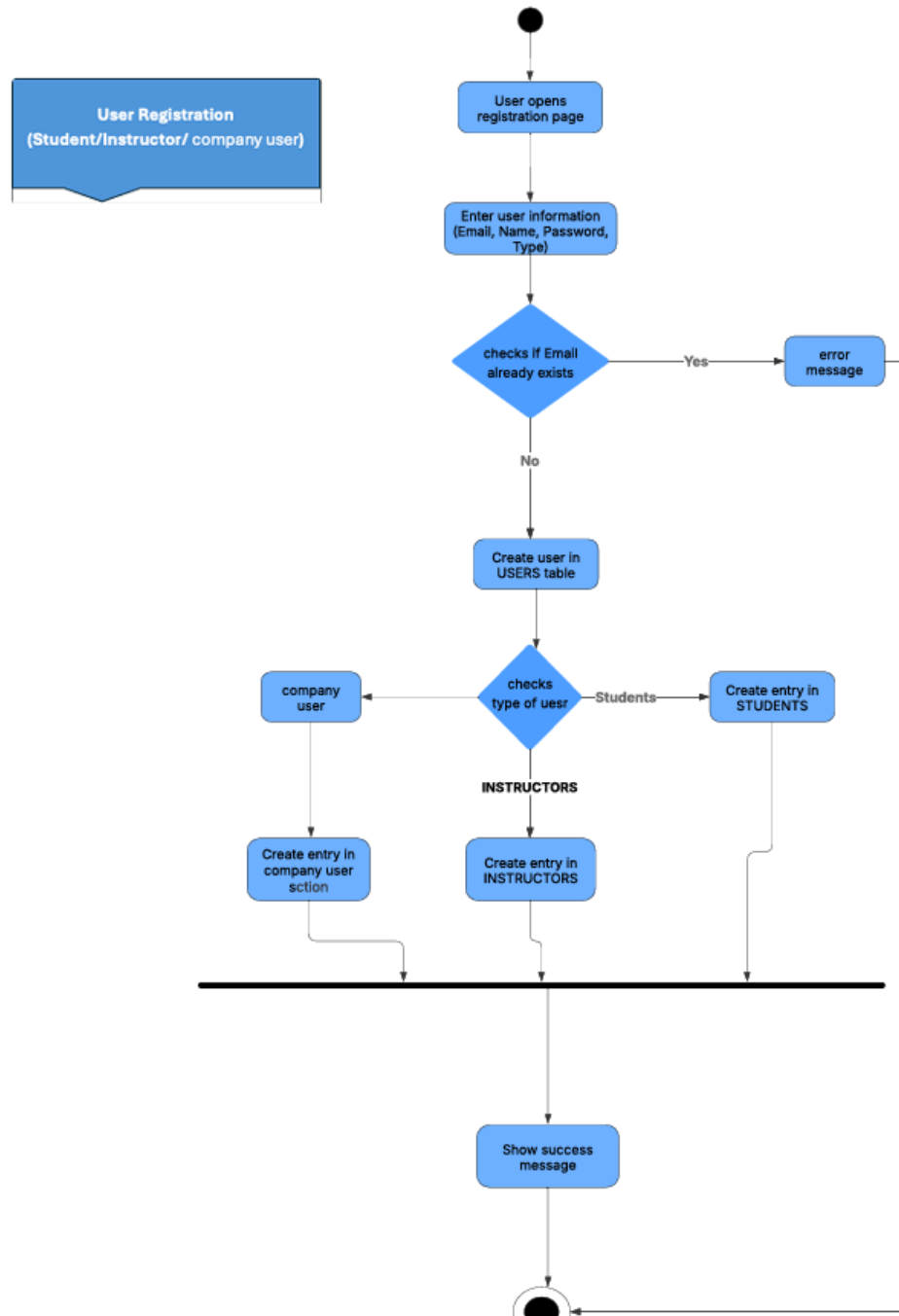


Figure 11. Activity Diagram (Registration)

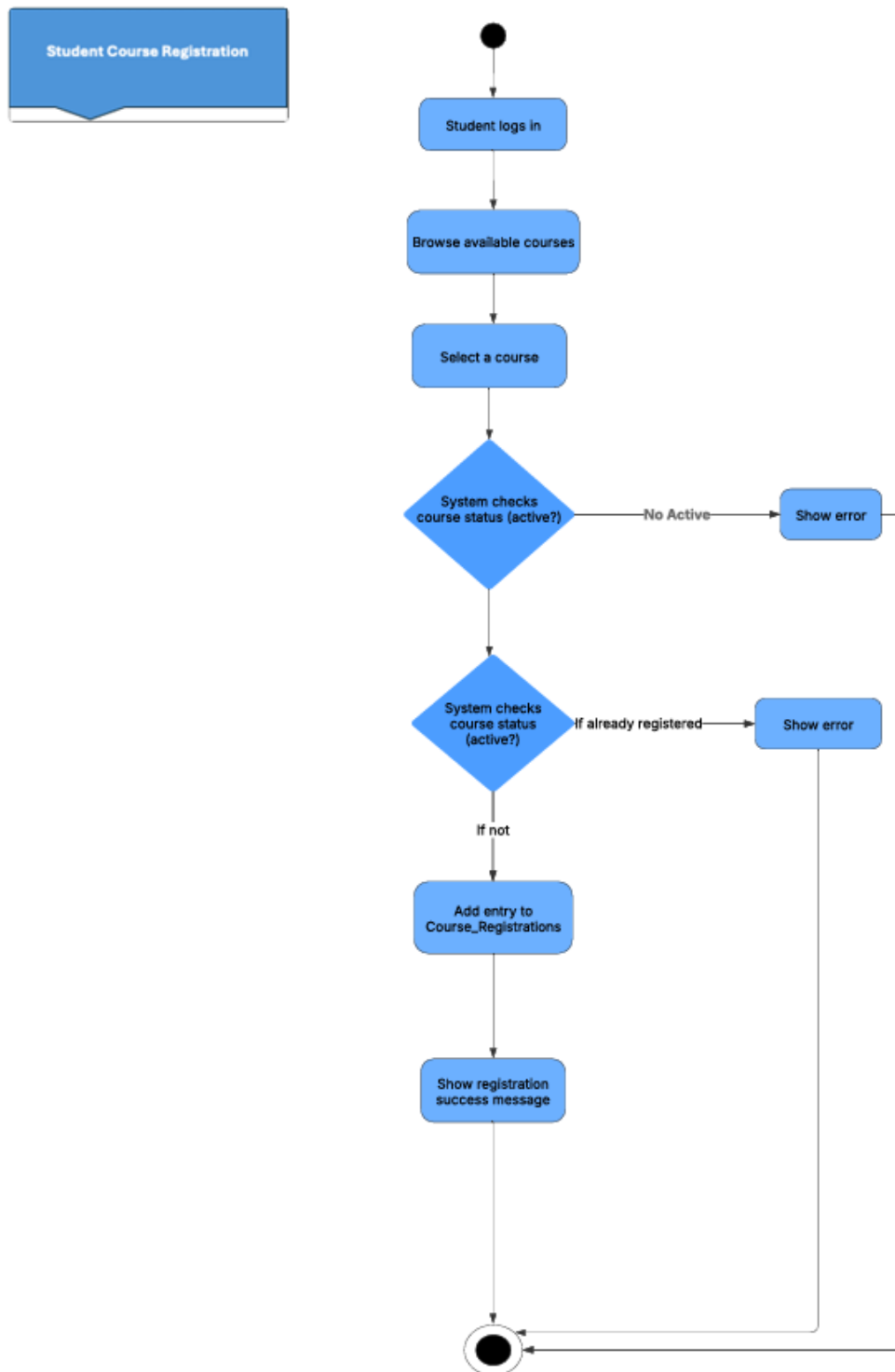


Figure 12. Activity Diagram (Course Registration)

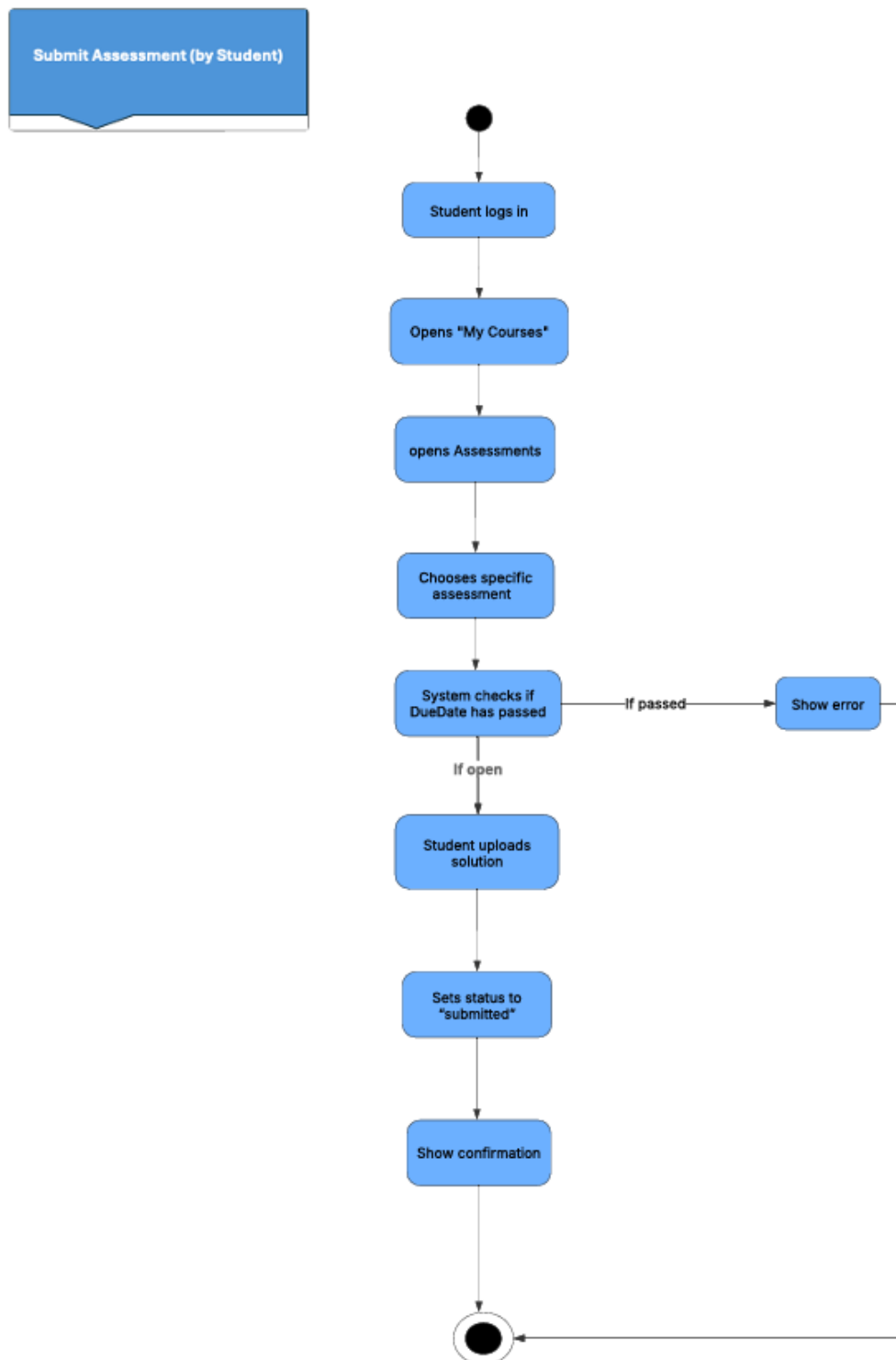


Figure 13. Activity Diagram (Submit Assignment)

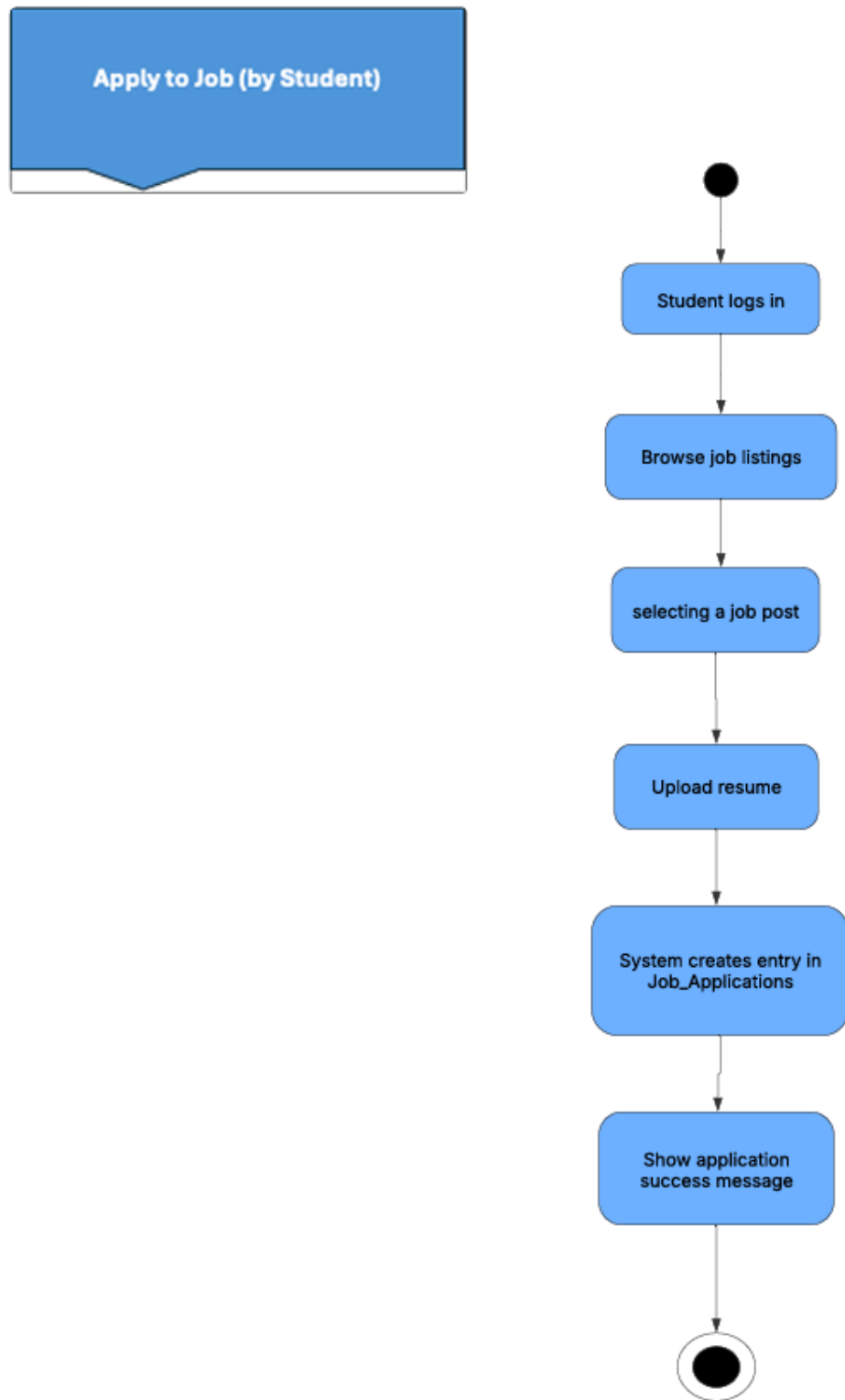


Figure 14. Activity Diagram (Apply to job)

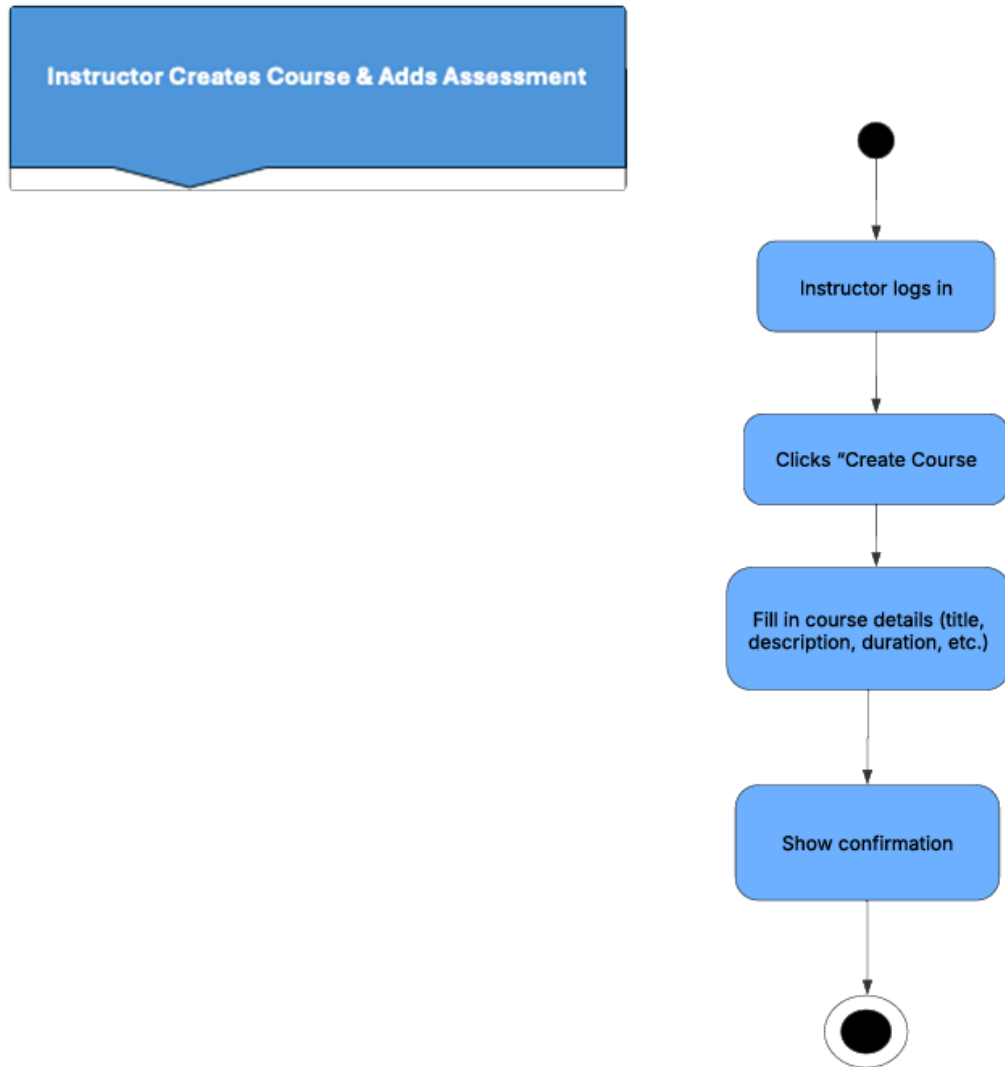


Figure 15. Activity Diagram (Create Course)

Add Assessment

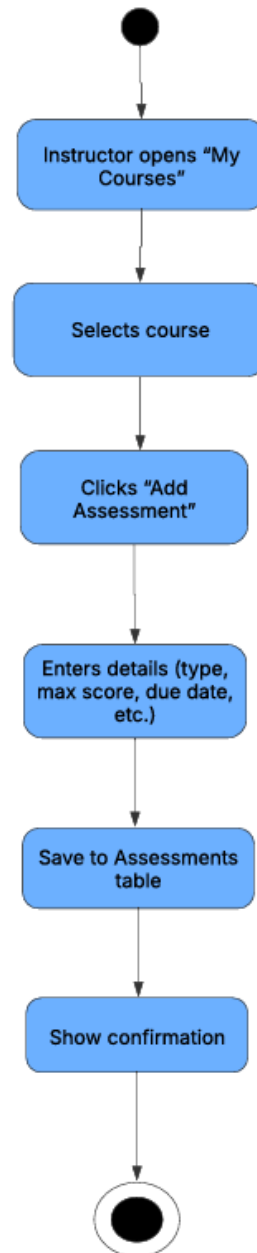


Figure 16. Activity Diagram (Add Assessment)

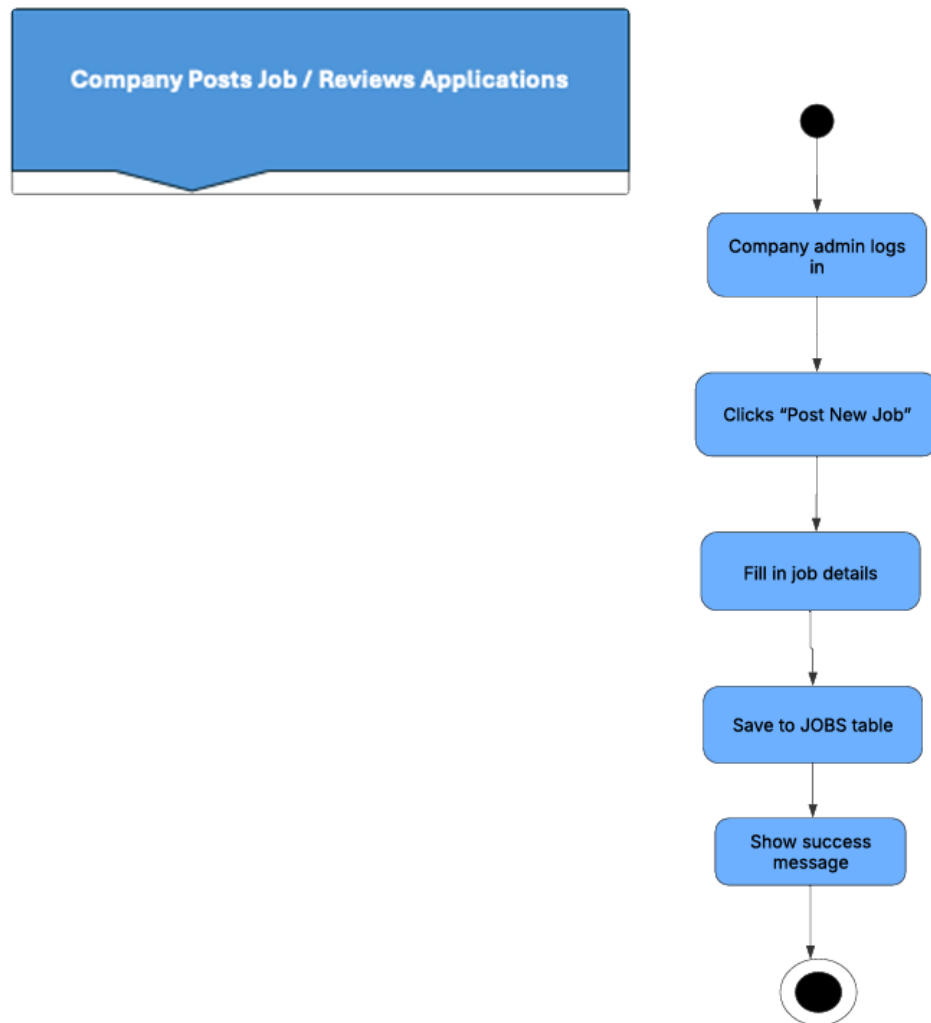


Figure 17. Activity Diagram (Post Job)

Reviews Applications

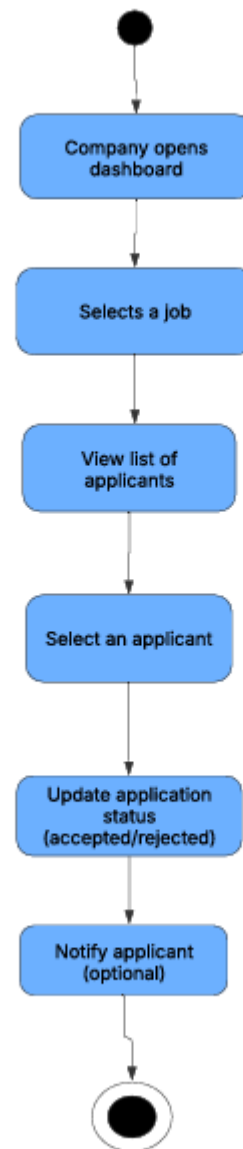


Figure 18. Activity Diagram (Review Job)

C h a p t e r F o u r

4. SYSTEM DESIGN

- 4.1. Introduction (Aim of the chapter)
- 4.2. Class Diagram
- 4.3. Database structure ERD
- 4.4. Database Mapping

Chapter Four: System Design

4.1 Introduction:

Systems Design is the process of defining the architecture, components, modules, interfaces, and data for a system to satisfy specified requirements. It involves translating user requirements into a detailed blueprint that guides the implementation phase. The goal is to create a well-organized and efficient structure that meets the intended purpose while considering factors like scalability, maintainability, and performance.

Mastering Systems Design is crucial for anyone looking to build robust and scalable systems. Our comprehensive Systems Design course provides you with the knowledge and skills to excel in this area. Through practical examples and expert insights, you will learn how to effectively translate user requirements into detailed designs that can be successfully implemented.

4.2 Class Diagram:

Class diagrams are fundamental to the object modeling process and model the static structure of a system. Depending on the complexity of a system, you can use a single class diagram to model an entire system, or you can use several class diagrams to model the components of a system. Class diagrams are the blueprints of your system or subsystem. You can use class diagrams to model the objects that make up the system, to display the relationships between the objects, and to describe what those objects do and the services that they provide.

Class diagrams are useful in many stages of system design. In the analysis stage, a class diagram can help you to understand the requirements of your problem domain and to identify its components. In an object-oriented software project, the class diagrams that you create during the early stages of the project contain classes that often translate into actual software classes and objects when you write code. Later, you can refine your earlier analysis and conceptual models into class diagrams that show the specific parts of your system, user interfaces, logical implementations, and so on.

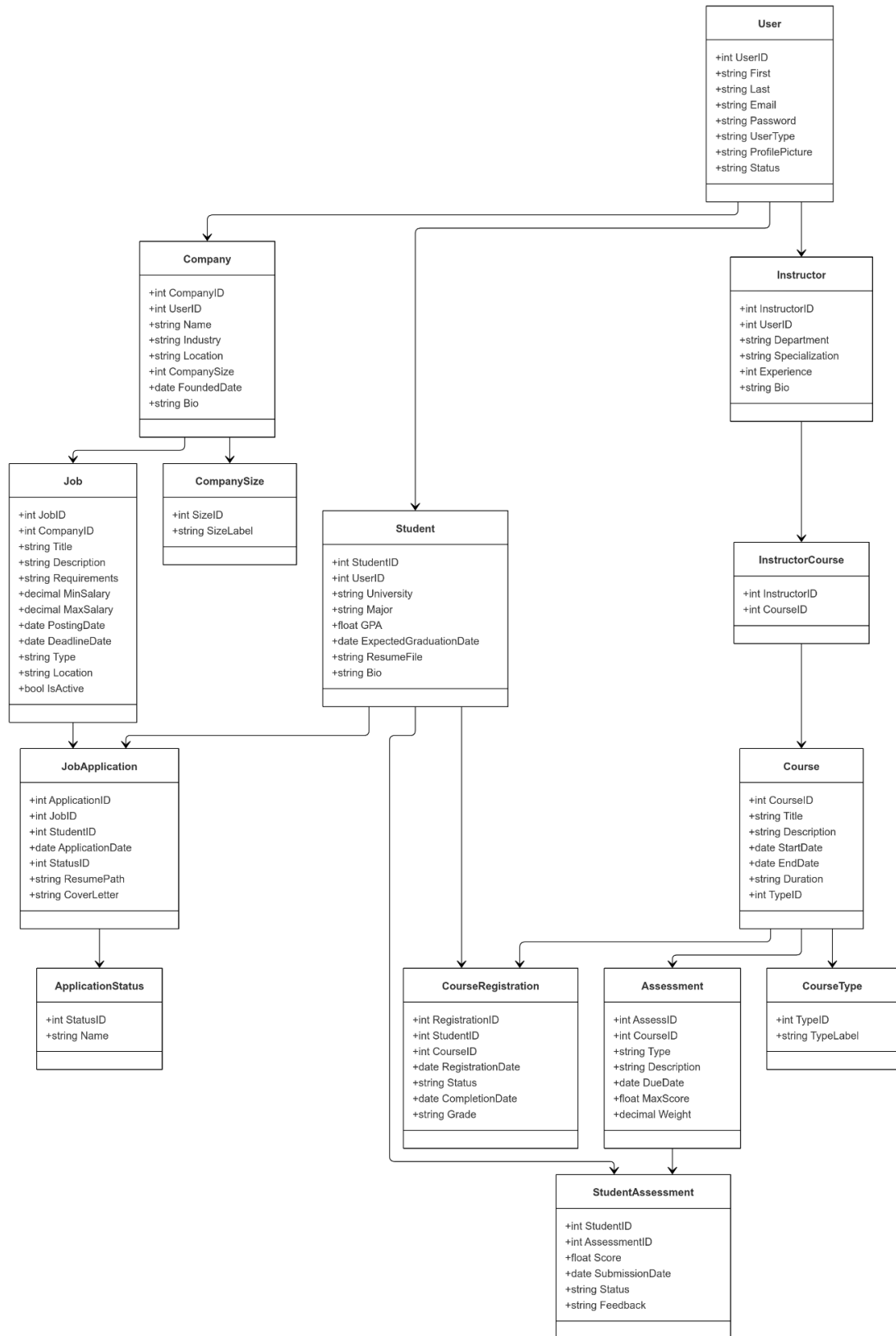


Figure 19. Class Diagram

4.3 Database Structure (ERD):

ERD stands for Entity Relationship Diagram. People also refer to these diagrams as ER diagrams or Entity Relationship Models.

An ERD visualizes the relationships between entities such as people, things, or concepts in a database. It often also depicts the attributes of these entities.

ER diagrams can help businesses document existing databases, which aids in troubleshooting logic or deployment problems, spotting inefficiencies, and improving processes—especially when undertaking business process re-engineering.

Additionally, ERDs are used to design and model new databases, ensuring engineers can identify any logic or design flaws before implementation in production.

- Document an existing database structure
- Debug, troubleshoot, and analyze
- Design a new database
- Gather design requirements
- Business process re-engineering (BPR)

When documenting a system or process, examining the system from multiple perspectives increases the overall understanding of that system.

ERD diagrams are commonly used alongside data flow diagrams to display the contents of a data store. They help visualize how data is connected in a general way and are particularly useful for constructing relational databases.

The traditional Chen ERD resembles a flowchart with connected symbols representing entities, relationships, and attributes. Since its introduction by Chen in the 1970s, various alternative cardinality notations have been proposed, but the basic symbols and their connections remain largely consistent.

4.4 Database Mapping:

Data mapping is the process of matching fields from one database to another. It's the first step to facilitate data migration, data integration, and other data management tasks.

Data mapping is crucial to the success of many data processes. One misstep in data mapping can ripple throughout your organization, leading to replicated errors, and ultimately, to inaccurate analysis.

Nearly every enterprise will, at some point, move data between systems. Different systems store similar data in different ways, so to move and consolidate data for analysis or other tasks, a roadmap is needed to ensure the data gets to its destination accurately.

For processes like data integration, data migration, data warehouse automation, data synchronization, automated data extraction, or other data management projects, quality in data mapping will determine the quality of the data to be analyzed for insights.

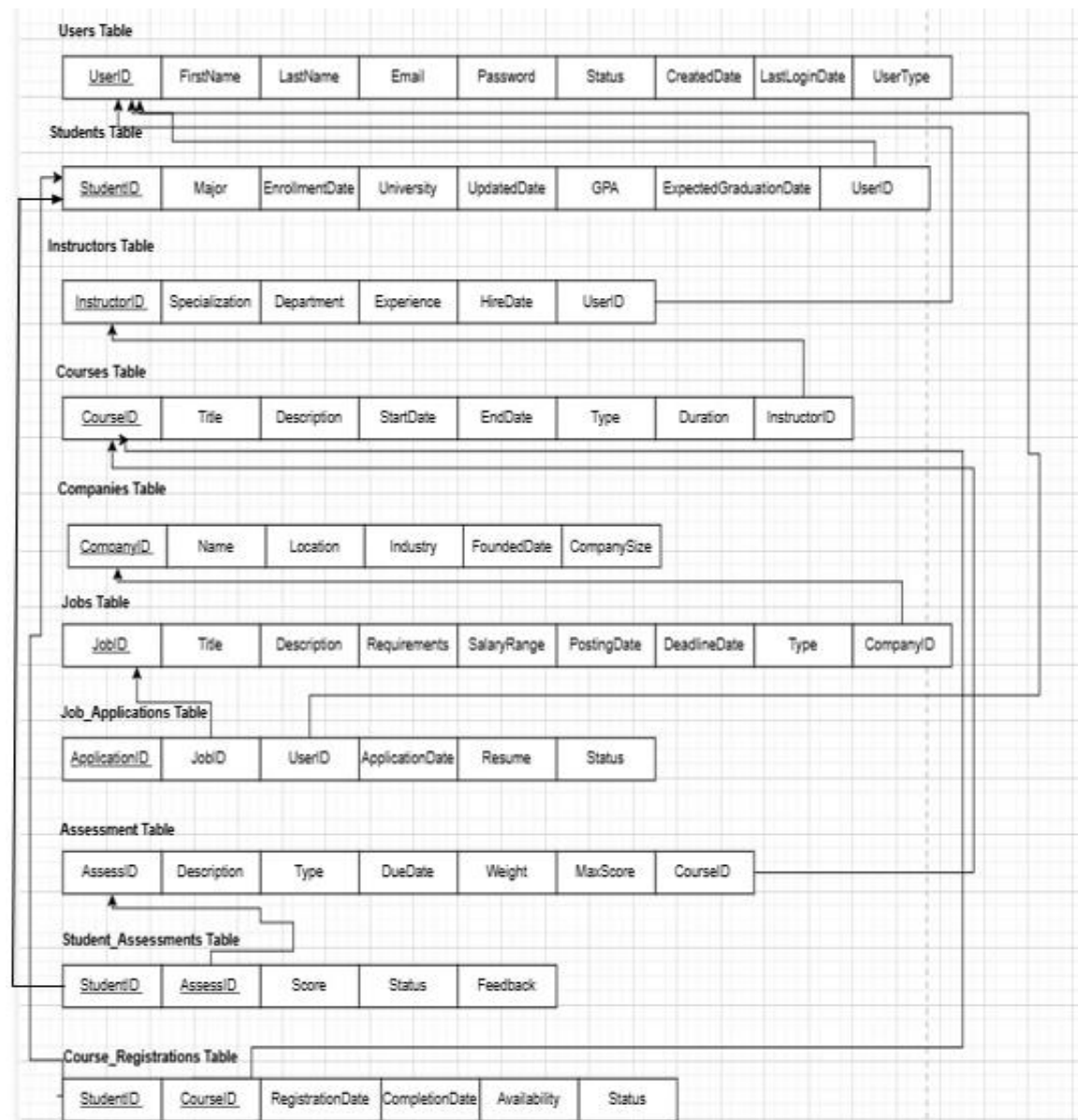


Figure 21. Mapping

Chapter Five

5. SYSTEM IMPLEMENTATION

- 5.1. Introduction (Aim of the chapter)**
- 5.2. System Screenshots**

Chapter Five: System Implementation

5.1 Introduction

The implementation plan facilitates the execution of a plan, idea, model, design, specification, or policy by presenting clear implementation steps that need to follow, after we have carefully planned our project and show you its business plan and system analysis, you will be ready to start the project implementation phase. The implementation phase involves putting the project plan into action which you can interact with.

5.2 System screenshots

the following are the system pages:

5.2.1 Home page

The home page consists of navigation bar, Hero section, Why Choose Our Platform? section, CTA section, and footer.

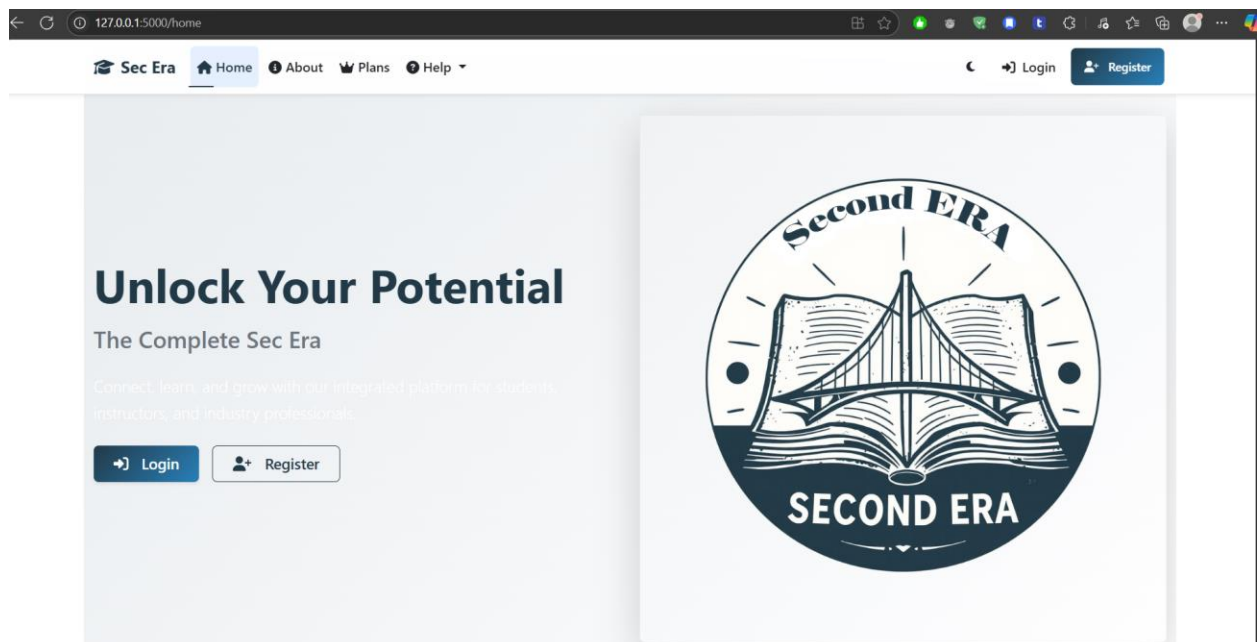


Figure 22. Home page (Hero Section)

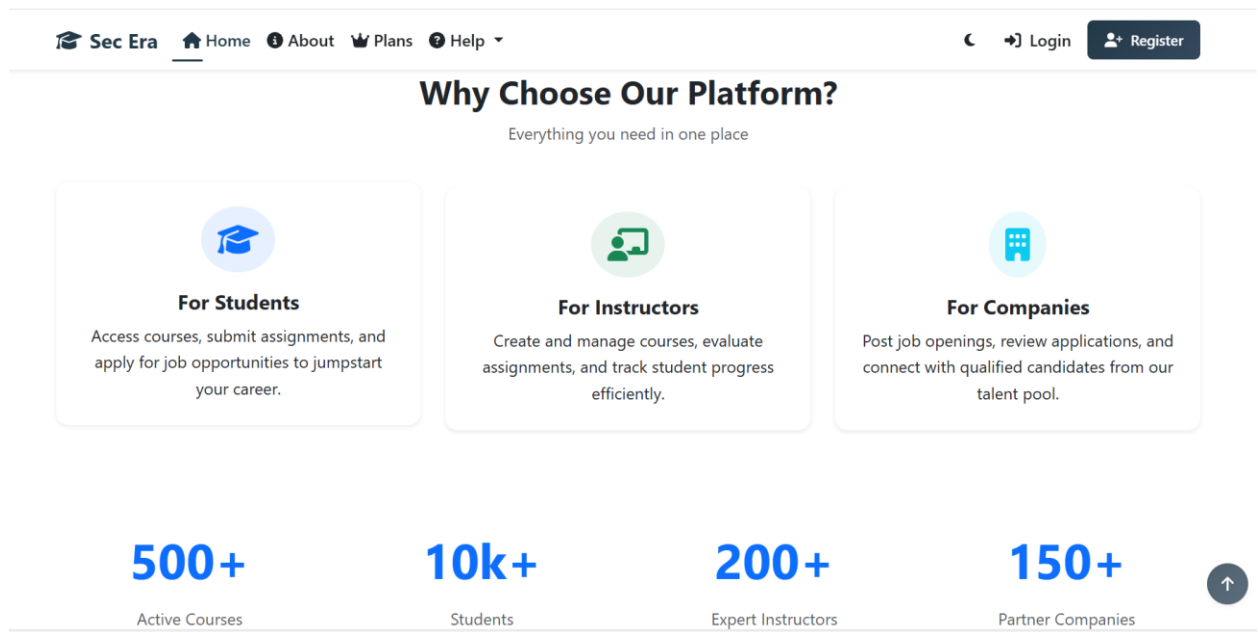


Figure 23. Home page (Info Section)

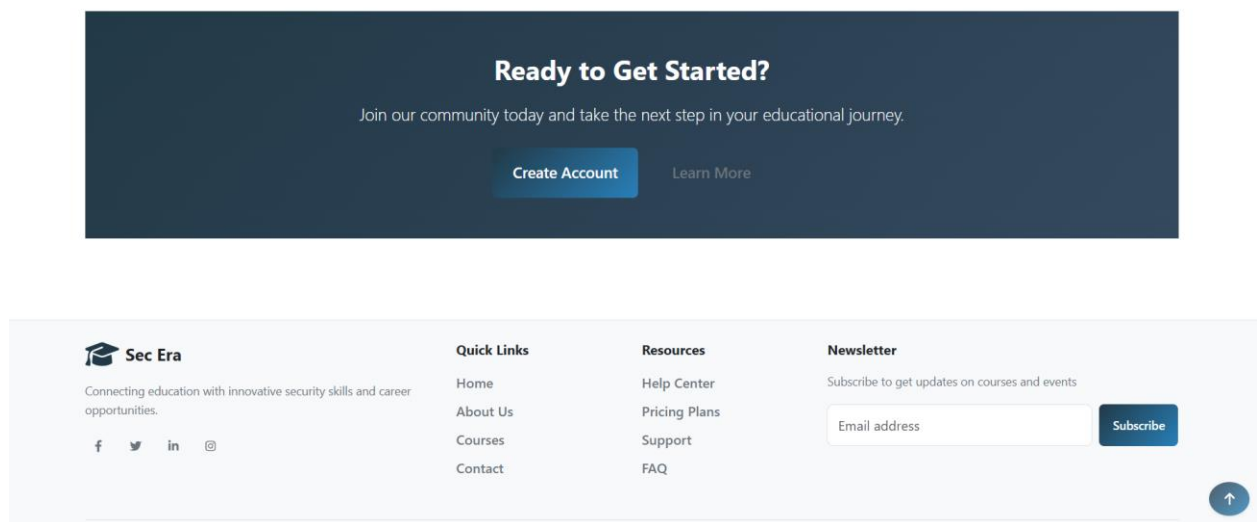


Figure 24. Home page (CTA Section, Footer)

5.2.2 About Page

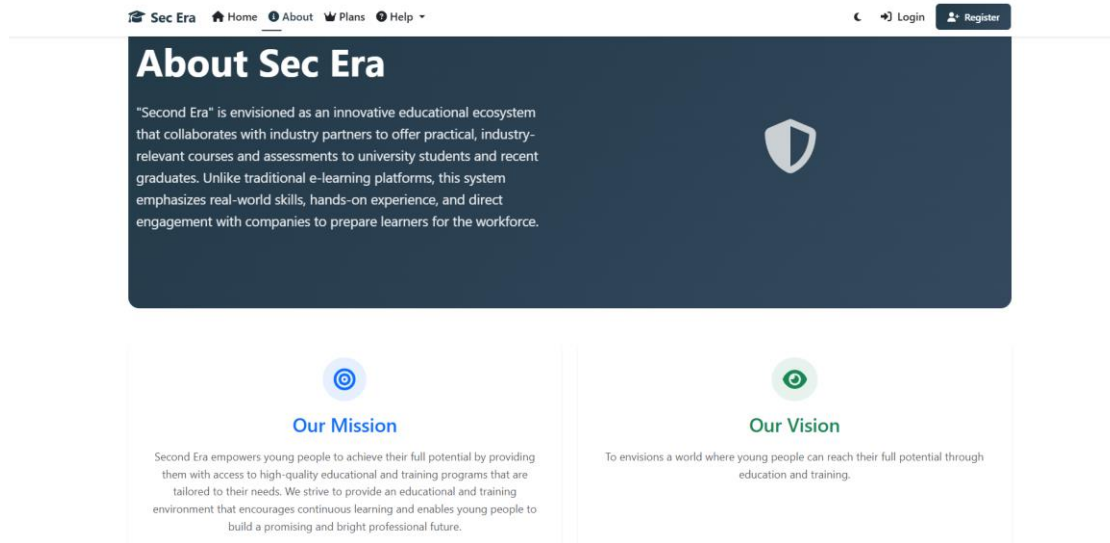


Figure 25. About Page (1)

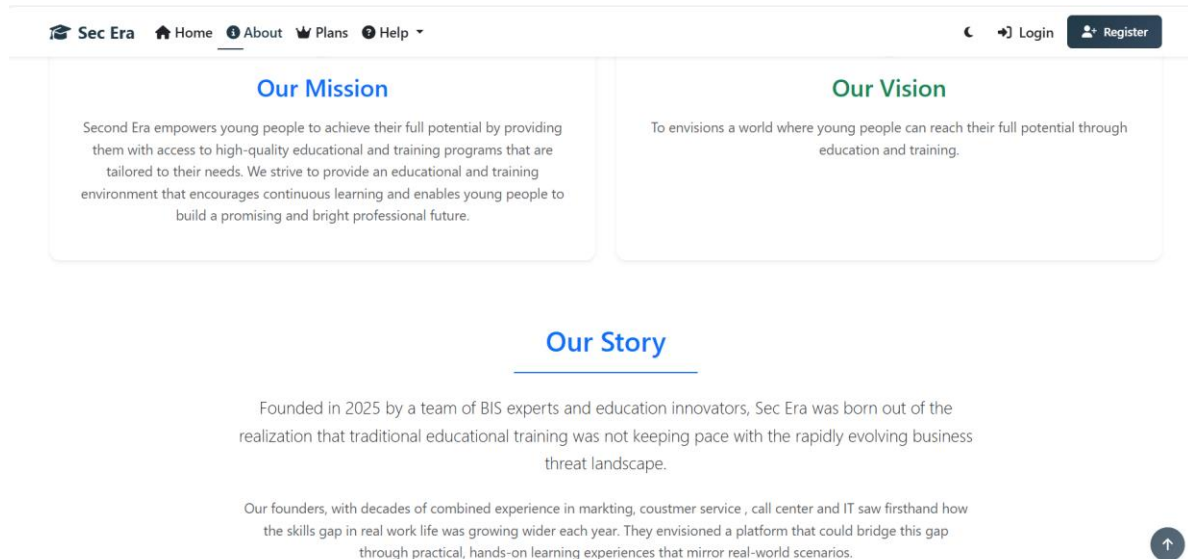


Figure 26. About Page (2)

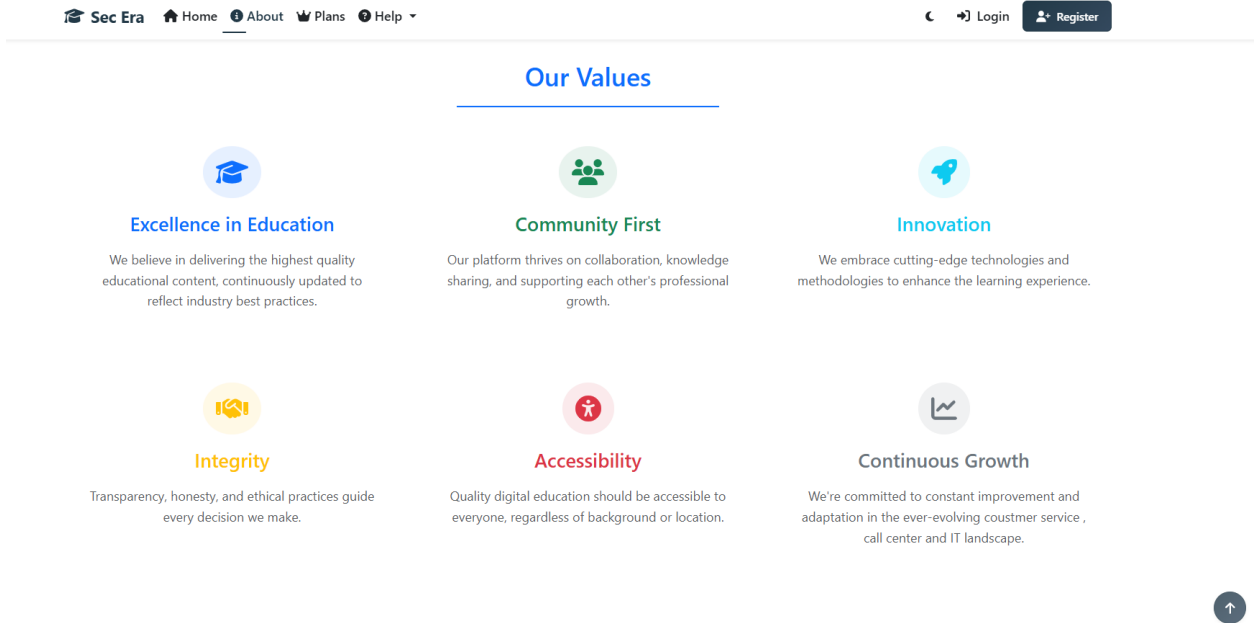


Figure 27. About Page (3)

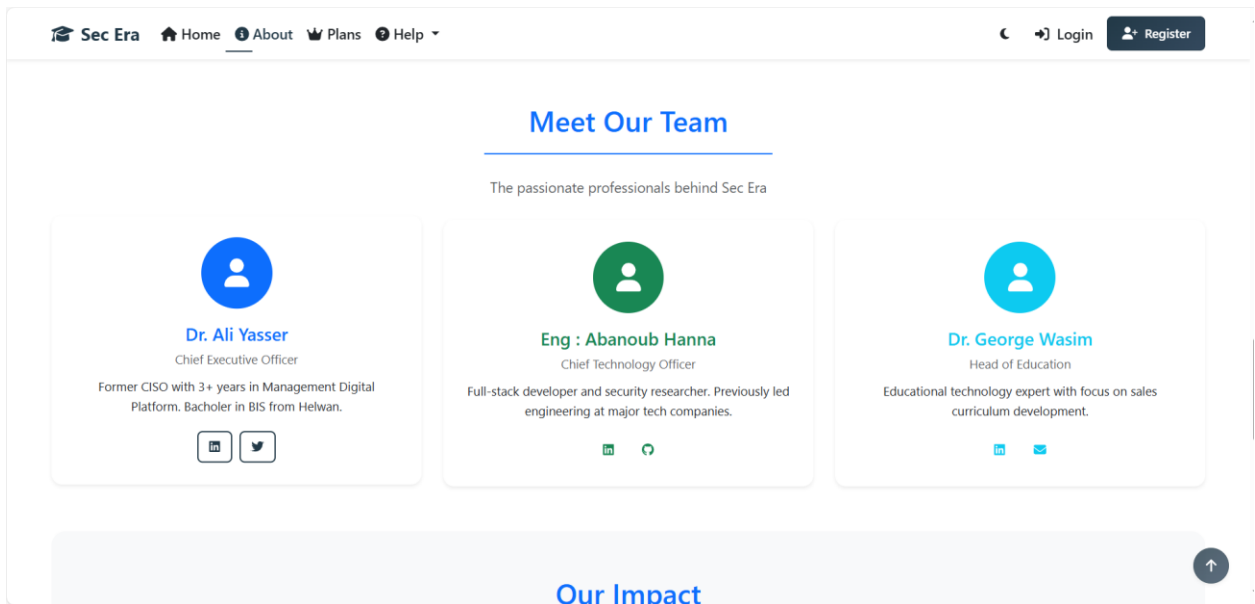


Figure 28. About Page (4)

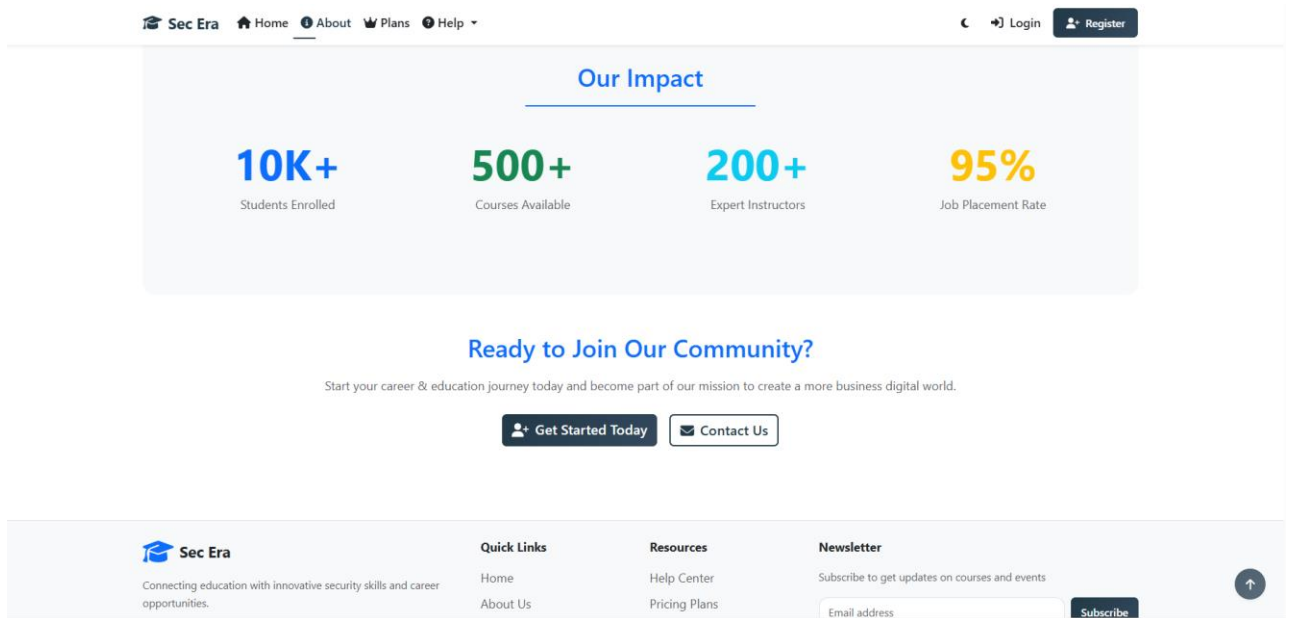


Figure 29. About Page (5)

5.2.3 Plans Page

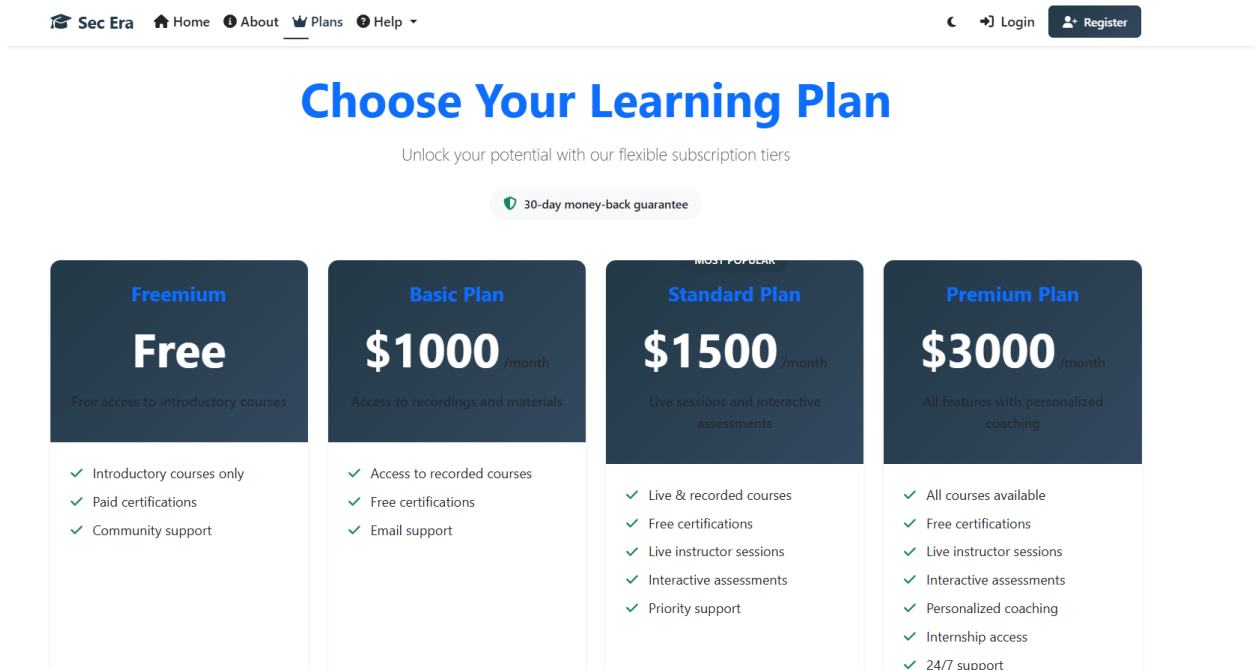


Figure 30. Plans Page (1)

[Home](#)
[About](#)
[Plans](#)
[Help](#)

[Login](#)
[Register](#)

[Get Started](#)

Feature Comparison					
Features	Freemium	Basic Plan	Standard Plan	Premium Plan	Premium Annual
Course Access	Limited	Recordings	Live + Recorded	All Access	All Access
Certifications	\$	✓	✓	✓	✓
Live Sessions	×	×	✓	✓	✓
Personal Coaching	×	×	×	✓	✓
Internship Access	×	×	×	✓	✓

Frequently Asked Questions

Can I switch plans anytime?

How do installments work for the Premium Annual plan?

Figure 31. Plans Page (2)

5.2.4 Contact & Support Page

[Home](#)
[About](#)
[Plans](#)
[Help](#)

[Login](#)
[Register](#)

Contact Support

Get help from our expert support team

24 Hour Response

Average response time

Expert Team

Cybersecurity professionals

98% Satisfaction

Customer satisfaction rate

Send Us a Message

Full Name *

Email Address *

Category

General Question

Priority

Normal - Standard issue

Subject *

Quick Help

Before contacting support, try these quick solutions:

- Check our FAQ
- Clear browser cache
- Check system status

Direct Contact

Figure 32. Contact & Support Page (1)

Page 94 of 142

Figure 33 Contact & Support Page (2)

5.2.5 Help Page

Figure 34. Help Page (FAQ)

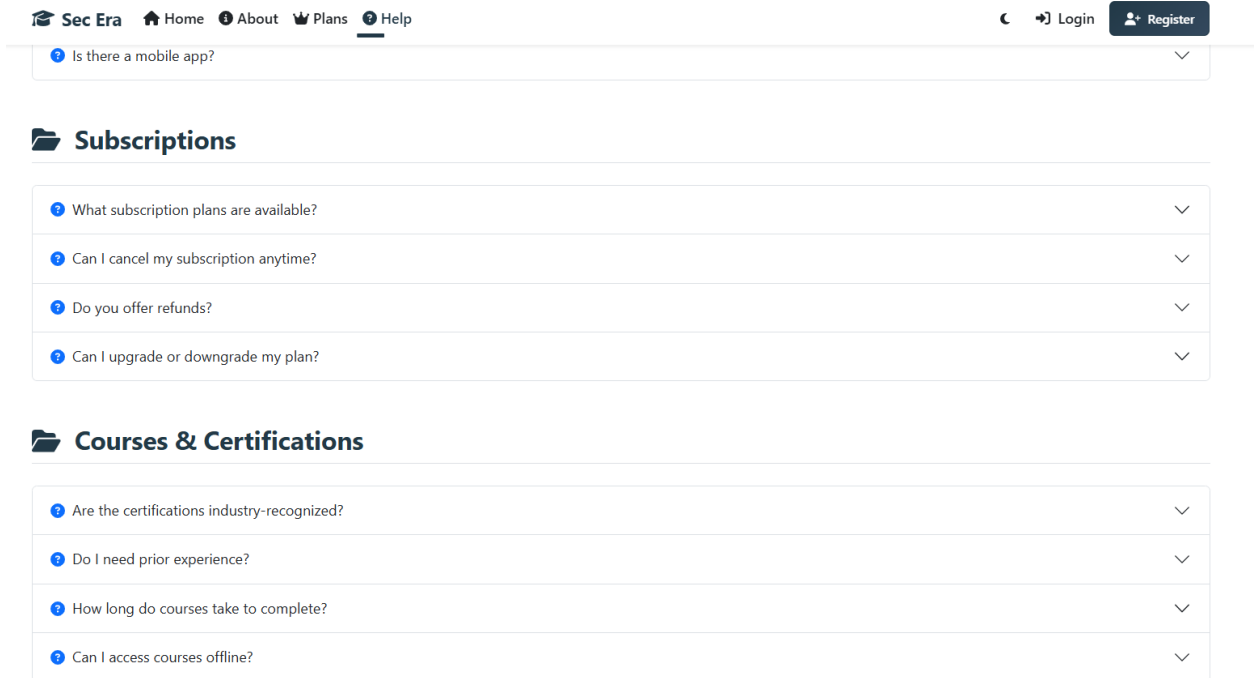


Figure 35. Help Page (FAQ)

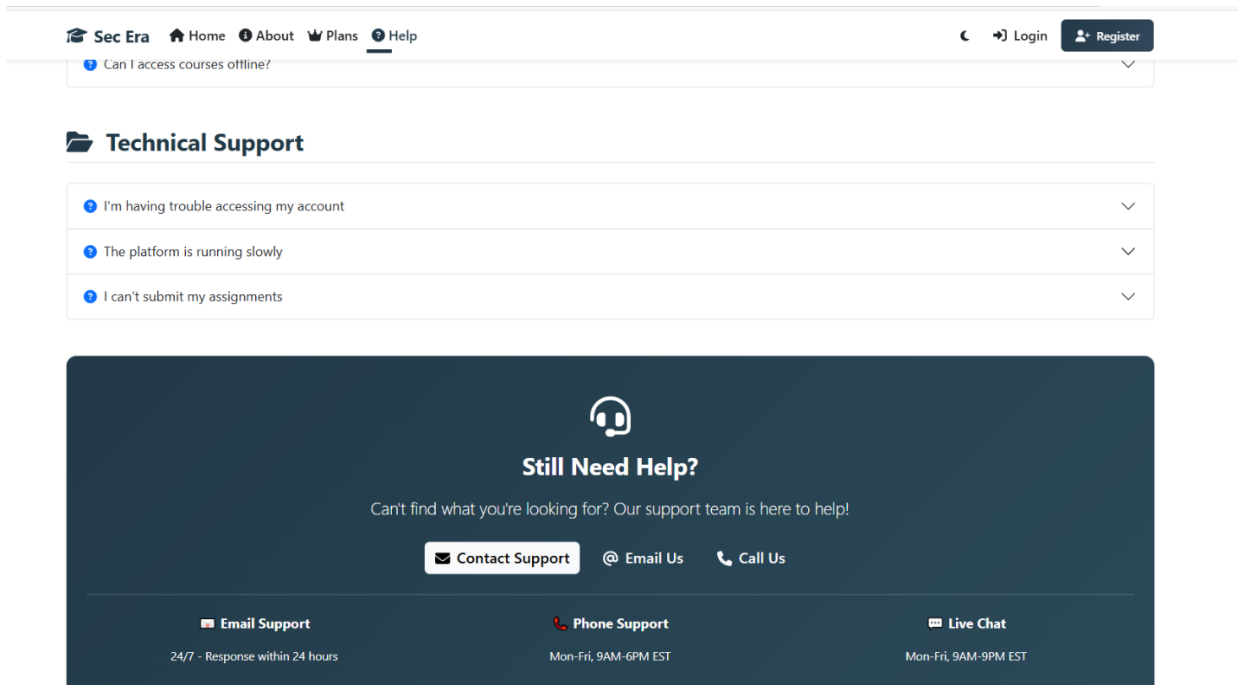


Figure 36. Help Page (FAQ)

5.2.6 Register Page

Sec Era Home About Plans Help

Login Register

Create Your Account

Basic Information

First Name
Ahmed

Last Name
Ali

Email
ahmedali@gmail.com

Password
.....

Password strength: Good

User Type
Select User Type

Create Account

Already have an account? Login here

Figure 37. Register Page

5.2.6.1 Register as Student:

Sec Era Home About Plans Help

Login Register

Password
.....

Password strength: Good

User Type
Student

Student Information

University
Helwan

Major
BIS

GPA
3

Expected Graduation Date
06/30/2025

Short Bio (optional)

Resume File (optional)

Create Account

Already have an account? [Login here](#)

Figure 38. Register Page (Student)

5.2.6.2 Register as Instructor:

The screenshot displays the 'Register as Instructor' page in the 'Sec Era' application. The page features a top navigation bar with links for 'Home', 'About', 'Plans', and 'Help', and buttons for 'Login' and 'Register'. The main content area is divided into two sections: 'Basic Information' and 'Instructor Information'.

Basic Information:

- First Name: Abanoub
- Last Name: Mikhail
- Email: abanoub@gmail.com
- Password: (masked with dots)
- Password strength: Good
- User Type: Instructor

Instructor Information:

- Department: IT
- Specialization: Cyber Security
- Years of Experience: 2
- Short Bio (optional): (text area)

Figure 39. Register Page (Instructor)

5.2.6.3 Register as Company Representer:

Sec Era Home About Plans Help

Login Register

Alice Alixes

Email
itsolutions@gmail.com

Password
.....

Password strength: Good

User Type
Company Representative

Company Information

Company Name
IT Solutions

Industry
IT

Location
Maadi

Company Size
20

Founded Date
06/18/2021

Short Bio (optional)

Create Account

Activate Window
Go to Settings to activate

Figure 40 . Register Page (Company Representer)

5.2.7 login Page:

5.2.7.1 Public & student (After Registration)

Sec Era Home About Plans Help

Login Register

Welcome Back

Please login to your account

Email Address

Password

Forgot Password?

Login

Don't have an account? Sign Up

Or login with

Google Facebook

Figure 41. login Page (Public & student (After Registration))

5.2.7.2 Instructor & Company Representer (wait until admin approve) (After Registration)

Sec Era Home About Plans Help

Login Register

Welcome Back

Please login to your account

Your account is pending approval. Please wait for admin approval.

Email Address

abanoub@gmail.com

Password

Enter your password

Forgot Password?

Login

Don't have an account? Sign Up

Or login with

Google Facebook

Figure 42. login Page (Instructor & Company Representer (wait until admin approve) (After Registration))

5.2.8 Dashboard Page:

5.2.8.1 Student

 **Sec Era**

[Dashboard](#) [My Courses](#) [Assignments](#) [Jobs](#) [Enroll](#) [My Plan](#) [Help](#)

 **A** Ahmed Ali

Welcome, Ahmed Ali

Profile Information
Email: ahmedali@gmail.com
User Type: Student
Studentid: 4
University: Helwan
Major: BIS
Gpa: 3.0
Expectedgraduationdate: 2025-06-30
Resume file: None
Bio: None
Subscription tier: freemium
Subscription status: active
Subscription start: None
Subscription end: None
Created at: 2025-06-12 11:41:36
Updated at: 2025-06-12 11:41:36

Quick Links

Courses
Assignments
Job Listings

 **Sec Era**
Connecting education with innovative security skills and career opportunities.

Quick Links

Home
About Us

Resources

Help Center
Pricing Plans

Newsletter
Subscribe to get updates on courses and events
 [Subscribe](#)

Figure 43. Dashboard Page (Student)

5.2.8.2 Instructor:

 **Sec Era**

[Dashboard](#) [My Courses](#) [Assignments](#) [Help](#)

 **A** **Abanoub Mikhail**

Welcome, Abanoub Mikhail

Profile Information

Email: abanoub@gmail.com
User Type: Instructor
Instructorid: 4
Department: IT
Specialization: Cyber Security
Experience: 2
Bio: None
Qualifications: None
Approvalstatus: Approved
Approvedby: 1
Approvedat: 2025-06-12 12:12:08
Rejectionreason: None
Createdat: 2025-06-12 11:58:10
Updatedat: 2025-06-12 12:12:08

Quick Links

[Courses](#)[Assignments](#)

 **Sec Era**

Connecting education with innovative security skills and career opportunities.

[f](#) [t](#) [in](#) [@](#)

Quick Links

[Home](#)[About Us](#)[Courses](#)[Contact](#)

Resources

[Help Center](#)[Pricing Plans](#)[Support](#)[FAQ](#)

Newsletter

Subscribe to get updates on courses and events

[Subscribe](#)

Figure 44. Dashboard Page (Instructor)

5.2.8.3 Company Representer:

 Sec Era

[Dashboard](#)

[Manage Jobs](#)

[Help](#)

 A Alice Alixes

Welcome, Alice Alixes

Profile Information

Email: itsolutions@gmail.com

User Type: Company

Companyid: 2

Name: IT Soutions

Industry: IT

Location: Maadi

Companysize: 20

Foundeddate: 2021-06-18

Bio: None

Website: None

Verificationdocument: None

Approvalstatus: Approved

Approvedby: 1

Approvedat: 2025-06-12 12:28:35

Rejectionreason: None

Createdat: 2025-06-12 12:27:09

Updatedat: 2025-06-12 12:28:35

Quick Links

Courses

Assignments

 Sec Era

Connecting education with innovative security skills and career opportunities.

[f](#)

[t](#)

[in](#)

[@](#)

Quick Links

Home

About Us

Courses

Control

Resources

Help Center

Pricing Plans

Support

FAQ

Newsletter

Subscribe to get updates on courses and events

Email address

Subscribe

Activate Windows

Go to Settings to activate Windows

Figure 45. Dashboard Page (Company Representer)

Page 105 of 142

5.2.9 Home Page (After Login):

5.2.9.1 Student

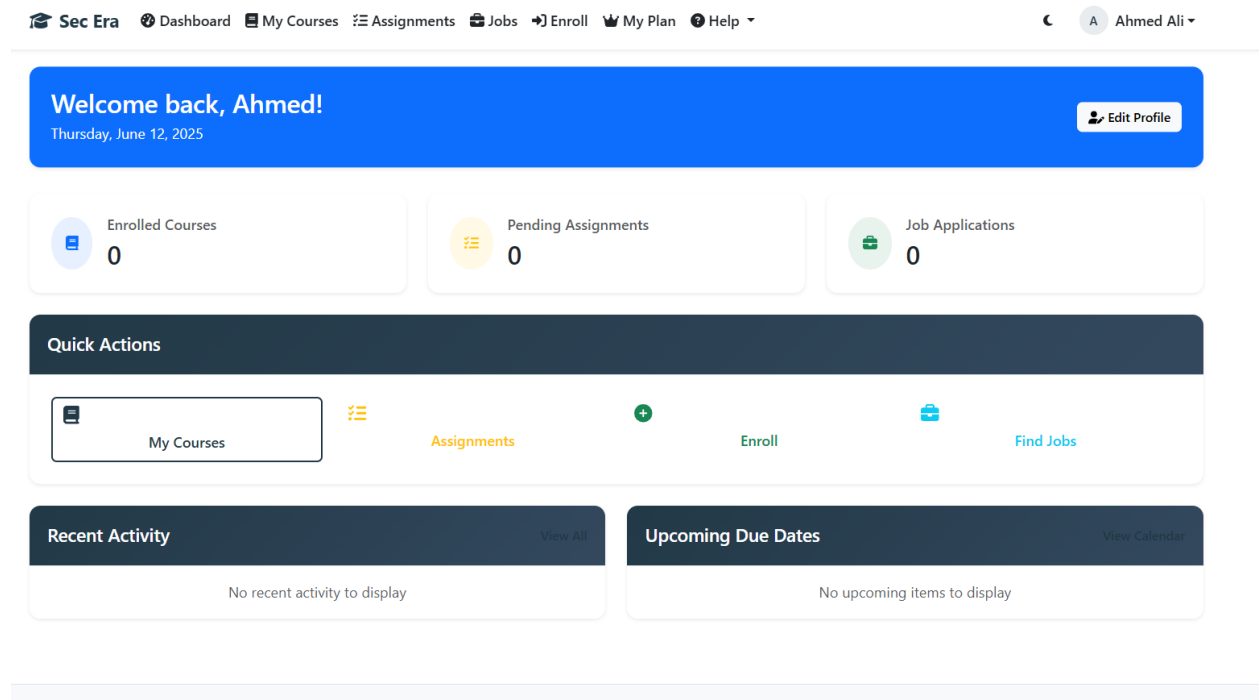


Figure 46. Home Page (After Login) (Student)

5.2.9.2 Instructor:

 **Sec Era**

[Dashboard](#) [My Courses](#) [Assignments](#) [Help](#)

 A **Abanoub Mikhail**

Welcome back, Abanoub!
Thursday, June 12, 2025

 **Edit Profile**

 **Active Courses**
0

 **Enrolled Students**
0

 **Pending Reviews**
0

Quick Actions

 **My Courses**

 **Grade Work**

 **Create Course**

 **New Assignment**

Recent Activity [View All](#)

No recent activity to display

Scheduled Classes [View Calendar](#)

No upcoming items to display

 **Sec Era**
Connecting education with innovative security skills and career opportunities.

Quick Links
[Home](#)
[About Us](#)

Resources
[Help Center](#)
[Pricing Plans](#)

Newsletter
Subscribe to get updates on courses and events
 Subscribe

Figure 47. Home Page (After Login) (Instructor)

5.2.9.3 Company Representer:

The screenshot shows the Sec Era dashboard for a company representer, Alice Alix. The top navigation bar includes the Sec Era logo, links to Dashboard, Manage Jobs, and Help, along with a user profile icon and the name Alice Alix. A blue welcome banner at the top says "Welcome back, Alice!" with the date Thursday, June 12, 2025, and an "Edit Profile" button. Below the banner are three summary cards: "Active Job Listings" with a count of 0, "New Applications" with a count of 0, and "Candidates Hired" with a count of 0. A "Quick Actions" section follows, featuring a "My Listings" button, a "Post New Job" button, a "Review Applications" button, and a "Find Talent" button. Below this are two sections: "Recent Activity" and "Closing Job Listings", both showing "No recent activity to display" and "No upcoming items to display" respectively. The footer contains the Sec Era logo and tagline, "Quick Links" (Home, About Us), "Resources" (Help Center, Pricing Plans), a "Newsletter" subscription form, and an "Activate Windows" watermark.

Sec Era | Dashboard | Manage Jobs | Help | Alice Alix

Welcome back, Alice!
Thursday, June 12, 2025 | [Edit Profile](#)

Active Job Listings
0

New Applications
0

Candidates Hired
0

Quick Actions

[My Listings](#) | [Post New Job](#) | [Review Applications](#) | [Find Talent](#)

Recent Activity | [View All](#)
No recent activity to display

Closing Job Listings | [View Calendar](#)
No upcoming items to display

Sec Era
Connecting education with innovative security skills and career opportunities.

Quick Links
[Home](#)
[About Us](#)

Resources
[Help Center](#)
[Pricing Plans](#)

Newsletter
Subscribe to get updates on courses and events
 [Subscribe](#)

Activate Windows
Go to Settings to activate Windows

Figure 48. Home Page (After Login) (Company Representer)

5.2.10 Student Only Pages:

5.2.10.1 Enroll Page

 **Sec Era**  Dashboard  My Courses  Assignments  Jobs  **Enroll**  My Plan  Help

Ahmed Ali

Course Enrollment

HTML Fundamentals: Build the Web from Scratch

Course Description: Dive into the foundation of the web with this hands-on HTML (HyperText Markup Language) course, designed for absolute beginners and aspiring web developers. In this course, you'll learn how to structure websites using HTML5 — the core language behind every webpage on the internet. From creating simple web pages to laying out content with semantic elements, you'll gain the skills needed to build clean, accessible, and well-organized HTML documents. No prior coding experience is required — just curiosity and a willingness to learn! What You'll Learn: The structure and syntax of HTML How to create and organize content with headings, paragraphs, lists, and links Embedding images, videos, and other media Building forms for user input Using semantic HTML to improve SEO and accessibility Best practices for clean, maintainable code Who This Course Is For: Beginners with no prior coding experience Students exploring web development Professionals looking to understand how websites are built Anyone curious about how the internet works Course Outcomes: By the end of this course, you'll be able to create your own HTML-based webpages, understand how web browsers render content, and be ready to move on to CSS and JavaScript for styling and interactivity.

Instructor: Alex Beta

Enroll

 **Sec Era**
Connecting education with innovative security skills and career

Quick Links
[Home](#)

Resources
[Help Center](#)

Newsletter
[Subscribe to get updates on courses and events](#)

Figure 49. Enroll Page

5.2.10.2 My Courses Page:

 Sec Era  Dashboard  My Courses  Assignments  Jobs  Enroll  My Plan  Help

A Ahmed Ali

Courses

HTML Fundamentals: Build the Web from Scratch

Course Description: Dive into the foundation of the web with this hands-on HTML (HyperText Markup Language) course, designed for absolute beginners and aspiring web developers. In this course, you'll learn how to structure websites using HTML5 — the core language behind every webpage on the internet. From creating simple web pages to laying out content with semantic elements, you'll gain the skills needed to build clean, accessible, and well-organized HTML documents. No prior coding experience is required — just curiosity and a willingness to learn! What You'll Learn: The structure and syntax of HTML How to create and organize content with headings, paragraphs, lists, and links Embedding images, videos, and other media Building forms for user input Using semantic HTML to improve SEO and accessibility Best practices for clean, maintainable code Who This Course Is For: Beginners with no prior coding experience Students exploring web development Professionals looking to understand how websites are built Anyone curious about how the internet works Course Outcomes: By the end of this course, you'll be able to create your own HTML-based webpages, understand how web browsers render

Figure 50. My Courses Page

5.2.10.3 Course Details Page:

Sec Era

Dashboard

My Courses

Assignments

Jobs

Enroll

My Plan

Help

A

Ahmed Ali

HTML Fundamentals: Build the Web from Scratch

Instructor:

Course Description: Dive into the foundation of the web with this hands-on HTML (HyperText Markup Language) course, designed for absolute beginners and aspiring web developers. In this course, you'll learn how to structure websites using HTML5 — the core language behind every webpage on the internet. From creating simple web pages to laying out content with semantic elements, you'll gain the skills needed to build clean, accessible, and well-organized HTML documents. No prior coding experience is required — just curiosity and a willingness to learn! What You'll Learn: The structure and syntax of HTML How to create and organize content with headings, paragraphs, lists, and links Embedding images, videos, and other media Building forms for user input Using semantic HTML to improve SEO and accessibility Best practices for clean, maintainable code Who This Course Is For: Beginners with no prior coding experience Students exploring web development Professionals looking to understand how websites are built Anyone curious about how the internet works Course Outcomes: By the end of this course, you'll be able to create your own HTML-based webpages, understand how web browsers render content, and be ready to move on to CSS and JavaScript for styling and interactivity.

Course Details

Start Date: 2025-06-15

End Date: 2025-06-30

Type:

Duration: 20

Assessments

Create a Personal Webpage Using HTML

Due: 2025-06-07

[View Details](#)

??? **HTML Course Assignment 1: Build Your First Personal Web Page** **Assignment Title:** "Create a Personal Webpage Using HTML" **Objective:** Apply your understanding of basic HTML tags and structure by creating a simple, well-organized personal web page. --- ## ??? **Instructions:** 1. **Create an HTML file** named 'index.html'. 2. Your webpage should include the following elements: * A title in the '<title>' tag * A main heading using '<h1>' * At least two paragraphs ('<p>') describing yourself * A list ('' or '') of your hobbies or favorite things * An image of your choice using the '' tag * A hyperlink to an external website (e.g., your favorite blog, YouTube channel, etc.) using '<a>' * A footer with your name and the current year 3. Use proper HTML document structure: ""html <!DOCTYPE html> <html> <head> <title>Your Name - Personal Page</title> </head> <body> <!-- Your content here --> </body> </html> "" --- ## ? **Requirements:** * Use semantic HTML where possible * File should be named 'index.html' * All tags must be properly opened and closed * Indent code for readability --- ## ??? **Submission Guidelines:** * Submit your 'index.html' file via [your platform's submission system or email]. * Make sure all resources (e.g., images) are included if you use local files. * Deadline: [Insert Date Here] --- ## ??? **Grading Criteria:** * Criteria | Points | | ----- | ----- | | Correct HTML structure | 10 | | Use of required elements | 10 | | Code readability and indentation | 5 | | Creativity and completeness | 5 | | **Total** | **30** |

MAX SCORE: 100.0

WEIGHT: 10%

Figure 51. Course Details Page

Page 111 of 142

5.2.10.4 Assessment Details Page:

Sec Era

Dashboard

My Courses

Assignments

Jobs

Enroll

My Plan

Help

A Ahmed Ali

Create a Personal Webpage Using HTML

Course: HTML Fundamentals: Build the Web from Scratch

??? **HTML Course Assignment 1: Build Your First Personal Web Page** **Assignment Title:** *Create a Personal Webpage Using HTML* **Objective:** *Apply your understanding of basic HTML tags and structure by creating a simple, well-organized personal web page.* --- ### ??? **Instructions:** *1. **Create an HTML file** named 'index.html'. 2. Your webpage should include the following elements: * A title in the '<title>' tag * A main heading using '<h1>' * At least two paragraphs ('<p>') describing yourself * A list ('' or '') of your hobbies or favorite things * An image of your choice using the '' tag * A hyperlink to an external website (e.g., your favorite blog, YouTube channel, etc.) using '<a>' * A footer with your name and the current year 3. Use proper HTML document structure: '<html> <!DOCTYPE html> <html> <head> <title>Your Name - Personal Page</title> </head> <body> <!-- Your content here --> </body> </html>' --- ### ? Requirements: * Use semantic HTML where possible * File should be named 'index.html' * All tags must be properly opened and closed * Indent code for readability --- ### ??? **Submission Guidelines:** * Submit your 'index.html' file via \[your platform's submission system or email]. * Make sure all resources (e.g., images) are included if you use local files. * Deadline: \[Insert Date Here] --- ### ??? **Grading Criteria:** * Criteria | Points | |-----| | Correct HTML structure | 10 | | Use of required elements | 10 | | Code readability and indentation | 5 | | Creativity and completeness | 5 | | **Total** | 30** |

Assessment Details

Due Date: 2025-06-07

Max Score: 100.0

Weight: 10%

Submit Assessment

Your Answer

Figure 52. Assessment Details Page

5.2.10.5 Find Job Page

Sec Era Dashboard My Courses Assignments **Jobs** Enroll My Plan Help

A Ahmed Ali

Job Listings

Frontend Developer
RTA

We are looking for a talented and passionate Frontend Developer to join our team! In this role, you will be responsible for creating beautiful, responsive, and user-friendly web interfaces. You'll work closely with designers, backend developers, and product managers to bring modern web applications to life. Key Responsibilities: Build and maintain modern web interfaces using HTML, CSS, JavaScript, and frameworks like React or Vue.js. Collaborate with UX/UI designers to implement responsive and accessible designs. Optimize applications for maximum speed and scalability. Work with RESTful APIs and backend developers to integrate data services. Debug and troubleshoot UI/UX issues and improve usability.

[View Details](#) [NOT APPLIED](#)

Sec Era
Connecting education with innovative security skills and career opportunities.

f t in @

Quick Links
Home
About Us
Courses
Contact

Resources
Help Center
Pricing Plans
Support
FAQ

Newsletter
Subscribe to get updates on courses and events

[Subscribe](#)

Figure 53. Find Job Page

5.2.10.6 Payment Page

Sec Era Dashboard My Courses Assignments Jobs Enroll My Plan Help

P pop alice

UPGRADING TO

Current Plan
Freemium
FREE

New Plan
Standard Plan
\$1500/month

What You'll Get:

✓ **Course Access:** Live_and_recorded

🔒 **Live Sessions:** Included

🌟 **Certifications:** Included

📞 **Support:** Priority

Payment Options

Payment Method

☒ Credit/Debit Card ☐ PayPal

[← Change Plan](#) [→ Proceed to Payment](#)

Standard Plan
Live sessions and interactive assessments

Total: **\$1500.00**

30-Day Money-Back Guarantee
Cancel anytime within 30 days for a full refund

Need Help?
Our team is here to help you choose the right plan

[Contact Support](#) [View FAQ](#)

Figure 54. Payment Page

5.2.10.6 Job Details Page:

Sec Era

Dashboard

My Courses

Assignments

Jobs

Enroll

My Plan

Help

A Ahmed Ali

Frontend Developer

RTA

Cairo

IT

Apply Now

Job Description

We are looking for a talented and passionate Frontend Developer to join our team! In this role, you will be responsible for creating beautiful, responsive, and user-friendly web interfaces. You'll work closely with designers, backend developers, and product managers to bring modern web applications to life. Key Responsibilities: Build and maintain modern web interfaces using HTML, CSS, JavaScript, and frameworks like React or Vue.js. Collaborate with UX/UI designers to implement responsive and accessible designs. Optimize applications for maximum speed and scalability. Work with RESTful APIs and backend developers to integrate data services. Debug and troubleshoot UI/UX issues and improve usability.

Job Details

Type: Full-time

Posted: 2025-06-11

Deadline: 2025-06-30

Requirements

- Proficiency in HTML, CSS, JavaScript
-
- Experience with React, Vue.js, or other modern frameworks
-

Figure 55. Job Details Page

5.2.10.6.1 Apply for Job Section:

The screenshot shows a web application interface for applying for a job. The main page is titled "Frontend Developer" and includes a sidebar with "RTA", "Cairo", "IT", and a profile icon. The main content area is divided into three sections: "Job Description", "Requirements", and "Job Details".

Job Description: We are looking for a talented and passionate Frontend Developer to join our team! In this role, you will be responsible for creating beautiful, responsive, and user-friendly web interfaces. You'll work closely with designers, backend developers, and product managers to bring modern web applications to life. Key Responsibilities: Build and maintain modern web interfaces using HTML, CSS, JavaScript, and frameworks like React or Vue.js. Collaborate with UX/UI designers to implement responsive and accessible designs. Optimize applications for maximum speed and scalability. Work with RESTful APIs and backend developers to integrate data services. Debug and troubleshoot UI/UX issues and improve usability.

Requirements:

- ✓ Proficiency in HTML, CSS, JavaScript
- ✓
- ✓ Experience with React, Vue.js, or other modern frameworks

Job Details:

- Type:** Full-time
- Posted:** 2025-06-11
- Deadline:** 2025-06-30

A modal window titled "Apply for Frontend Developer" is open, showing a "Resume / Cover Letter" section with a "Choose File" button, a "No file chosen" message, and an "Upload" button. At the bottom of the modal are "Close" and "Submit Application" buttons.

Figure 56. Apply for Job Section

5.2.11 Instructor Only Pages:

5.2.11.1 Create course page

Sec Era Dashboard My Courses Assignments Help

Abanoub Mikhail

Create a New Course

Title
Cyber Security (Penetration Testing)

Description
40 hours total: 60 % lab work, 40 % instructor-led demos and reviews
Live online or on-site; recordings and lab VPN access included
Basic Linux/Windows command-line familiarity recommended; no prior hacking experience required

Start Date
07/10/2025

End Date
10/31/2025

Duration
3 month

Create Course

Sec Era
Connecting education with innovative security skills and career opportunities.

Quick Links
Home
About Us
Courses

Resources
Help Center
Pricing Plans
Support

Newsletter
Subscribe to get updates on courses and events
email address **Subscribe**

Figure 57. Create course page

5.2.11.2 My Courses Page:

 **Sec Era**  Dashboard  My Courses  Assignments  Help ▾

 **A** Abanoub Mikhail ▾

guidelines Reconnaissance & OSINT: mapping targets without touching them Scanning & enumeration with Nmap, Masscan, and custom scripts Exploiting common web, network, and system vulnerabilities (OWASP Top 10, misconfigurations, privilege-escalation paths) Post-exploitation: persistence, credential dumping, lateral movement Wireless & cloud attack fundamentals Writing clear, executive-level and technical reports with remediation advice Hands-on labs & tools Metasploit, Burp Suite, Impacket, BloodHound, Mimikatz, Wireshark, and custom Python/Bash exploits in a safe, isolated lab that simulates corporate infrastructure. Format & requirements 40 hours total: 60 % lab work, 40 % instructor-led demos and reviews Live online or on-site; recordings and lab VPN access included Basic Linux/Windows command-line familiarity recommended; no prior hacking experience required

[View Details](#)

 **Sec Era**

Connecting education with innovative security skills and career opportunities.

Quick Links

[Home](#)
[About Us](#)
[Courses](#)

Resources

[Help Center](#)
[Pricing Plans](#)
[Support](#)

Newsletter

Subscribe to get updates on courses and events

[Subscribe](#)

Figure 58. My Courses Page (Instructor)

5.2.11.3 New Assignment Page:

Create New Assessment

Assessment Title

Cyber Security (Penetration Testing)

Assessment Description

Report Penetration Testing For Web site related portswigger and machine windows

Due Date

10/10/2025

Select Course

Cyber Security (Penetration Testing)

Maximum Score

100

The maximum points available for this assessment.

Weight (%)

90

The percentage weight of this assessment in the final course grade.

Create Assessment

Cancel

Activate Wind
Go to Settings to a

Figure 59. New Assignment Page

Sec Era

Dashboard

My Courses

Assignments

Help

Abanoub Mikhail

Create New Assessment

Assessment created successfully!

Figure 60. New Assignment Notification

5.2.11.4 Courses Details Page:

Sec Era

Dashboard

My Courses

Assignments

Help

Abanoub Mikhail

Cyber Security (Penetration Testing)

Instructor:

This practical, beginner-to-intermediate course teaches you how real-world attackers break into modern networks and how defenders detect and stop them. Across guided labs and mini-CTF challenges, you'll learn to plan and execute a full penetration-testing engagement—from initial reconnaissance to final report. What you'll learn Scoping, rules of engagement, and legal/ethical guidelines Reconnaissance & OSINT: mapping targets without touching them Scanning & enumeration with Nmap, Masscan, and custom scripts Exploiting common web, network, and system vulnerabilities (OWASP Top 10, misconfigurations, privilege-escalation paths) Post-exploitation: persistence, credential dumping, lateral movement Wireless & cloud attack fundamentals Writing clear, executive-level and technical reports with remediation advice Hands-on labs & tools Metasploit, Burp Suite, Impacket, BloodHound, Mimikatz, Wireshark, and custom Python/Bash exploits in a safe, isolated lab that simulates corporate infrastructure. Format & requirements 40 hours total: 60 % lab work, 40 % instructor-led demos and reviews Live online or on-site; recordings and lab VPN access included Basic Linux/Windows command-line familiarity recommended; no prior hacking experience required

Course Details

Start Date: 2025-07-10

End Date: 2025-10-31

Type:

Duration: 3 month

Assessments

Cyber Security (Penetration Testing)

Report Penetration Testing For Web site related portswiger and machine windows

MAX SCORE: 100.0 **WEIGHT:** 90%

Due: 2025-10-10

[View Details](#)

Activate Wind
Go to Settings to a

Figure 61. Courses Details Page (Instructor)

5.2.12 Company Representer Only Page:

5.2.12.1 Post New Job Page:

Sec Era

Dashboard

Manage Jobs

Help

Alice Alixes

Post a New Job

Job Title *

IT Support

Job Description *

Provide first-line technical support to end-users, ensuring that workstations, applications, and network services run smoothly. Troubleshoot incidents, maintain hardware/software inventory, and escalate complex issues to senior engineers while keeping detailed ticket records.

Provide a detailed description of the job responsibilities and expectations.

Requirements

Windows 10/11 troubleshooting; familiarity with macOS basics.
Active Directory Users & Computers, Group Policy basics.
Ticketing platforms (e.g., Jira Service Management, Freshservice, Zendesk).
Remote assistance tools (TeamViewer, AnyDesk, Microsoft Quick Assist).
Fundamental networking (TCP/IP, DHCP, DNS) and Wi-Fi configuration.
List the required skills, experience, and qualifications. Enter each requirement on a new line.

Job Type

Full-time

Location

onsite

Minimum Salary

\$ 8000

Maximum Salary

\$ 10000

Application Deadline *

08/01/2025

Cancel

Post Job

Sec Era

Connecting education with innovative security skills and career opportunities.

Quick Links

Home

About Us

Resources

Help Center

Privacy Policy

Newsletter

Subscribe to get updates on courses and events

Figure 62. Post New Job Page

5.2.12.2 My Listing Job Page:

Job Listings

[Post New Job](#)

IT Support
Provide first-line technical support to end-users, ensuring that workstations, applications, and network services run smoothly. Troubleshoot incidents, maintain hardware/software inventory, and escalate complex issues to senior engineers while keeping detailed ticket records.

[View Details](#) **APPLICATIONS: 0**

Sec Era
Connecting education with innovative security skills and career opportunities.

Quick Links
[Home](#)
[About Us](#)
[Courses](#)
[Contact](#)

Resources
[Help Center](#)
[Pricing Plans](#)
[Support](#)
[FAQ](#)

Newsletter
Subscribe to get updates on courses and events
 [Subscribe](#)

Activate Windows
Go to Settings to activate Windows.

Figure 63. My Listing Job Page

5.2.12.3 View application jobs:

Applications for Frontend Developer
Posted on 2025-06-11 [← Back to Jobs](#)

1 Applications Received

Alex Alpha
[View Cover Letter](#) [Update Status](#)
✉ Student1@mail.com
ACCEPTED Applied on 2025-06-10

Sec Era
Connecting education with innovative security skills and career opportunities.

Quick Links
[Home](#)
[About Us](#)
[Courses](#)
[Contact](#)

Resources
[Help Center](#)
[Pricing Plans](#)
[Support](#)
[FAQ](#)

Newsletter
Subscribe to get updates on courses and events
 [Subscribe](#)

Activate Windows
Go to Settings to activate Windows.

Figure 64. View application jobs

5.2.13 Admin Pages:

5.2.13.1 Dashboard

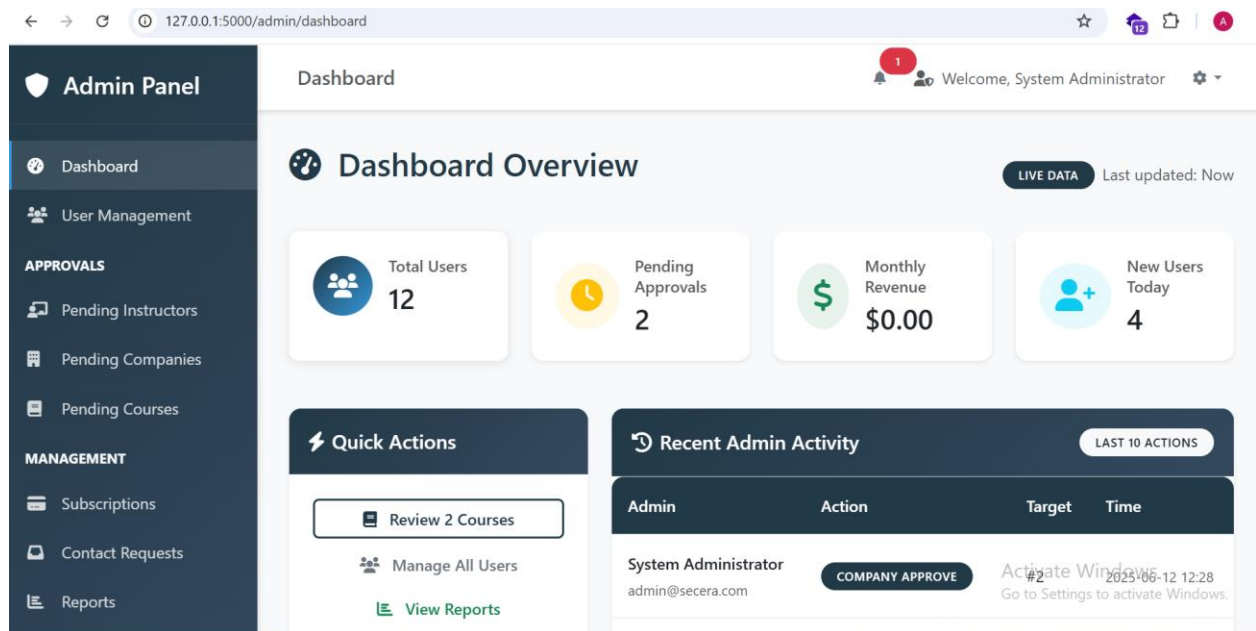


Figure 65. Admin (Dashboard)

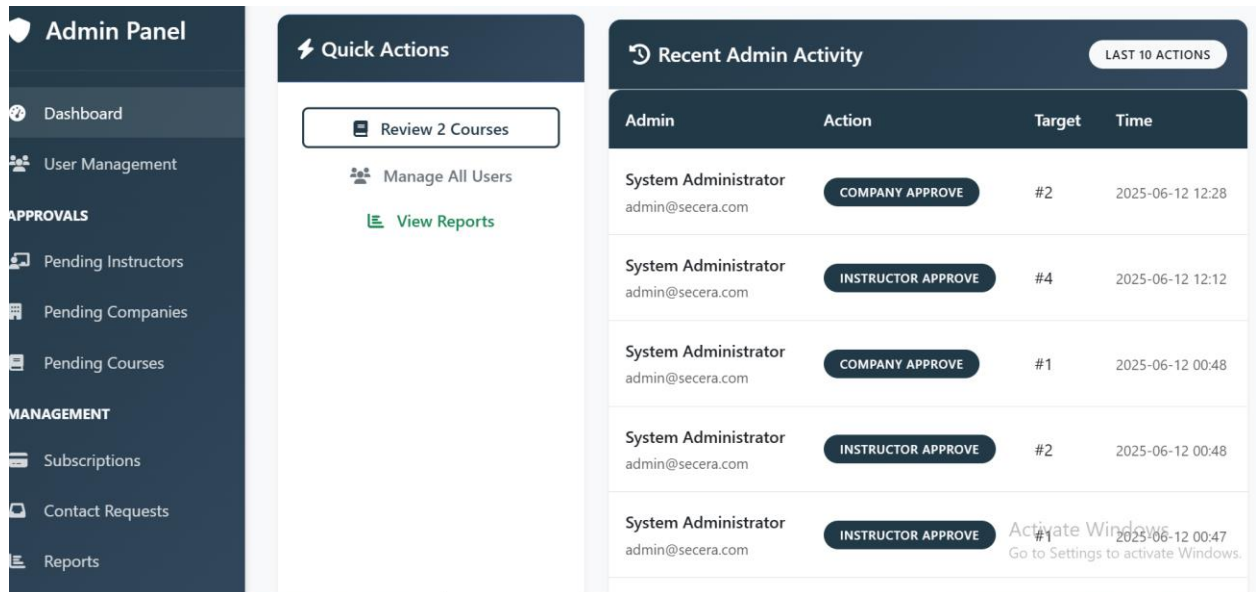


Figure 66. Admin (Dashboard)

5.2.13.2 Reports Page:

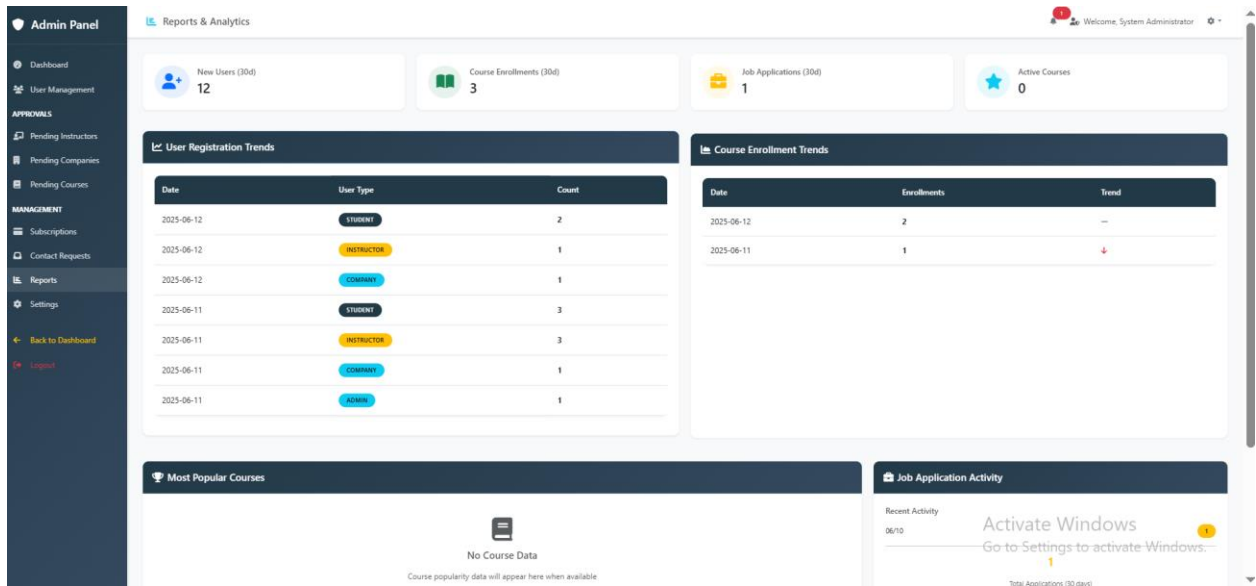


Figure 67. Admin (Reports Page)

5.2.13.3 User Management Page:

The screenshot shows the 'User Management' page in an admin panel. The left sidebar contains navigation links for 'Admin Panel', 'Dashboard', 'User Management', 'APPROVALS' (Pending Instructors, Pending Companies, Pending Courses), and 'MANAGEMENT' (Subscriptions, Contact Requests, Reports, Settings). The main content area has a header 'User Management' with a notification bell and 'Welcome, System Administrator'. Below the header, there are filters for 'Search Users' (Search by name or email), 'User Type' (Students), and 'Status' (Inactive). A 'Filter' button is present. The 'Users List' section shows 'No users found' with a message 'Try adjusting your search filters'. At the bottom right, there is a 'Activate Windows' watermark.

Admin Panel

Dashboard

User Management

APPROVALS

Pending Instructors

Pending Companies

Pending Courses

MANAGEMENT

Subscriptions

Contact Requests

Reports

Settings

Back to Dashboard

Logout

User Management

1 Welcome, System Administrator

User Management

0 USERS SHOWN Export

Search Users

Search by name or email

User Type

Students

Status

Inactive

Filter

Users List

No users found

Try adjusting your search filters

Activate Windows

Go to Settings to activate Windows.

Figure 68. User Management Page

The screenshot shows the 'User Management' page with filters set to 'Companies' and 'Active'. The 'Users List' table displays two users: Alice Alixes and Alex Delta. Both are of type 'COMPANY', status 'ACTIVE', and approval 'APPROVED'. Alice Alixes has extra info 'IT Soutions' and joined on 2025-06-12. Alex Delta has extra info 'RTA' and joined on 2025-06-11. Both have an 'Actions' link. The 'Activate Windows' watermark is present at the bottom right.

Admin Panel

Dashboard

User Management

APPROVALS

Pending Instructors

Pending Companies

Pending Courses

MANAGEMENT

Subscriptions

Contact Requests

Reports

Settings

Back to Dashboard

Logout

User Management

1 Welcome, System Administrator

User Management

2 USERS SHOWN Export

Search Users

Search by name or email

User Type

Companies

Status

Active

Filter

Users List

User	Type	Status	Approval	Extra Info	Joined	Actions
Alice Alixes itsolutions@gmail.com	COMPANY	ACTIVE	APPROVED	IT Soutions	2025-06-12	Actions
Alex Delta com1@mail.com	COMPANY	ACTIVE	APPROVED	RTA	2025-06-11	Actions

Activate Windows

Go to Settings to activate Windows.

Figure 69. User Management Page

5.2.13.4 Subscription Page:

Admin Panel

Dashboard

User Management

APPROVALS

Pending Instructors

Pending Companies

Pending Courses

MANAGEMENT

Subscriptions

Contact Requests

Reports

Settings

Back to Dashboard

Subscription Management

1

Welcome, System Administrator

Total Subscribers

0

Monthly Revenue

\$0.00

Active Plans

5

Recent Payments

0

Subscription Plans

+ Add Plan

Plan Name	Price	Billing Cycle	Features	Status	Actions
Freemium Free access to introductory courses	FREE	MONTHLY	View Features	ACTIVE	Actions ▾
Basic Plan Access to recordings and materials	EGP1000.00	MONTHLY	View Features	ACTIVE	Actions ▾
Standard Plan Live sessions and interactive assessments	EGP1500.00	MONTHLY	View Features	ACTIVE	Actions ▾
Premium Plan All features with personalized coaching	EGP3000.00	MONTHLY	View Features	ACTIVE	Actions ▾

Figure 70. Admin (Subscription Page)

5.2.13.5 Pending Pages

5.2.13.5.1 Company Representer:

The screenshot displays the 'Pending Company Approvals' section of the Admin Panel. On the left is a dark sidebar with navigation links: Admin Panel, Dashboard, User Management, APPROVALS (with sub-links for Pending Instructors, Pending Companies, and Pending Courses), and MANAGEMENT (with sub-links for Subscriptions, Contact Requests, Reports, and Settings). A 'Back to Dashboard' link is at the bottom of the sidebar. The main content area has a header 'Pending Company Approvals' with a notification badge showing '2'. Below this, a card for 'Cheses' is shown with a 'PENDING REVIEW' status. The card contains the following information: Contact Person: Mohamed Abdo, Email: abdo@gmail.com, Location: Alex, Company Size: 50, Company Name: Cheses, Founded: 2019, Industry: Foods, and Applied: 2025-06-12 13:03. To the right of this information is a 'Quick Actions' panel with a green 'Approve' button, a red 'Reject' button, and a blue 'View Details' link. At the bottom right of the page, there is a watermark that says 'Activate Windows Go to Settings to activate Windows.'

Figure 71. Pending Pages (Company Representer)

5.2.13.5.2 Instructor:

The screenshot displays the 'Pending Instructor Approvals' section of the Admin Panel. The layout is similar to Figure 71, with the same sidebar on the left. The main content area has a header 'Pending Instructor Approvals' with a notification badge showing '3'. Below this, a card for 'POP ALEX' is shown with a 'PENDING REVIEW' status. The card contains the following information: Email: alexpop@gmail.com, Experience: 3, Department: Costumer Service & Call Center, Applied: 2025-06-12 13:06, Specialization: Costumer Service & Call Center, and Phone: Not provided. To the right of this information is a 'Quick Actions' panel with a green 'Approve' button, a red 'Reject' button, and a blue 'View Details' link. At the bottom right of the page, there is a watermark that says 'Activate Windows Go to Settings to activate Windows.'

Figure 72. Pending Pages (Instructor)

5.2.13.5.2 Reject Section:

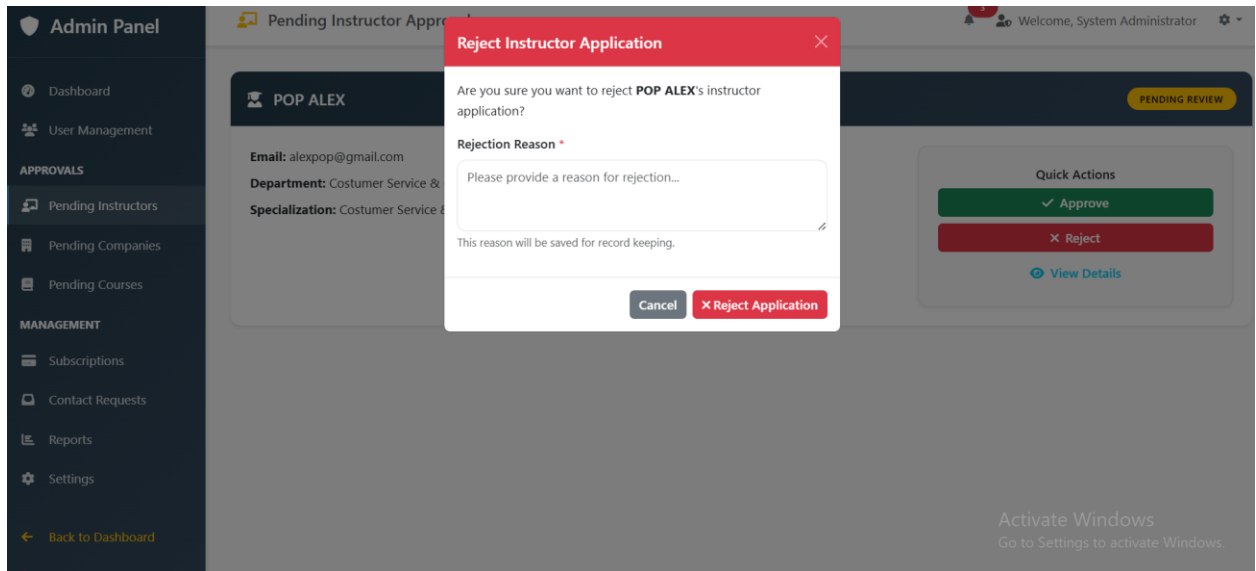


Figure 73. Pending Pages (Reject Section)

5.2.14 System code & database

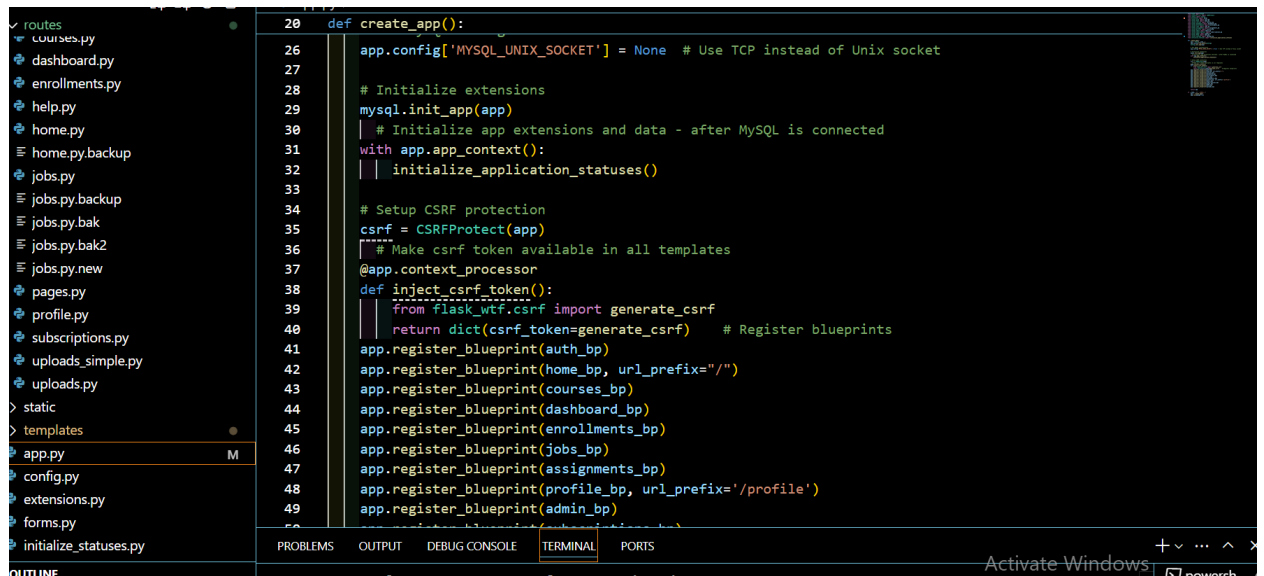
5.2.14.1 Code backend



```
EXPLORER
BIS-TEAM-102-GRADU...
> _old
> _report
> _test
> .vscode
> database
> env
> routes
  > _pycache_
  admin.py
  assignments.py
  auth.py
  auth.py.bak
  courses.py
  dashboard.py
  enrollments.py
  help.py
  home.py
  home.py.backup
  jobs.py
  jobs.py.backup
  jobs.py.bak
  jobs.py.bak2
  jobs.py.new
  pages.py

routes > admin.py > ...
9 import json
10 from werkzeug.security import generate_password_hash
11
12 admin_bp = Blueprint('admin', __name__, url_prefix='/admin')
13
14 @admin_bp.before_request
15 def check_admin():
16     """Ensure only admin users can access admin routes"""
17     if 'user_id' not in session or session.get('user_type') != 'admin':
18         # For API endpoints, return JSON error instead of redirect
19         if request.path.startswith('/admin/api/'):
20             return jsonify({'error': 'Admin authentication required', 'authenticated': False}), 401
21         flash('Access denied. Admin privileges required.', 'danger')
22         return redirect(url_for('auth.login'))
23
24 @admin_bp.route('/')
25 @admin_bp.route('/dashboard')
26 def dashboard():
27     """Admin dashboard with overview statistics"""
28     try:
29         # Get platform statistics
30         stats = {}
31
32         # User statistics
33         stats['total_users'] = get_record('SELECT COUNT(*) as count FROM users')['count']
34
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
127.0.0.1 - - [12/Jun/2025 13:14:16] "GET /admin/api/notifications HTTP/1.1" 200 -
```

Figure 74. Code backend



```
routes
  courses.py
  dashboard.py
  enrollments.py
  help.py
  home.py
  home.py.backup
  jobs.py
  jobs.py.backup
  jobs.py.bak
  jobs.py.bak2
  jobs.py.new
  pages.py
  profile.py
  subscriptions.py
  uploads_simple.py
  uploads.py
  static
  templates
  app.py M
  config.py
  extensions.py
  forms.py
  initialize_statuses.py

20 def create_app():
26     app.config['MYSQL_UNIX_SOCKET'] = None # Use TCP instead of Unix socket
27
28     # Initialize extensions
29     mysql.init_app(app)
30     # Initialize app extensions and data - after MySQL is connected
31     with app.app_context():
32         initialize_application_statuses()
33
34     # Setup CSRF protection
35     csrf = CSRFProtect(app)
36     # Make csrf token available in all templates
37     @app.context_processor
38     def inject_csrf_token():
39         from flask_wtf.csrf import generate_csrf
40         return dict(csrf_token=generate_csrf) # Register blueprints
41     app.register_blueprint(auth_bp)
42     app.register_blueprint(home_bp, url_prefix="/")
43     app.register_blueprint(courses_bp)
44     app.register_blueprint(dashboard_bp)
45     app.register_blueprint(enrollments_bp)
46     app.register_blueprint(jobs_bp)
47     app.register_blueprint(assignments_bp)
48     app.register_blueprint(profile_bp, url_prefix='/profile')
49     app.register_blueprint(admin_bp)
```

Figure 75 Code backend

```

routes > auth.py > ...
1  from flask import render_template, request, redirect, url_for, flash, session, Blueprint
2  from werkzeug.security import generate_password_hash, check_password_hash
3  from forms import LoginForm, RegisterForm
4  from models import get_record, execute_query
5  from permissions import login_required
6  from extensions import mysql
7
8
9  auth_bp = Blueprint('auth', __name__)
10
11 @auth_bp.route('/register', methods=['GET', 'POST'])
12 def register():
13     form = RegisterForm()
14     if form.validate_on_submit():
15         # Check if email already exists
16         existing = get_record('SELECT * FROM users WHERE Email = %s', (form.email.data,))
17         if existing:
18             flash('Email already registered.', 'danger')
19             return render_template('register.html', form=form)
20
21         # Determine approval status and user status based on user type
22         if form.user_type.data == 'student':
23             approval_status = 'Approved'
24             user_status = 'Active'
25         else: # instructor or company
26             approval_status = 'Pendine'

```

127.0.0.1 - - [12/Jun/2025 13:15:46] "GET /admin/api/stats HTTP/1.1" 200 -

Figure 76 Code backend

5.2.14.2 Code frontend:

```

templates > base.html > html > head
1  <!DOCTYPE html>
2  <html lang="en" data-bs-theme="light">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <meta name="description" content="Sec Era - Connecting education with innovative security skills and c
7      <meta name="keywords" content="education, cybersecurity, courses, learning, professional development">
8      <meta name="author" content="Sec Era">
9
10     <!-- Open Graph / Social Media Meta Tags -->
11     <meta property="og:title" content="{% block og_title %}Sec Era{% endblock %}">
12     <meta property="og:description" content="{% block og_description %}Connecting education with innovativ
13     <meta property="og:type" content="website">
14     <meta property="og:url" content="{{ request.url }}">
15     <meta property="og:image" content="{{ url_for('static', filename='img/og-image.png', _external=True) }}">
16
17     <title>{% block title %}Sec Era{% endblock %}</title>
18
19     <!-- Favicon and PWA -->
20     <link rel="manifest" href="{{ url_for('static', filename='manifest.json') }}">
21     <link rel="apple-touch-icon" sizes="180x180" href="{{ url_for('static', filename='img/apple-touch-icon
22     <link rel="icon" type="image/png" sizes="32x32" href="{{ url_for('static', filename='img/favicon-32x32
23     <link rel="icon" type="image/png" sizes="16x16" href="{{ url_for('static', filename='img/favicon-16x16
24
25     <!-- Preload Critical Resources -->
26     <link rel="preload" href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" as=

```

127.0.0.1 - - [12/Jun/2025 13:16:35] "GET /admin/api/notifications HTTP/1.1" 200 -

Figure 77 Code frontend

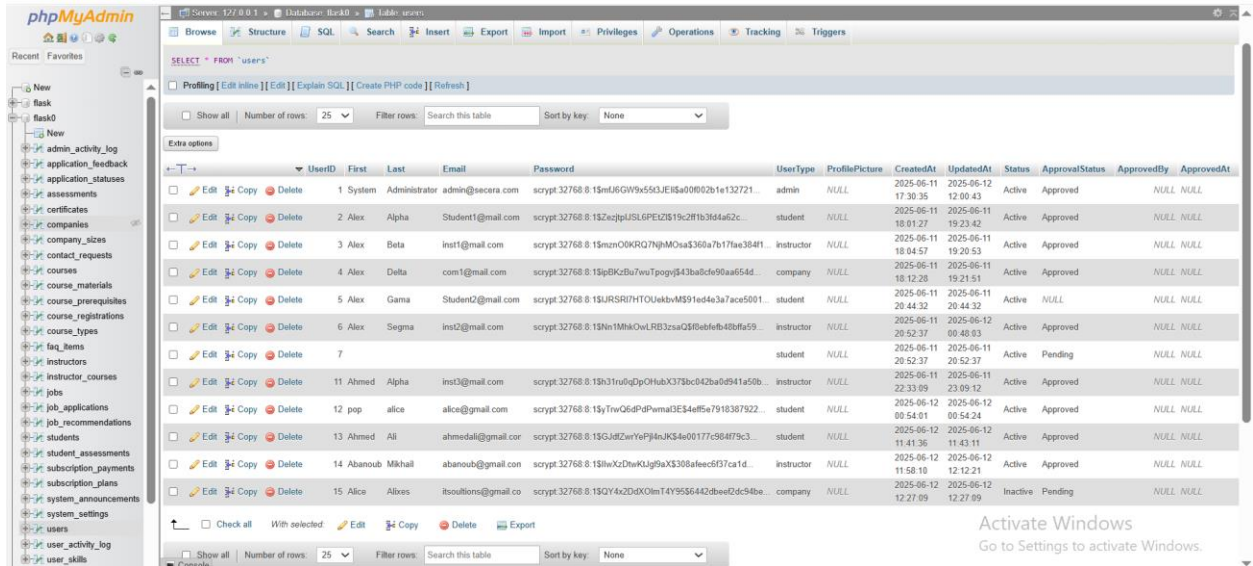
```
1 {% extends "base.html" %}
2
3 {% block content %}
4 <div class="container">
5   <div class="row mt-4">
6     <div class="col-md-12">
7       <h1>Welcome to our Learning Platform</h1>
8       <div class="card-deck mt-4">
9         <div class="card">
10           <div class="card-body">
11             <h5 class="card-title">For Students</h5>
12             <p class="card-text">Access courses, assignments, and connect with instructors.</p>
13           </div>
14         </div>
15         <div class="card">
16           <div class="card-body">
17             <h5 class="card-title">For Instructors</h5>
18             <p class="card-text">Create courses, manage assignments, and interact with student
19           </div>
20         </div>
21         <div class="card">
22           <div class="card-body">
23             <h5 class="card-title">For Companies</h5>
24             <p class="card-text">Post job opportunities and connect with talented students.</p>
25           </div>
26         </div>
27       </div>
28     </div>
29   </div>
30 </div>
31 {% endblock %}
```

Figure 78 Code frontend

```
1 {% extends "base.html" %}
2
3 {% block title %}Login - Education System{% endblock %}
4
5 {% block extra_css %}
6 <style>
7   .login-container {
8     max-width: 450px;
9     margin: 4rem auto;
10    padding: 2rem;
11    background-color: #ffffff;
12    border-radius: 10px;
13    box-shadow: 0 0 20px rgba(0, 0, 0, 0.1);
14  }
15
16  .login-header {
17    text-align: center;
18    margin-bottom: 2rem;
19  }
20
21  .login-header h2 {
22    color: #333;
23    font-weight: 600;
24  }
25
26  .form-group {
```

Figure 79 Code frontend

5.2.14.3 Database:



The screenshot displays the phpMyAdmin web interface. On the left is a sidebar with a tree view of database tables, including 'admin_activity_log', 'application_statuses', 'assessments', 'certificates', 'companies', 'company_sizes', 'contact_requests', 'courses', 'course_materials', 'course_prerequisites', 'course_registrations', 'course_types', 'faq_items', 'instructors', 'instructor_courses', 'jobs', 'job_applications', 'job_recommendations', 'students', 'student_assessments', 'subscription_payments', 'subscription_plans', 'system_announcements', 'system_settings', 'users', 'user_activity_log', and 'user_skills'. The main panel shows the 'users' table selected. At the top, there's a SQL query editor with the text 'SELECT * FROM `users`'. Below this are tabs for 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', 'Privileges', 'Operations', 'Tracking', and 'Triggers'. The 'Structure' tab is active, showing the table's columns and their data types. Below the column list, there's a table of 15 rows of user data. Each row has icons for 'Edit', 'Copy', and 'Delete' actions. At the bottom right, there's a watermark that says 'Activate Windows Go to Settings to activate Windows.'

	UserID	First	Last	Email	Password	UserType	ProfilePicture	CreatedAt	UpdatedAt	Status	ApprovalStatus	ApprovedBy	ApprovedAt
☐	1	System	Administrator	admin@secera.com	script.32768.8.15mU6GW9x55GJEi8a0902b1e132721...	admin	NULL	2025-06-11 17:30:35	2025-06-12 12:00:43	Active	Approved	NULL	NULL
☐	2	Alex	Alpha	Student1@mail.com	script.32768.8.11ZecjplUSLGPxiZ619c291b364a62c...	student	NULL	2025-06-11 18:01:27	2025-06-11 19:23:42	Active	Approved	NULL	NULL
☐	3	Alex	Beta	inst1@mail.com	script.32768.8.15mznO0KRQ7NpMOsa360a7b17faw384f1...	instructor	NULL	2025-06-11 19:04:57	2025-06-11 19:20:53	Active	Approved	NULL	NULL
☐	4	Alex	Delta	com1@mail.com	script.32768.8.15pBKzBu7wu7pog9\$43ba8c96aa654d...	company	NULL	2025-06-11 18:12:28	2025-06-11 19:21:51	Active	Approved	NULL	NULL
☐	5	Alex	Gamma	Student2@mail.com	script.32768.8.15JRSRlHTOUakbM991e04a3a7ace5001...	student	NULL	2025-06-11 20:44:32	2025-06-11 20:44:32	Active	NULL	NULL	NULL
☐	6	Alex	Sigma	inst2@mail.com	script.32768.8.15Nn1MhKQvLRB3zuaQ580eb4fb48bfa59...	instructor	NULL	2025-06-11 20:52:37	2025-06-12 00:48:03	Active	Approved	NULL	NULL
☐	7					student	NULL	2025-06-11 20:52:37	2025-06-11 20:52:37	Active	Pending	NULL	NULL
☐	11	Ahmed	Alpha	inst3@mail.com	script.32768.8.15h31ru0qDpCHubX378bc042ba0d941a50b...	instructor	NULL	2025-06-11 22:33:09	2025-06-11 23:09:12	Active	Approved	NULL	NULL
☐	12	pop	alice	alice@gmail.com	script.32768.8.15yTnvQ6dPpPema1E34ef5e7918387822...	student	NULL	2025-06-12 00:54:01	2025-06-12 00:54:24	Active	Approved	NULL	NULL
☐	13	Ahmed	Ali	ahmedali@gmail.com	script.32768.8.15GJdZerYeFPj4nJK54e00177c584f79c3...	student	NULL	2025-06-12 11:41:36	2025-06-12 11:43:11	Active	Approved	NULL	NULL
☐	14	Abanoub	Mikhail	abanoub@gmail.com	script.32768.8.15lhwXzDhwKJg9uX3308afec6f37ca1d...	instructor	NULL	2025-06-12 11:58:10	2025-06-12 12:12:21	Active	Approved	NULL	NULL
☐	15	Alice	Alises	itsoultions@gmail.co	script.32768.8.15QY4x2DdXOlmT4Y956402dhw02dc94be...	company	NULL	2025-06-12 12:27:09	2025-06-12 12:27:09	Inactive	Pending	NULL	NULL

Figure 80. Database

Chapter Six

6. TESTING METHODOLOGY

- 6.1. Introduction (Aim of the chapter)
- 6.2. Testing methodology
- 6.3. Login validation
- 6.4. Sign up validation
- 6.5. Main functions validation

Chapter Six: Testing Methodology

6.1 Introduction

System testing is the level of testing that validates the complete and fully integrated software product. The purpose of a system test is to evaluate the end-to-end system specification. Usually, the software is only one element of a larger computer-based system. Ultimately, the software is interfaced with other software/hardware systems. System testing is a series of different tests whose sole purpose is to exercise the full computer-based system.

6.2 Testing Methodology

In this project, agile methodology for testing has been selected for the following reasons:

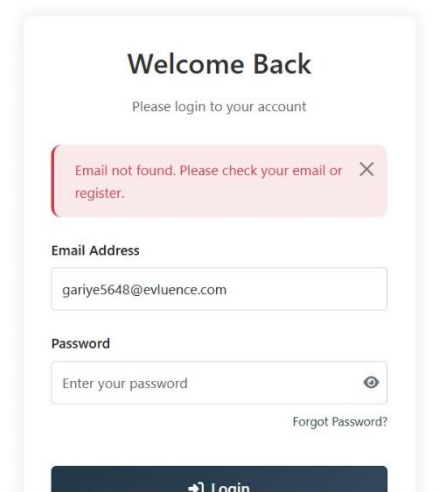
- An agile test is an incremental model where tests are executed at the end of each iteration or increment.
- It can cover the sphere of tests and software development, and marketing.
- The outcomes are better with agile methodology when an experienced and strong product manager can make rapid decisions.
- Besides, the complete app is ideally tested at the end of the project. There is a lesser threat in the development procedure with the agile technique as every team member understands what has or has not been completed.
- Use of **server-side and client-side validations**.
- Application of **security measures** like CSRF protection and password hashing.
- Implementation of **data type validations** and business logic constraints.
- Integration of **database-level constraints** supporting validation integrity.
- Use of enhanced frontend components for better user experience.

6.3 Login Validation

If the user enters a wrong email and password on the login page, a message indicating the wrong email and password will appear.

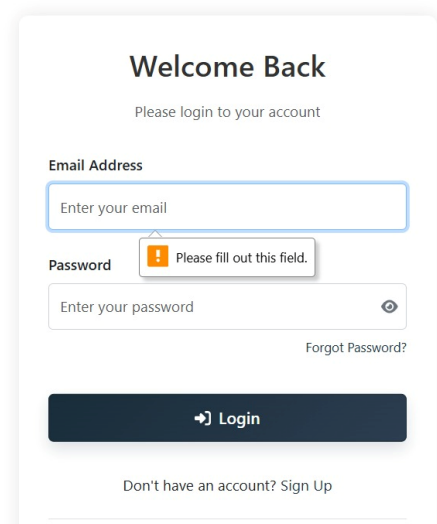
6.3.1 Validations related to the login process include:

- **Form Validations (LoginForm):**
 - *email*: DataRequired(), Email(), Length(max=255)
 - *password*: DataRequired()
- **Authentication & Authorization Validations:**
 - Server-side checks to confirm:
 - Email exists in database
 - Password verification (hashed comparison)
 - Account approval status (Pending/Approved)
 - Account activity status (Active/Inactive)
 - Permission decorators like `@login_required` ensure user authentication.
- **Security Validations:**
 - CSRF protection applied to login form.
 - Secure password handling with hashing.
- **Frontend Validations:**
 - Real-time validation on email format and required fields.



The image shows a login form titled "Welcome Back" with the subtitle "Please login to your account". A red error message box at the top states: "Email not found. Please check your email or register." Below this, the "Email Address" field contains the text "gariye5648@evluence.com". The "Password" field is empty and has a toggle icon. A "Forgot Password?" link is visible. At the bottom is a dark blue "Login" button with a right arrow icon.

Figure 81. Login Validation



The image shows the same login form as Figure 81, but with real-time validation feedback. The "Email Address" field has a blue border and a tooltip that says "Please fill out this field." The "Password" field also has a tooltip that says "Please fill out this field." and a toggle icon. A "Forgot Password?" link is visible. At the bottom is a dark blue "Login" button with a right arrow icon. Below the button is a link: "Don't have an account? Sign Up".

Figure 82. Login Validation

6.4 Sign Up Validation

6.4.1 Validations covering the user registration process include:

- **Form Validations (RegisterForm):**
 - Common fields: *first_name*, *last_name*, email, password, *user_type*
 - Student-specific: university, major, GPA (0-4), expected graduation date, bio, resume
 - Instructor-specific: department, specialization, experience (≥ 0), bio, qualifications
 - Company-specific: *company_name*, industry, location, *company_size* (≥ 1), founded date, bio, website
- **Business Logic Validations:**
 - User-type specific validations ensure data relevance (e.g., GPA range for students, positive company size for companies).
- **Security Validations:**
 - Password minimum length enforced (≥ 6).
 - Passwords hashed securely.
 - Email formats strictly validated.
- **Frontend Validations:**
 - Conditional fields appear based on selected user type.
 - Real-time feedback for errors like invalid GPA or weak password.

Basic Information

First Name ⓘ Last Name ⓘ

Email ⓘ
Please enter a valid email address.

Password ⓘ
Password must be at least 6 characters long.

Password strength: None

User Type ⓘ
Select User Type

Create Account

Already have an account? [Login here](#)

Figure 83. Sign Up Validation

Email
testpop

Invalid email address.

Figure 85. Sign Up Validation

Instructor Information

Department: 234edmqiwop ✓ Specialization: cwetg34vxfv4 ✓

Years of Experience: 344e357 ⓘ

Figure 84. Sign Up Validation

6.5 Main Function Validation

The following are the main function validations:

6.5.1 Validations encompassing core application features beyond login and sign up:

- **Form Validations:**
 - *ProfileForm*: Validates profile updates (name, email, bio).
 - *PasswordChangeForm*: Ensures new passwords meet security standards and confirmation matches.
 - *CourseForm*: Validates course details (title, dates with logical sequence, pricing positive).
 - *JobForm*: Checks job posting details (salary ranges with $\text{max} \geq \text{min}$, experience).
 - *AssessmentForm*: Validates assessments (due dates in future, positive scoring).
 - *ContactForm*: Ensures all required contact fields are provided.
- **Business Logic Validations:**
 - Enforces constraints like salary range correctness, course date logic, and assessment scoring rules.
- **Frontend Validations:**
 - Real-time validations across all forms for immediate user feedback.
- **Database Validation Layer:**
 - SQL constraints to enforce data integrity supporting above validations.

6.5.2 Jumping Attack Validation

- A user cannot access website pages without logging in.
- A logged-in user cannot access the login page again; attempting to do so will redirect them to their respective home page (Student, Instructor, or Company Representer).
- If a non-logged-in user tries to access a restricted page, they will be redirected to the login page.

C h a p t e r S e v e n

7. CONCLUSION AND FUTURE WORK

7.1. Introduction (Aim of the chapter)

7.2. Conclusion

- A summary of our key arguments and results.
- A short discussion of the implications of our research.
- Summarize and reflect on our research process
- Show what new knowledge we have contributed to our field.

7.3. Future Work

Chapter Seven Conclusion and Future Work

7.1 Introduction (Aim of the Chapter)

This chapter aims to encapsulate the key achievements of the developed educational platform, reflect on the development process, and outline potential future enhancements. It highlights the significance of integrating practical training with academic learning and explores how the current system can evolve to better serve students, educational institutions, and industry partners.

7.2 Conclusion

The project successfully designed and implemented a comprehensive platform that bridges the gap between theoretical education and practical workplace skills. By collaborating with companies, the system offers final-year students and recent graduates specialized courses and hands-on assessments, effectively reducing internship durations and costs while enhancing their readiness for the labor market.

The key outcomes demonstrate that the platform can streamline skill development, facilitate effective collaboration between academia and industry, and provide a scalable solution adaptable to various educational fields. The system's features—such as user management for students, instructors, company representatives, and administrators—contribute to operational efficiency and improved user engagement.

Throughout the development process, careful analysis of user requirements, system design, and rigorous testing ensured the platform's reliability and usability. This project contributes to the field by illustrating an integrated approach to educational management systems that combine content delivery, practical assessments, and industry collaboration, aligning with modern trends in digital education.

The implications of this research emphasize the critical role of technology in transforming traditional education into a more practical, industry-relevant experience. It underscores the importance of continuous innovation to meet the evolving needs of students and industries, ultimately fostering a more competent and job-ready workforce.

7.3 Future Work

Building upon the current system, several enhancements are envisioned to expand its functionality and impact:

- **Artificial Intelligence Integration:** Incorporate AI algorithms to provide personalized learning paths, automate assessments, and generate detailed analytics for students' performance and engagement, enabling more tailored educational experiences.
- **Mobile Application Development:** Develop a dedicated mobile app to increase platform accessibility for students, instructors, and industry representatives, facilitating learning and collaboration on-the-go.
- **Expanded and Specialized Course Content:** Introduce new courses in emerging fields such as Artificial Intelligence, Data Science, Cybersecurity, and Digital Marketing, aligning with industry demands.
- **Real-Time Industry Collaboration Tools:** Integrate live communication features like video conferencing, instant messaging, and collaborative document editing to enhance interaction between students and companies.
- **Gamification and Certification:** Implement gamified elements—such as badges, leaderboards, and achievement milestones—to motivate learners and provide recognized certifications that boost employability.
- **Enhanced Data Privacy and Security:** Strengthen security measures to ensure compliance with data protection laws and safeguard user information against potential breaches.
- **Advanced Analytics Dashboard:** Develop comprehensive dashboards for administrators and industry partners to monitor progress, engagement metrics, and outcome evaluations, supporting data-driven decision-making.
- **Integration with Existing Educational Platforms:** Explore options to connect with other Learning Management Systems (LMS) and industry databases to broaden the system's interoperability and reach.

These future features aim to make the platform more dynamic, user-centric, and adaptable to the rapidly changing educational landscape and industry needs, ensuring its sustainability and scalability.

References

- <https://www.trade.gov/country-commercial-guides/egypt-information-and-communications-technology-and-digital-economy>
- <https://itida.gov.eg/English/Programs/Industry-Outlook/Pages/default.aspx>
- https://mcit.gov.eg/en/ICT_Strategy
- <https://app.diagrams.net/>
- <https://laravel.com/docs/9.x/>
- <https://stackoverflow.com/>
- <https://www.w3schools.com>
- <https://www.mermaidchart.com/>
- <https://www.godaddy.com/en/domainsearch/find?domainToCheck=Second.Era>
- <https://tradingeconomics.com/egypt/social-security-rate>
- <https://taxsummaries.pwc.com/egypt/individual/other-taxes#:~:text=Social%20security%20contributions&text=The%20employee%20contribution%20percentage%20is,starting%20from%201%20January%202021.>